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(54) **ORGANOMETALLIC COMPOUND AND
ORGANIC LIGHT-EMITTING DEVICE
INCLUDING THE SAME**

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H10K 50/11 (2023.01)

(Continued)

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(2023.02); **H10K 50/155** (2023.02); **H10K**
50/16 (2023.02); **H10K 2101/10** (2023.02)

(58) **Field of Classification Search**

CPC **C07F 15/0033**; **H01L 51/0085**; **H10K**
50/155; **H10K 50/16**; **H10K 50/11**; **H10K**
2101/10

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,580,961 B2 11/2013 Chi et al.
8,795,853 B2 8/2014 Oshiyama et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 1777663 A 5/2006
CN 105801627 A 7/2016
(Continued)

OTHER PUBLICATIONS

Kuei, et al. Advanced Materials, 2016, vol. 28, pp. 2795-2800 (Year:
2016).*

(Continued)

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(57)

ABSTRACT

An organometallic compound is represented by Formula 1.
An organic light-emitting device includes: a first electrode;
a second electrode facing the first electrode; an organic layer
between the first electrode and the second electrode and
including an emission layer; and at least one of the organo-
metallic compound represented by Formula 1.

21 Claims, 4 Drawing Sheets

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190

150

110

(2023.01)

References Cited

FOREIGN PATENT DOCUMENTS

CN	109251223	A	1/2019
CN	109285956	A	1/2019
KR	10-2017-0080453	A	7/2017
KR	10-2018-0109735	A	10/2018
WO	WO 2006/082742	A	8/2006

Kuo, et al. *Advanced Materials*, 2017, vol. 29, 1702464 (Year: 2017).*

Kuo, et al., *Advanced Science*, 2018, vol. 5, pp. 1800846-1 through 1800846-7 (Year: 2018).*

Chen, et al., *Inorganic Chemistry*, 2010, vol. 49, pp. 823-832 (Year: 2010).*

Kuei, et al., *Advanced Materials*, 2016, 28, 2795-2800 (Year: 2016).*

Breivogel, et al. *Eur. J. Inorg. Chem.*, 2014, 288-295 (Year: 2014).*

* cited by examiner

FIG. 1

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150
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FIG. 2

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150
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FIG. 3

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FIG. 4

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ORGANOMETALLIC COMPOUND AND ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to and the benefit of Korean Patent Application No. 10-2019-0064072, filed on May 30, 2019, in the Korean Intellectual Property Office, the entire content of which is incorporated herein by reference.

BACKGROUND

1. Field

One or more embodiments relate to an organometallic compound and an organic light-emitting device including the same.

2. Description of the Related Art

Organic light-emitting devices are self-emission devices that produce full-color images, and also have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of brightness, driving voltage, and response speed, as compared to other devices in the art.

The organic light-emitting device may include a first electrode on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode, which are sequentially on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transit (e.g., transition or relax) from an excited state to a ground state, thereby generating light.

SUMMARY

One or more embodiments include an organometallic compound and an organic light-emitting device including the same.

Additional aspects of embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

An aspect of an embodiment of the present disclosure provides an organometallic compound represented by Formula 1:



Formula 1

In Formula 1,

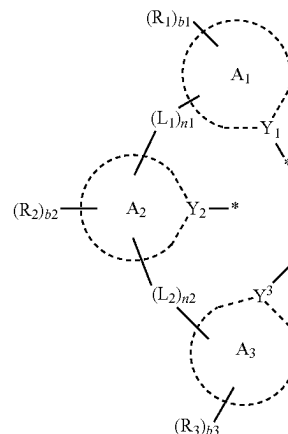
M may be selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm),

L_A may be a tridentate ligand represented by Formula 1A, and

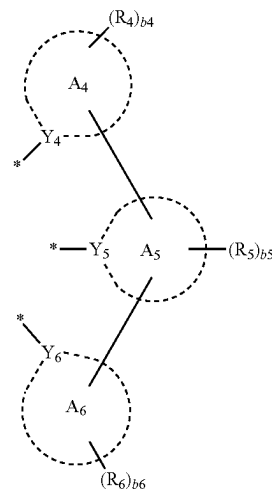
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L_B may be a tridentate ligand represented by Formula 1B:

Formula 1A



Formula 1B



In Formulae 1A and 1B,

ring A_1 may be a 5-membered ring including Y_1 , a C_5 - C_{60} carbocyclic ring in which a 5-membered ring including Y_1 is condensed, or a C_1 - C_{60} heterocyclic ring in which a 5-membered ring including Y_1 is condensed,

ring A_2 to ring A_6 may each independently be selected from a C_5 - C_{60} carbocyclic ring and a C_1 - C_{60} heterocyclic ring,

Y_1 to Y_6 may each independently be N or C,

L_1 and L_2 may each independently be selected from $*-O-*$, $*-S-*$, $*-C(R_7)(R_8)-*$, $*-C(R_7)=C(R_8)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R_7)-*$, $*-N(R_7)-*$, $*-P(R_7)-*$, $*-Si(R_7)(R_8)-*$, $*-P(R_7)(R_8)-*$, and $*-Ge(R_7)(R_8)-*$,

n_1 may be selected from 1, 2, and 3, and n_2 may be selected from 0, 1, 2, and 3,

R_1 to R_8 may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a

3

substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-\text{Si}(\text{Q}_1)(\text{Q}_2)(\text{Q}_3)$, $-\text{N}(\text{Q}_1)(\text{Q}_2)$, $-\text{B}(\text{Q}_1)(\text{Q}_2)$, $-\text{C}(=\text{O})(\text{Q}_1)$, $-\text{S}(=\text{O})_2(\text{Q}_1)$, and $-\text{P}(=\text{O})(\text{Q}_1)(\text{Q}_2)$, and two neighboring $\text{R}_1(\text{s})$ among $\text{R}_1(\text{s})$ in the number of b1, two neighboring $\text{R}_2(\text{s})$ among $\text{R}_2(\text{s})$ in the number of b2, two neighboring $\text{R}_3(\text{s})$ among $\text{R}_3(\text{s})$ in the number of b3, two neighboring $\text{R}_4(\text{s})$ among $\text{R}_4(\text{s})$ in the number of b4, two neighboring $\text{R}_5(\text{s})$ among $\text{R}_5(\text{s})$ in the number of b5, two neighboring $\text{R}_6(\text{s})$ among $\text{R}_6(\text{s})$ in the number of b6, and at least one of R_1 and R_7 , R_2 and R_7 , and R_3 and R_7 may optionally be linked to form a substituted or unsubstituted C_5 - C_{60} carbocyclic group or a substituted or unsubstituted C_1 - C_{60} heterocyclic group, b1 to b6 may each independently be an integer from 1 to 10,

* indicates a binding site to M,

at least one substituent of the substituted C_5 - C_{60} carbocyclic group, the substituted C_1 - C_{60} heterocyclic group, the substituted C_1 - C_{60} alkyl group, the substituted C_2 - C_{60} alkenyl group, the substituted C_2 - C_{60} alkynyl group, the substituted C_1 - C_{60} alkoxy group, the substituted C_3 - C_{10} cycloalkyl group, the substituted C_1 - C_{10} heterocycloalkyl group, the substituted C_3 - C_{10} cycloalkenyl group, the substituted C_1 - C_{10} heterocycloalkenyl group, the substituted C_6 - C_{60} aryl group, the substituted C_6 - C_{60} aryloxy group, the substituted C_6 - C_{60} arylthio group, the substituted C_1 - C_{60} heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from:

deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, and a C_1 - C_{60} alkoxy group;

a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, and a C_1 - C_{60} alkoxy group, each substituted with at least one selected from deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $-\text{Si}(\text{Q}_{11})(\text{C}_{212})(\text{Q}_{13})$, $-\text{N}(\text{Q}_{11})(\text{Q}_{12})$, $-\text{B}(\text{Q}_{11})(\text{Q}_{12})$, $-\text{C}(=\text{O})(\text{Q}_{11})$, $-\text{S}(=\text{O})_2(\text{Q}_{11})$, and $-\text{P}(=\text{O})(\text{Q}_{11})(\text{Q}_{12})$;

a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed

4

polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group;

a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $-\text{Si}(\text{Q}_{21})(\text{Q}_{22})(\text{Q}_{23})$, $-\text{N}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{B}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{C}(=\text{O})(\text{Q}_{21})$, $-\text{S}(=\text{O})_2(\text{Q}_{21})$, and $-\text{P}(=\text{O})(\text{Q}_{21})(\text{Q}_{22})$; and $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, and $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, and

Q_1 to Q_3 , Q_{11} to Q_{13} , Q_{21} to Q_{23} , and Q_{31} to Q_{33} may each independently be selected from hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

Another aspect of an embodiment of the present disclosure provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and including an emission layer; and at least one organometallic compound represented by Formula 1.

BRIEF DESCRIPTION OF THE DRAWINGS

These and/or other aspects of embodiments will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:

FIG. 1 is a schematic view of an organic light-emitting device according to an embodiment;

FIG. 2 is a schematic view of an organic light-emitting device according to another embodiment;

FIG. 3 is a schematic view of an organic light-emitting device according to another embodiment; and

FIG. 4 is a schematic view of an organic light-emitting device according to another embodiment.

DETAILED DESCRIPTION

Reference will now be made in more detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodi-

5

ments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described herein below, by referring to the figures, to explain aspects of embodiments of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

An organometallic compound according to an embodiment may be represented by Formula 1:

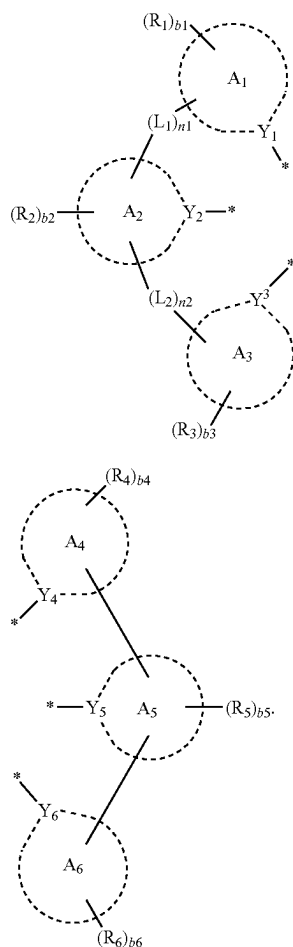
$$ML_A L_B$$

Formula 1

In Formula 1, M may be selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm).

In one embodiment, M may be iridium or rhodium.

In Formula 1, L_A may be a tridentate ligand represented by Formula 1A, and L_B may be a tridentate ligand represented by Formula 1B:

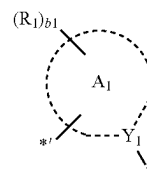


In Formula 1A, ring A_1 may be a 5-membered ring including Y_1 , a C_5-C_{60} carbocyclic ring in which the 5-membered

6

ring including Y_1 is condensed, or a C_1-C_{60} heterocyclic ring in which the 5-membered ring including Y_1 is condensed.

The 5-membered ring including Y_1 refers to a 5-membered ring including Y_1 as a ring-forming atom. The C_5-C_{60} carbocyclic ring in which the 5-membered ring including Y_1 is condensed refers to a polycyclic group in which the 5-membered ring including Y_1 and at least one C_5-C_{60} carbocyclic ring are condensed. The C_1-C_{60} heterocyclic ring in which the 5-membered ring including Y_1 is condensed refers to a polycyclic group in which the 5-membered ring including Y_1 and at least one C_1-C_{60} heterocyclic ring are condensed.

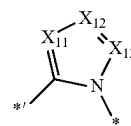


In Formula 1A, a group represented by may be selected from groups represented by Formulae A1-1 to A1-7:

Formula 1A

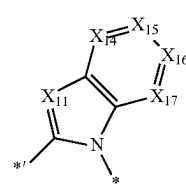
A1-1

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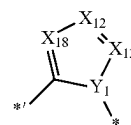
A1-2

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A1-3

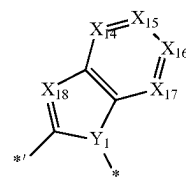
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Formula 1B

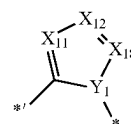
A1-4

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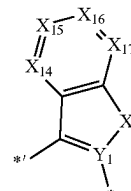
A1-5

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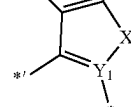


A1-6

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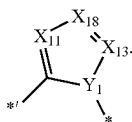
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-continued



In Formulae A1-1 to A1-7,

X_{11} may be $C(R_{11})$ or N, X_{12} may be $C(R_{12})$ or N, X_{13} may be $C(R_{13})$ or N, X_{14} may be $C(R_{14})$ or N, X_{15} may be $C(R_{15})$ or N, X_{16} may be $C(R_{16})$ or N, and X_{17} may be $C(R_{17})$ or N,

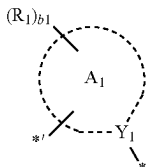
X_{18} may be $C(R_{18})(R_{19})$, $Si(R_{18})(R_{19})$, $N(C_{18})$, $P(C_{18})$, O, or S,

R_{11} to R_{19} may each independently be the same as defined in connection with R_1 , and

* indicates a binding site to M, and * indicates a binding site to $(L_1)_{a1}$.

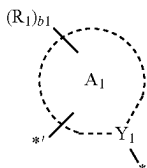
In one embodiment, in Formulae A1-1 to A1-7, R_{11} to R_{19} may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkyl group substituted with at least one —F, a C_1 - C_{10} alkyl group substituted with at least one —Cl, a C_1 - C_{10} alkyl group substituted with at least one —Br, a C_1 - C_{10} alkyl group substituted with at least one —I, a C_1 - C_{10} alkoxy group, a C_6 - C_{20} aryl group, and a C_1 - C_{20} heteroaryl group.

In one embodiment, a group represented by



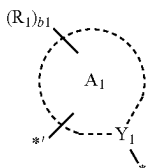
in Formula 1A may be represented by Formula A1-1, X_{11} may be $C(R_{11})$, X_{12} may be $C(R_{12})$, and X_{13} may be N.

In one embodiment, a group represented by



in Formula 1A may be represented by Formula A1-1, X_{11} may be N, X_{12} may be $C(R_{12})$, and X_{13} may be N.

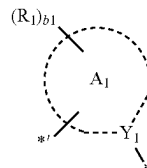
In one embodiment, a group represented by



in Formula 1A may be represented by Formula A1-1, X_{11} may be $C(R_{11})$, X_{12} may be $C(R_{12})$, X_{13} may be N, and at

8

least one of R_u and R_{12} may be selected from a cyano group, a C_1 - C_{10} alkyl group substituted with at least one —F, a C_1 - C_{10} alkyl group substituted with at least one —Cl, a C_1 - C_{10} alkyl group substituted with at least one —Br, and a C_1 - C_{10} alkyl group substituted with at least one —I. For example, at least one of R_u and R_{12} may be

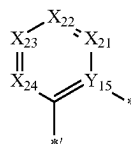


a cyano group or —CF₃. In another embodiment, a group represented by in Formula 1A may be represented by Formula A1-1, X_{11} may be N, X_{12} may be $C(R_{12})$, X_{13} may be N, and R_{12} may be selected from a cyano group, a C_1 - C_{10} alkyl group substituted with at least one —F, a C_1 - C_{10} alkyl group substituted with at least one —Cl, a C_1 - C_{10} alkyl group substituted with at least one —Br, and a C_1 - C_{10} alkyl group substituted with at least one —I. For example, R_{12} may be a cyano group or —CF₃.

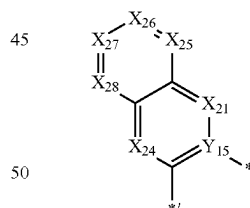
In Formulae 1A and 1B, ring A_2 to ring A_6 may each independently be selected from a C_5 - C_{60} carbocyclic ring and a C_1 - C_{60} heterocyclic ring.

In one embodiment, in Formulae 1A and 1B, ring A_3 , ring A_4 , and ring A_6 may each independently be selected from groups represented by Formulae 2-1(1) to 2-1(27), and ring A_2 and ring A_5 may each independently be selected from groups represented by Formulae 2-2(1) to 2-2(19):

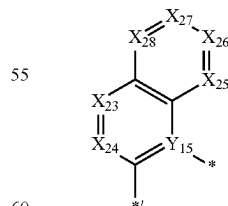
2-1(1)



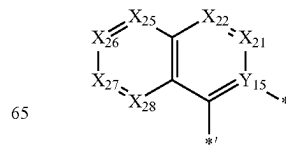
2-1(2)



2-1(3)

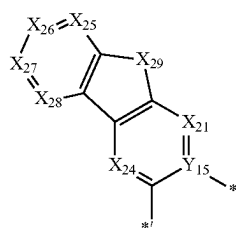
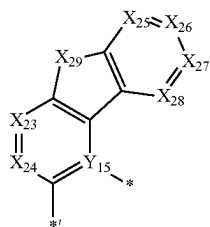
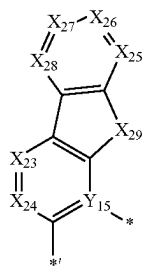
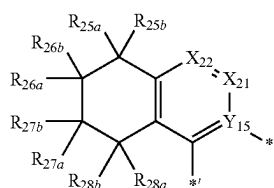
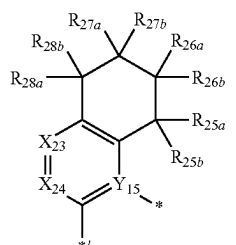
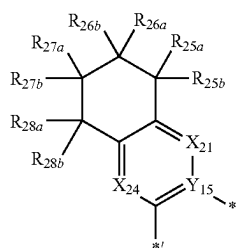


2-1(4)



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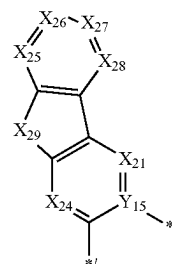
**10**

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2-1(5)

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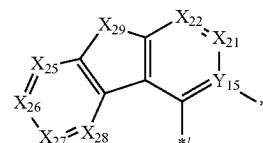
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2-1(11)

2-1(6) 15

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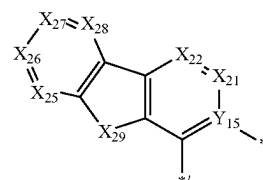


2-1(12)

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2-1(7)

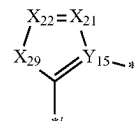
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2-1(13)

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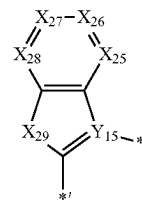
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2-1(14)

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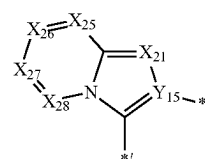
2-1(15)



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2-1(9)

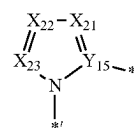
2-1(16)



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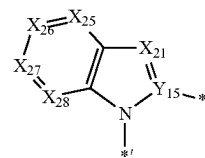
2-1(17)



2-1(10)

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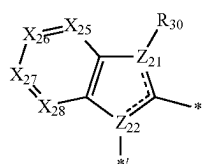
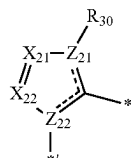
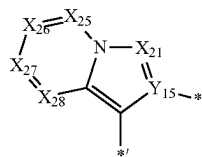
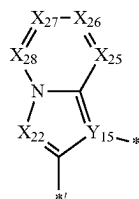
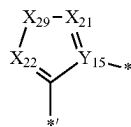
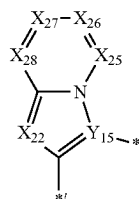
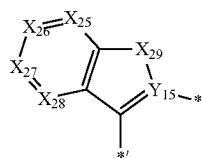
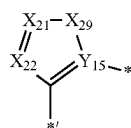
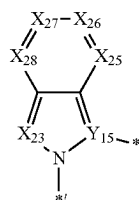
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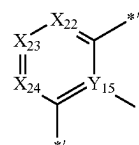


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2-1(19)

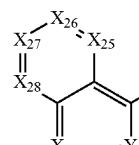
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2-2(1)

2-1(20)

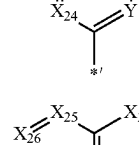
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2-2(2)

2-1(21)

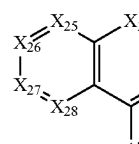
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2-2(3)

2-1(22)

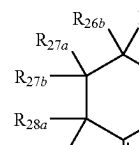
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2-2(4)

2-1(23)

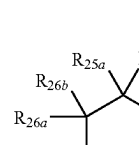
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2-2(5)

2-1(24)

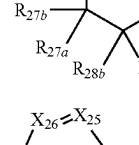
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2-2(6)

2-1(25)

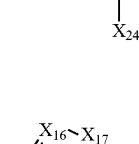
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2-2(7)

2-1(26)

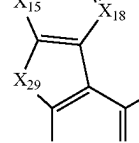
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2-2(8)

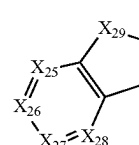
2-1(27)

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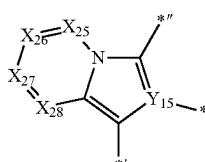
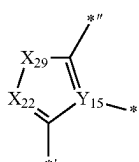
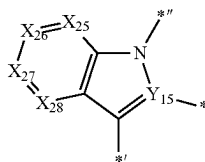
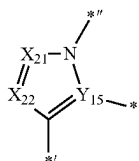
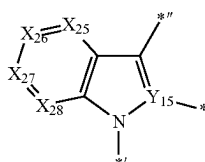
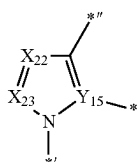
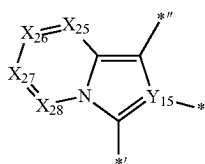
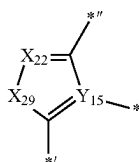
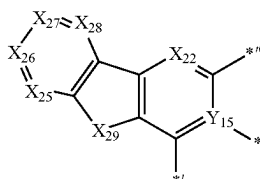
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13

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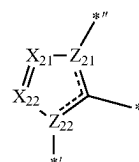


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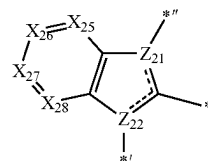
2-2(9)

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2-2(10) 10

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2-2(11)

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In Formulae 2-1(1) to 2-1(27) and 2-2(1) to 2-2(19),
Y₁₅ may be N or C bonded to M,
X₂₁ may be N or C(R₂₁), X₂₂ may be N or C(R₂₂), X₂₃
may be N or C(R₂₃), X₂₄ may be N or C(R₂₄), X₂₅ may
be N or C(R₂₅), X₂₆ may be N or C(R₂₆), X₂₇ may be
N or C(R₂₇), and X₂₈ may be N or C(R₂₈),
X₂₉ may be C(R_{29a})(R_{29b}), Si(R_{29a})(R_{29b}), N(R₂₉), O, or
S,

2-2(12)

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X₃₀ may be C(R_{30a})(R_{30b}), Si(R_{30a})(R_{30b}), N(R₃₀), O, or
S,

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Z₂₁ and Z₂₂ may each independently be N or P,
R₂₁ to R₃₀, R_{25a} to R_{30a} and R_{25b} to R_{30b} may each
independently be the same as defined in connection
with R₂ to R₆,

2-2(13)

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* indicates a binding site to M, and
*' and *'' each indicate a binding site to a neighboring
atom.

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In one embodiment, in Formulae 1A and 1B, ring A₂ and
ring A₃ may each independently be a group represented by
Formula 2-2(1), A₃ may be a group represented by Formula
2-1(1), and A₄ and A₆ may each independently be a group
represented by Formula 2-1(17) or 2-1(26).

2-2(14)

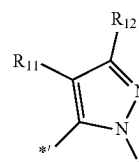
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In one embodiment, in Formula 1A, ring A₁ may be a
group represented by Formula A1-11 or A1-12, ring A₂
may be a group represented by Formula A2-11 or A2-12, and ring
A₃ may be a group represented by Formula A3-11, A3-12, or
A3-13:

2-2(15)

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A1-11

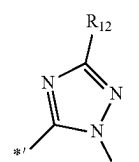


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2-2(16)

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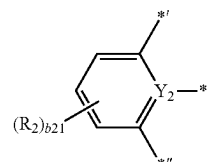
A1-12



2-2(17)

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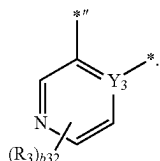
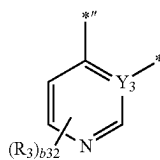
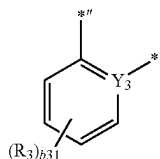
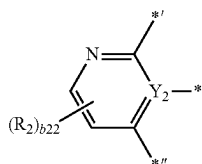
A2-11



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In Formulae A1-11, A1-12, A2-11, A2-12, and A3-11 to A3-13,

Y_2 , Y_3 , R_{11} , R_{12} , R_2 , and R_3 may each independently be the same as described herein,

b_{21} and b_{32} may each independently be an integer from 1 to 3,

b_{22} may be 1 or 2,

b_{31} may be an integer from 1 to 4,

$*$ indicates a binding site to M,

$*$ indicates a binding site to $(L_1)_{n1}$, and

$*$ indicates a binding site to $(L_2)_{n2}$.

In Formulae 1A and 1B, Y_1 to Y_6 may each independently be N or C.

In one embodiment, in Formula 1A, Y_{11} and Y_2 may each be N, and Y_3 may be C,

a bond between Y_{11} and M may be a covalent bond,

a bond between Y_2 and M may be a coordinate bond, and

a bond between Y_3 and M may be a covalent bond.

In one or more embodiment, in Formula 1B, Y_4 and Y_6 may each be C, and a bond between Y_4 and M and a bond between Y_6 and M may each be a coordinate bond.

In one or more embodiments, in Formula 1B, Y_4 and Y_6 may each be N, and a bond between Y_4 and M and a bond between Y_6 and M may each be a coordinate bond.

In one or more embodiment, in Formula 1B, Y_4 to Y_6 may each be C, Y_5 may be C, a bond between Y_4 and M and a bond between Y_6 and M may each be a coordinate bond, and a bond between Y_5 and M may be a covalent bond.

In one or more embodiments, in Formula 1B, Y_4 and Y_6 may each be N, Y_5 may be C, a bond between Y_4 and M and a bond between Y_6 and M may each be a coordinate bond, and a bond between Y_5 and M may be a covalent bond.

In Formula 1A, L_1 and L_2 may each independently be selected from $*-O-*$, $*-S-*$, $*-C(R_7)(R_8)-*$, $*-C(R_7)-*$, $*-C(R_7)-*$, $*-C(R_7)=C(R_8)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R_7)-*$,

16

$*-N(R_7)-*$, $*-P(R_7)-*$, $*-Si(R_7)(R_8)-*$, $*-P(R_7)(R_8)-*$, and $*-Ge(R_7)(R_8)-*$.

In Formula 1A, n_1 may be selected from 1, 2, and 3, and n_2 may be selected from 0, 1, 2, and 3. When n_2 is 0, a bond represented by $*(L_2)_{n2}-*$ may be a single bond, and ring A_2 and ring A_3 may be linked via a single bond.

In one embodiment, n_1 may be 1, and n_2 may be 0; or n_1 may be 1, and n_2 may be 1.

In one embodiment, n_1 may be 1, n_2 may be 0, and L_1 may be selected from $*-O-*$, $*-C(R_7)(R_8)-*$, $*-B(R_7)-*$, $*-N(R_7)-*$, and $*-Si(R_7)(R_8)-*$.

In one or more embodiments, n_1 may be 1, n_2 may be 1, and L_1 and L_2 may each independently be selected from $*-O-*$, $*-C(R_7)(R_8)-*$, $*-B(R_7)-*$, and $*-N(R_7)-*$.

For example, n_1 and n_2 may each be 1, and

L_1 may be $*-O-*$, and L_2 may be $*-O-*$,

L_1 may be $*-C(R_7)(R_8)-*$, and L_2 may be $*-O-*$,

L_1 may be $*-N(R_7)-*$, and L_2 may be $*-O-*$,

L_1 may be $*-S-*$, and L_2 may be $*-O-*$,

L_1 may be $*-B(R_7)-*$, and L_2 may be $*-O-*$,

L_1 may be $*-Si(R_7)(R_8)-*$, and L_2 may be $*-O-*$,

L_1 may be $*-O-*$, and L_2 may be $*-N(R_7)-*$,

L_1 may be $*-C(R_7)(R_8)-*$, and L_2 may be $*-N(R_7)-*$,

L_1 may be $*-N(R_7)-*$, and L_2 may be $*-N(R_7)-*$,

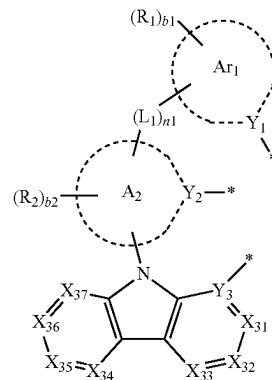
L_1 may be $*-S-*$, and L_2 may be $*-N(R_7)-*$,

L_1 may be $*-B(R_7)-*$, and L_2 may be $*-N(R_7)-*$, or

L_1 may be $*-Si(R_7)(R_8)-*$, and L_2 may be $*-N(R_7)-*$.

In one embodiment, in Formula 1, L_A may be a group represented by Formula 1C:

Formula 1C



In Formula 1C,

ring A_1 , ring A_2 , Y_1 to Y_3 , L_1 , n_1 , R_1 , R_2 , b_1 , and b_2 may each independently be the same as described herein,

X_{31} may be C(R_{31}) or N, X_{32} may be C(R_{32}) or N, X_{33} may be C(R_{33}) or N, X_{34} may be C(R_{34}) or N, X_{35} may be C(R_{35}) or N, X_{36} may be C(R_{36}) or N, and X_{37} may be C(R_{37}) or N,

R_{31} to R_{37} may each independently be the same as defined in connection with R_3 in Formula 1A, and

$*$ indicates a binding site to M.

In one embodiment, in Formula 1C, X_{31} may be C(R_{31}), X_{32} may be C(R_{32}) or N, X_{33} may be C(R_{33}), X_{34} may be C(R_{34}), X_{35} may be C(R_{35}) or N, X_{36} may be C(R_{36}), and X_{37} may be C(R_{37}).

17

R_{31} , R_{33} , R_{34} , R_{36} , and R_{37} may each independently be hydrogen,

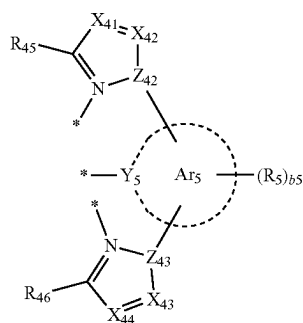
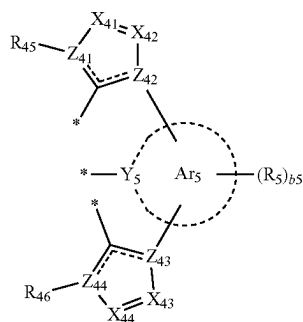
R_{32} and R_{35} may each independently be selected from hydrogen, a cyano group, a methyl group, $-\text{CF}_3$, $-\text{CCl}_3$, $-\text{CBr}_3$, $-\text{Cl}_3$, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, and a phenyl group.

In one or more embodiments, in Formula 1C, X_{31} may be $\text{C}(\text{R}_{31})$, X_{32} may be $\text{C}(\text{R}_{32})$ or N, X_{33} may be $\text{C}(\text{R}_{33})$, X_{34} may be $\text{C}(\text{R}_{34})$, X_{35} may be $\text{C}(\text{R}_{35})$ or N, X_{36} may be $\text{C}(\text{R}_{36})$, and X_{37} may be $\text{C}(\text{R}_{37})$,

R_{31} , R_{33} , R_{34} , R_{36} , and R_{37} may each independently be hydrogen, and

R_{32} and R_{35} may each independently be selected from a cyano group and $-\text{CF}_3$.

In one embodiment, in Formula 1, L_B may be a group represented by Formula 1B-11 or 1B-12:



In Formulae 1B-11 and 1B-12,

Y_5 , ring A_5 , R_5 , and b_5 may each independently be the same as described herein,

Z_{41} to Z_{44} may each independently be N or P,

X_{41} may be $\text{C}(\text{R}_{41})$ or N, X_{42} may be $\text{C}(\text{R}_{42})$ or N, X_{43} may be $\text{C}(\text{R}_{43})$ or N, and X_{44} may be $\text{C}(\text{R}_{44})$ or N,

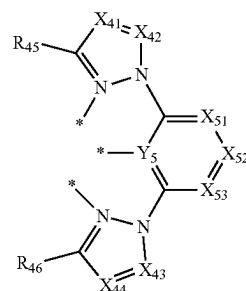
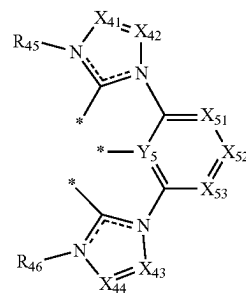
R_{41} to R_{46} may each independently be the same as defined in connection with R_4 and R_6 in Formula 1B, and

* indicates a binding site to M.

In one embodiment, in Formulae 1B-11 and 1B-12, Z_{41} to Z_{44} may each be N.

In one embodiment, in Formula 1, L_B may be a group represented by Formula 1B-21 or 1B-22:

18



In Formulae 1B-21 and 1B-22,

Y_5 , X_{41} to X_{44} , and R_{41} to R_{46} may each independently be the same as described herein,

X_{51} may be $\text{C}(\text{R}_{51})$ or N, X_{52} may be $\text{C}(\text{R}_{52})$ or N, and X_{53} may be $\text{C}(\text{R}_{53})$ or N,

R_{51} to R_{53} may each independently be the same as defined in connection with R_5 in Formula 1B, and

* indicates a binding site to M.

In one embodiment, in Formulae 1B-21 and 1B-22, X_{51} may be $\text{C}(\text{R}_{51})$, X_{52} may be $\text{C}(\text{R}_{52})$, and X_{53} may be $\text{C}(\text{R}_{53})$, and

R_{51} to R_{53} may each independently be selected from hydrogen, a cyano group, a methyl group, $-\text{CF}_3$, $-\text{CCl}_3$, $-\text{CBr}_3$, $-\text{Cl}_3$, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, and a phenyl group.

In Formulae 1A and 1B, R_1 to R_8 may each independently be selected from hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-\text{Si}(\text{Q}_1)(\text{Q}_2)(\text{Q}_3)$, $-\text{N}(\text{Q}_1)(\text{Q}_2)$, $-\text{B}(\text{Q}_1)(\text{Q}_2)$, $-\text{C}(=\text{O})(\text{Q}_1)$, $-\text{S}(=\text{O})_2(\text{Q}_1)$, and $-\text{P}(=\text{O})(\text{Q}_1)(\text{Q}_2)$.

19

Two neighboring $R_1(s)$ among $R_1(s)$ in the number of b1, two neighboring $R_2(s)$ among $R_2(s)$ in the number of b2, two neighboring $R_3(s)$ among $R_3(s)$ in the number of b3, two neighboring $R_4(s)$ among $R_4(s)$ in the number of b4, two neighboring $R_5(s)$ among $R_5(s)$ in the number of b5, two neighboring $R_6(s)$ among $R_6(s)$ in the number of b6, and at least one of R_1 and R_7 , R_2 and R_7 , and R_3 and R_7 may optionally be linked to form a substituted or unsubstituted C_5 - C_{60} carbocyclic group or a substituted or unsubstituted C_1 - C_{60} heterocyclic group.

In one embodiment, R_1 to R_8 in Formulae 1A and 1B may each independently be selected from:

hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group and a C_1 - C_{20} alkoxy group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a cyano group, a phenyl group, and a biphenyl group;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzo-fluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, $-Si(Q_1)(Q_2)(Q_3)$, $-N(Q_1)(Q_2)$, and $-B(Q_1)(Q_2)$; and

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzo-fluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridi-

20

nyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, and an azadibenzosilolyl group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzo-fluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, a biphenyl group, and a terphenyl group.

In one or more embodiments, R_1 to R_8 in Formulae 1A and 1B may each independently be selected from:

hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a cyano group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

21

a C₁-C₂₀ alkyl group and a C₁-C₂₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group;

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, —CF₃, —CCl₃, —CBr₃, —Cl₃, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group.

In one or more embodiments, R₁ to R₈ in Formulae 1A and 1B may each independently be selected from:

hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a methyl group, —CF₃, —CCl₃, —CBr₃, —Cl₃, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, and a tert-butyl group;

a phenyl group and a pyridinyl group; and

a phenyl group and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a methyl group, —CF₃, —CCl₃, —CBr₃, —Cl₃, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group.

In one embodiment, at least one selected from R₁(s) in the number of b1, R₂(s) in the number of b2, R₃(s) in the number of b3, R₄(s) in the number of b4, R₅(s) in the number of b5, and R₆(s) in the number of b6 may be an electron withdrawing group. For example, at least one selected from R₁(s) in the number of b1, R₂(s) in the number of b2, R₃(s) in the number of b3, R₄(s) in the number of b4, R₅(s) in the number of b5, and R₆(s) in the number of b6 may be selected from a cyano group, a C₁-C₁₀ alkyl group substituted with at least one —F, a C₁-C₁₀ alkyl group substituted with at least one —Cl, a C₁-C₁₀ alkyl group substituted with at least one —Br, and a C₁-C₁₀ alkyl group substituted with at least one —I.

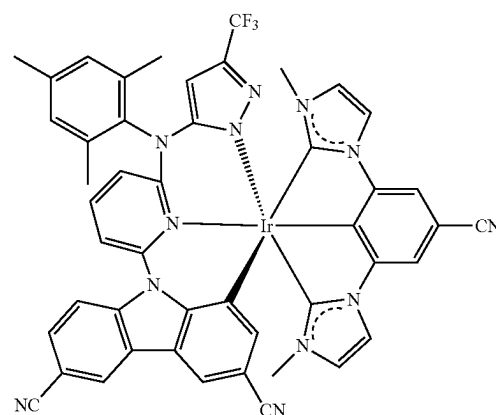
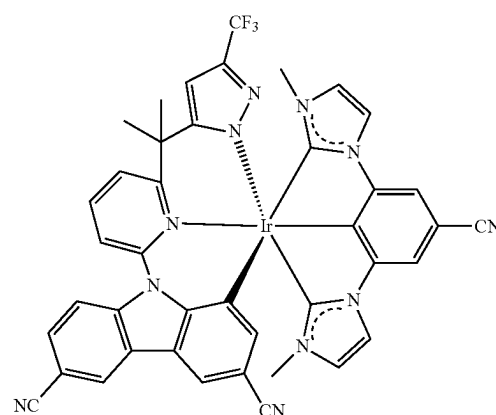
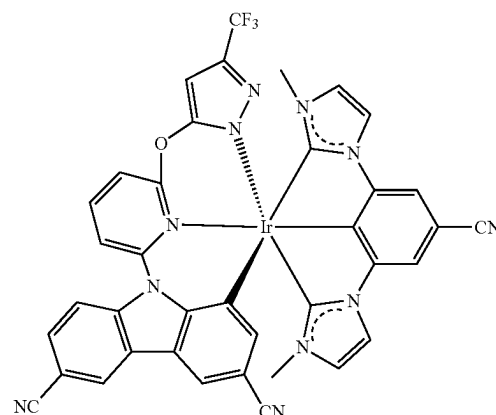
In one or more embodiments, at least one selected from R₁(s) in the number of b1, R₂(s) in the number of b2, R₃(s) in the number of b3, R₄(s) in the number of b4, R₅(s) in the number of b5, and R₆(s) in the number of b6 may be selected from a cyano group, —CF₃, —CCl₃, —CBr₃, and —Cl₃. For example, at least one selected from R₁(s) in the number of b1, R₂(s) in the number of b2, R₃(s) in the number of b3, R₄(s) in the number of b4, R₅(s) in the number of b5, and R₆(s) in the number of b6 may be selected from a cyano group and —CF₃.

When the organometallic compound represented by Formula 1 includes at least one electron withdrawing group such as a cyano group, an emission wavelength of the organometallic compound may be shifted to a shorter wavelength (blue shift), and thus, blue light having high color purity is emitted.

In Formulae 1A and 1B, b1 to b6 may each independently be an integer from 1 to 10.

22

In one embodiment, the organometallic compound may be selected from Compounds 1 to 66:



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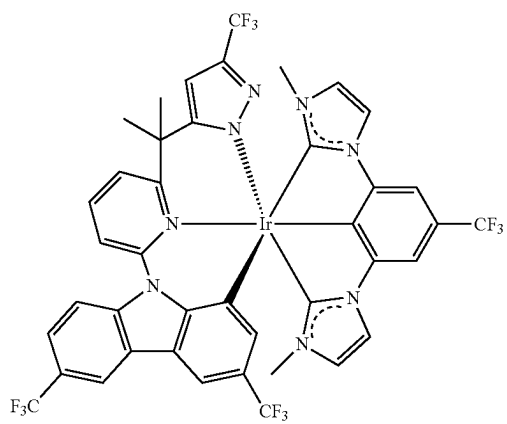
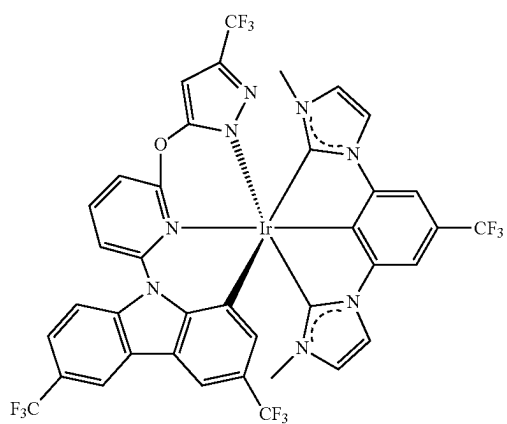
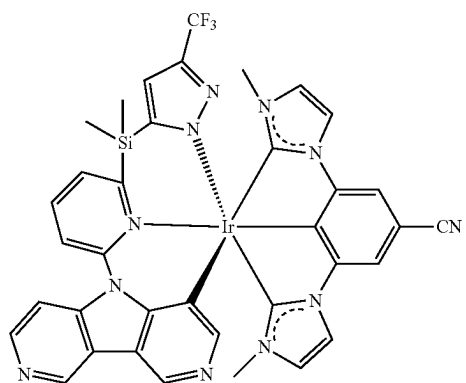


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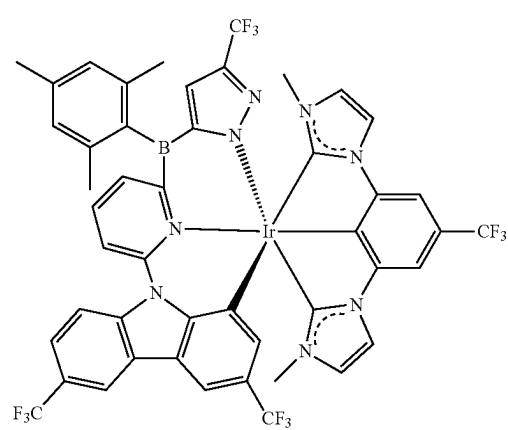
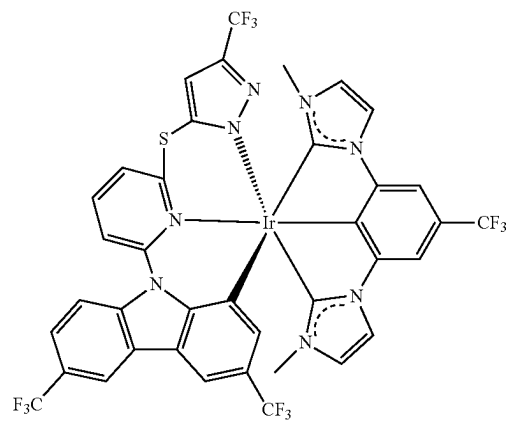
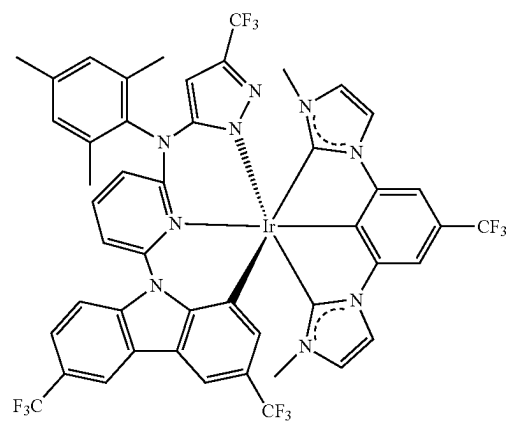


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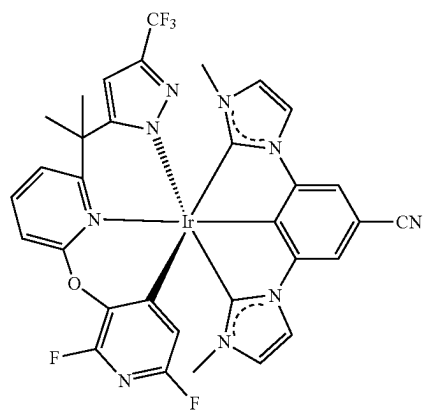
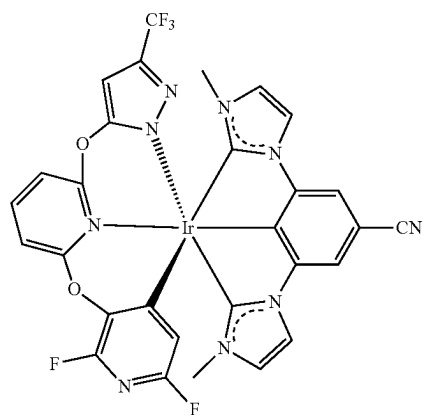
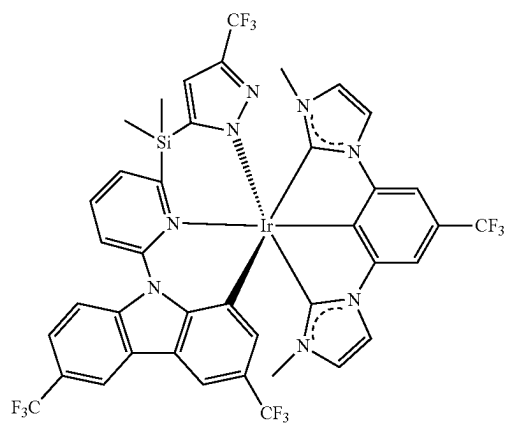
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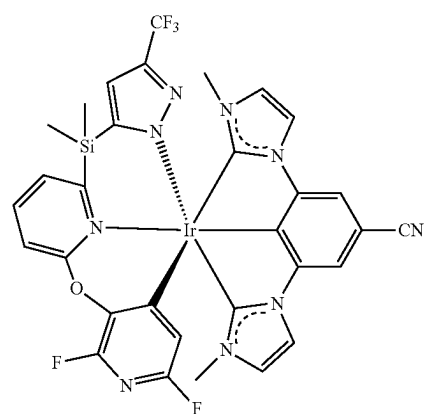
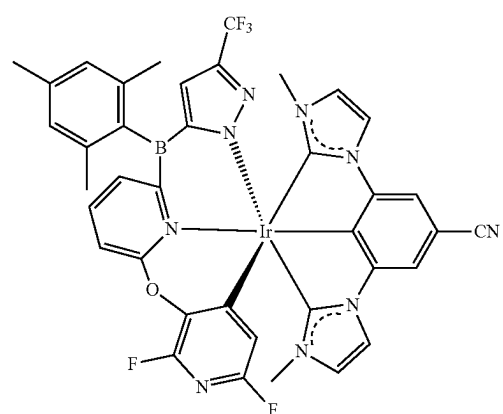
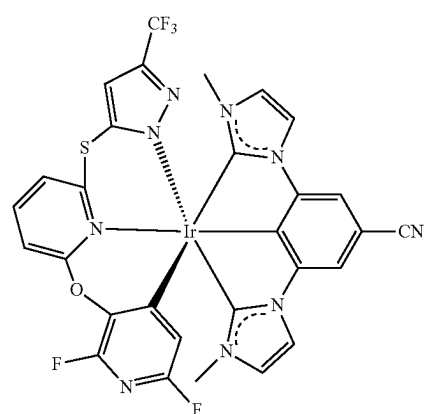
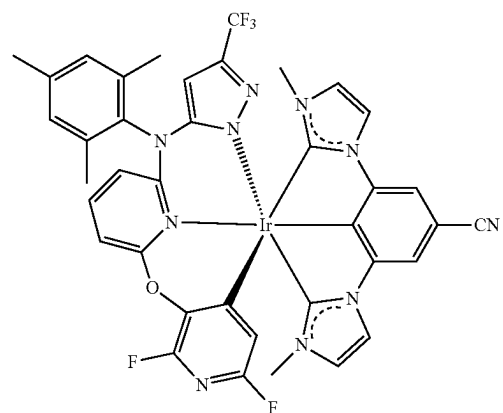
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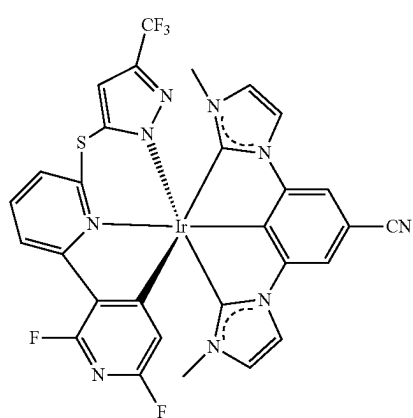
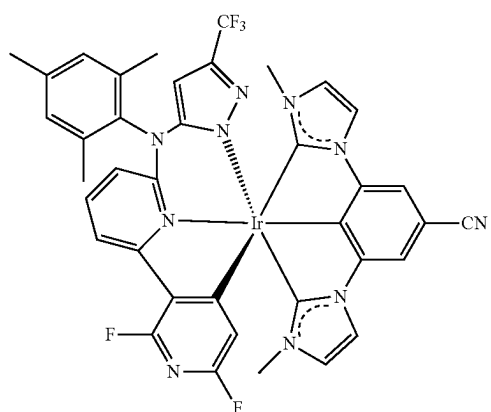
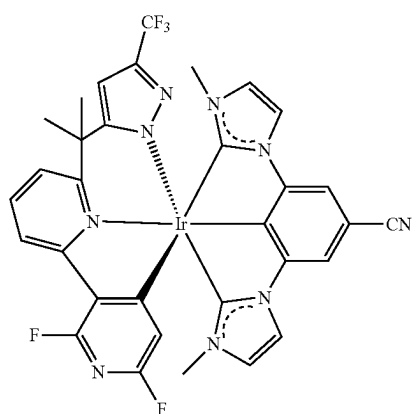
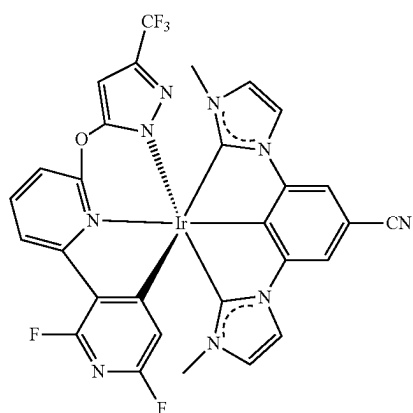
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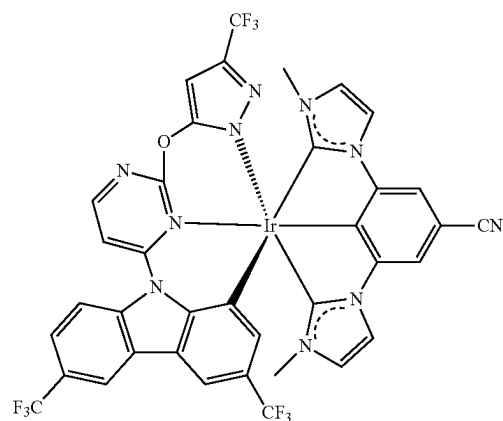
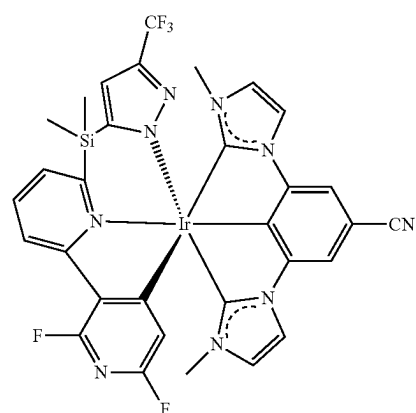
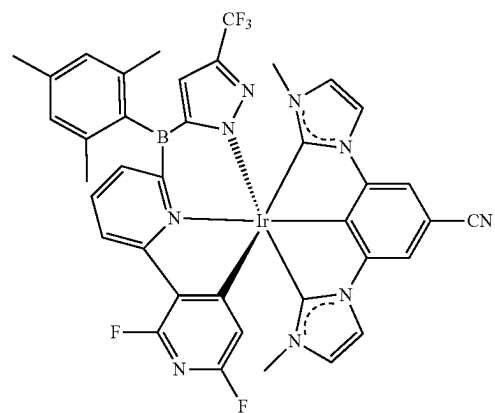
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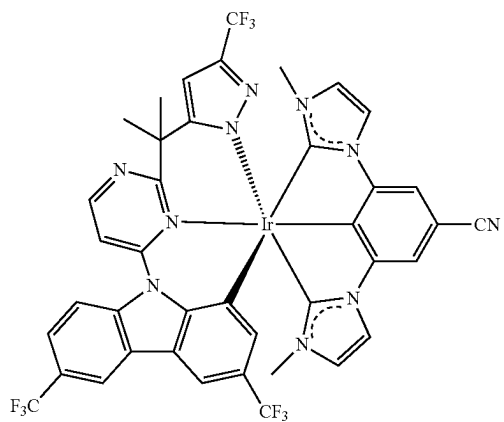
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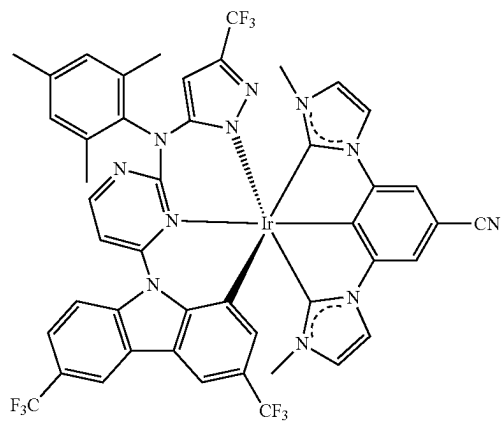
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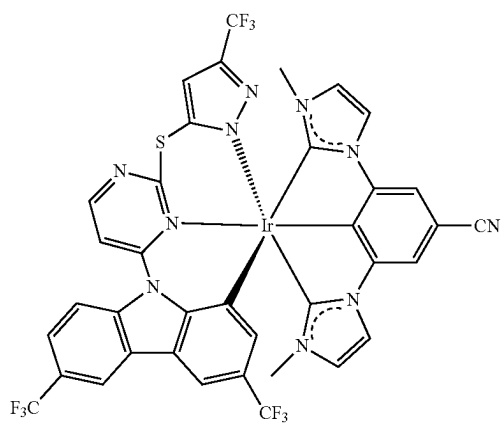
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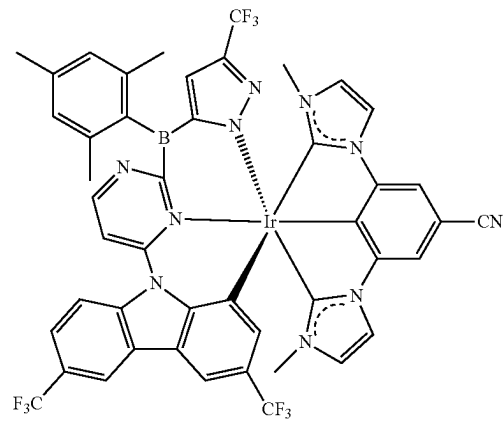


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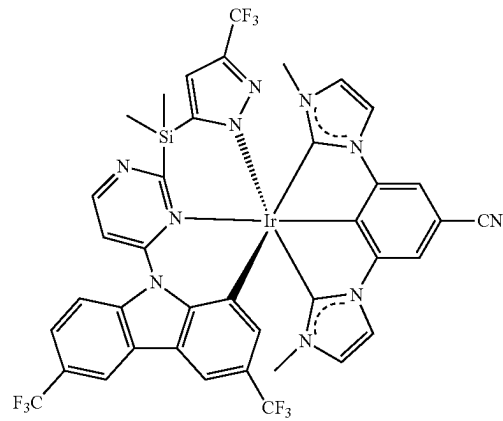
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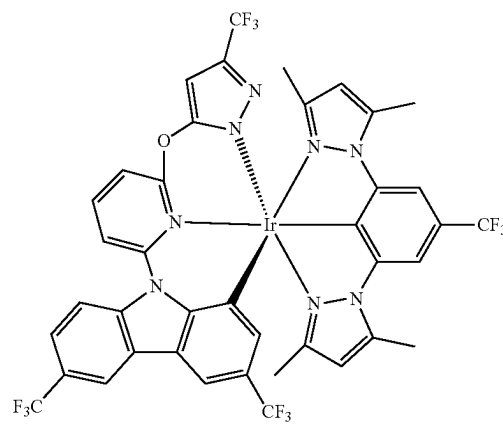


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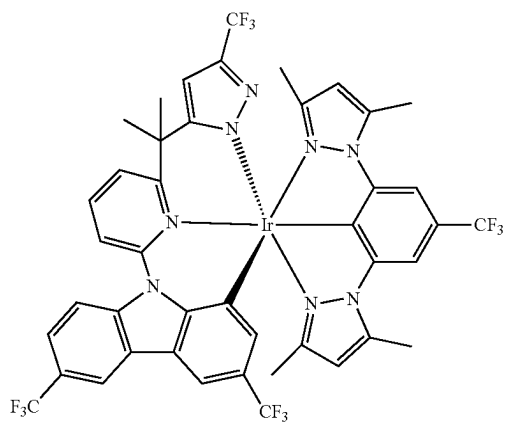
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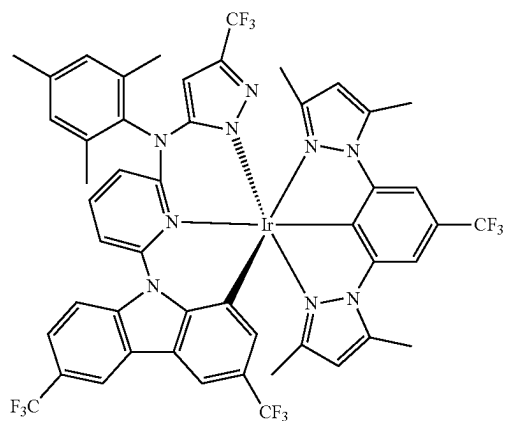
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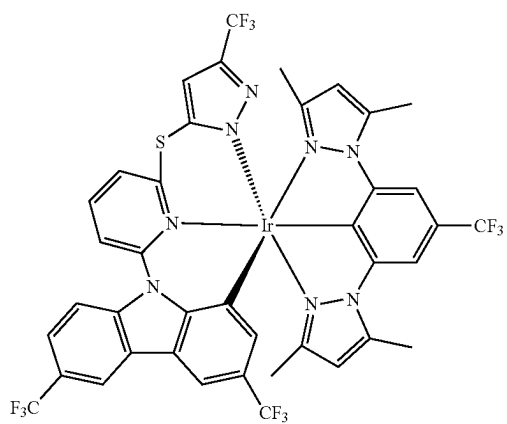
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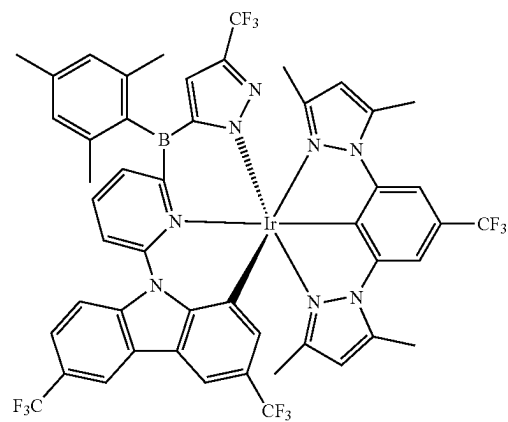
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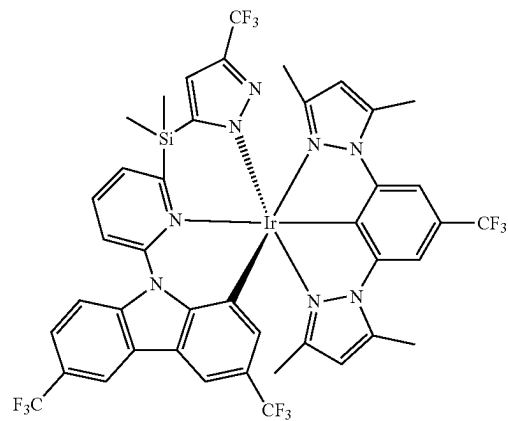
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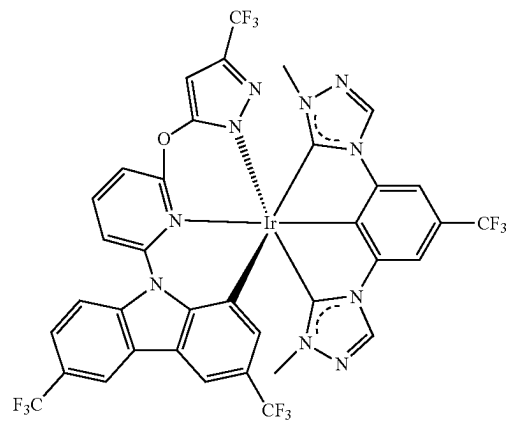
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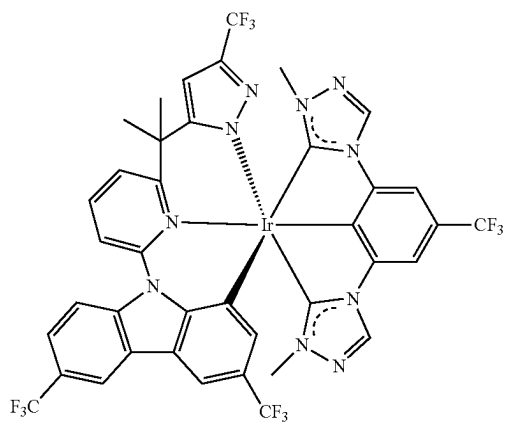


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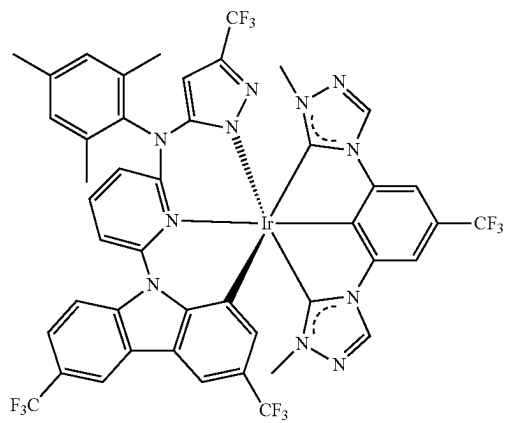
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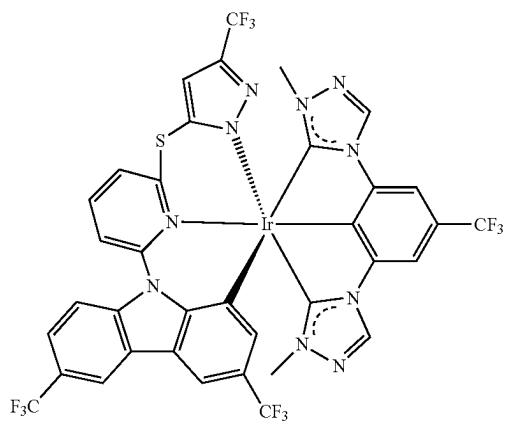
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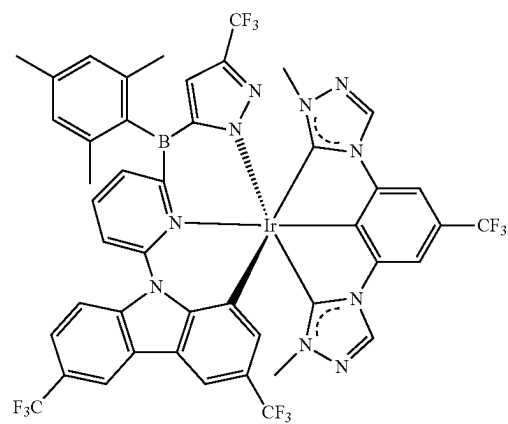
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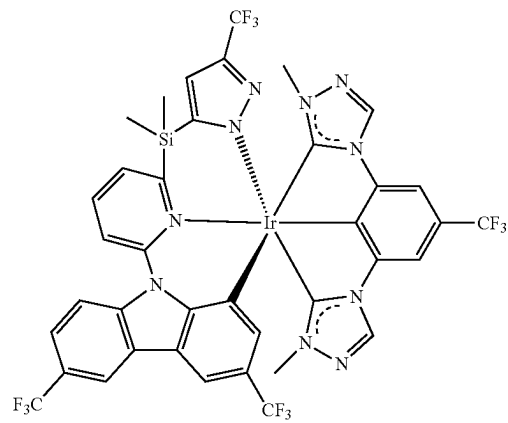
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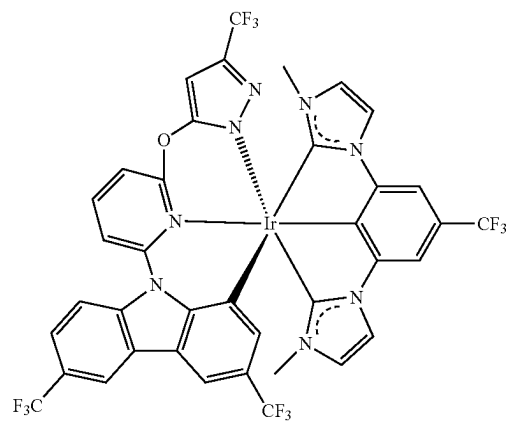
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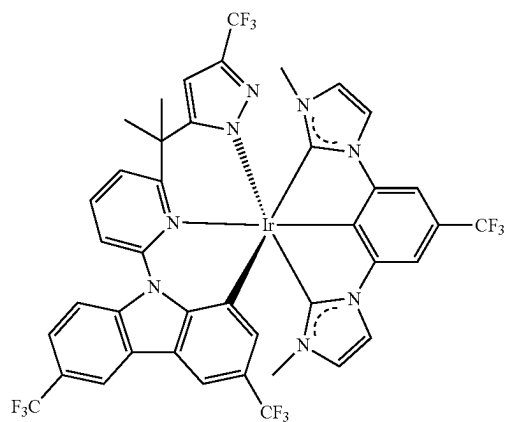


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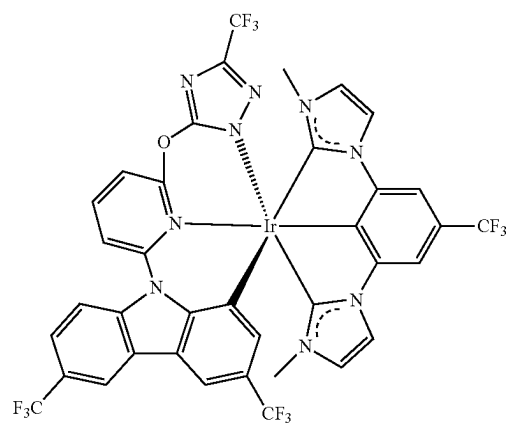
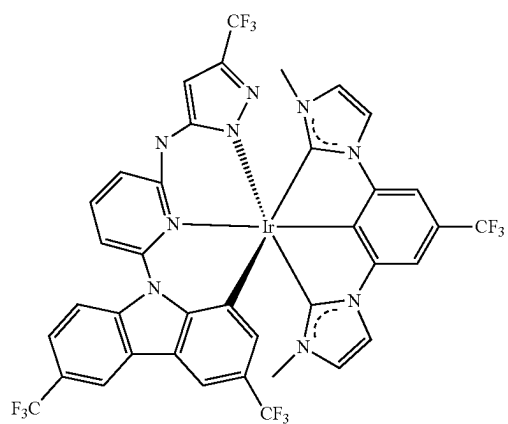
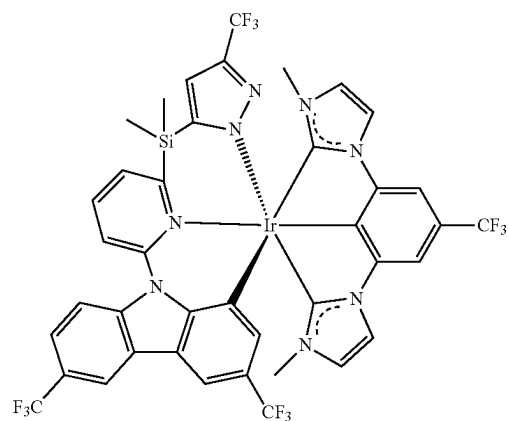
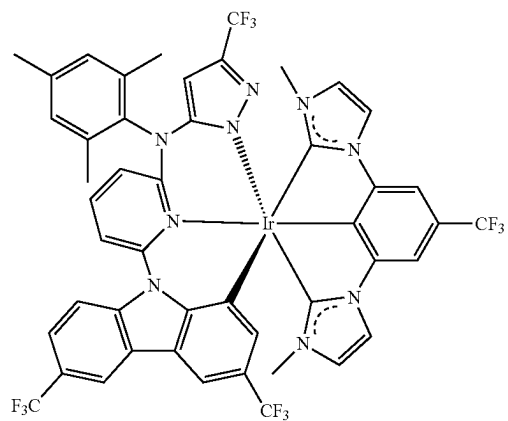
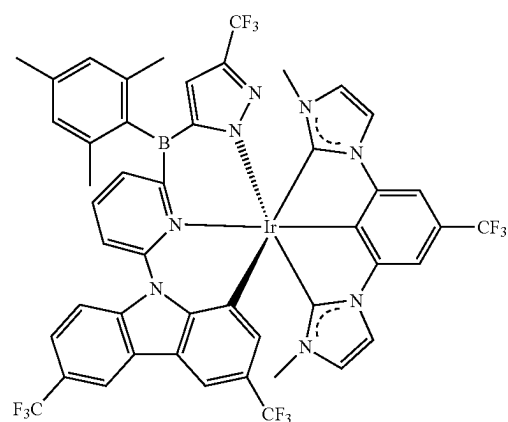
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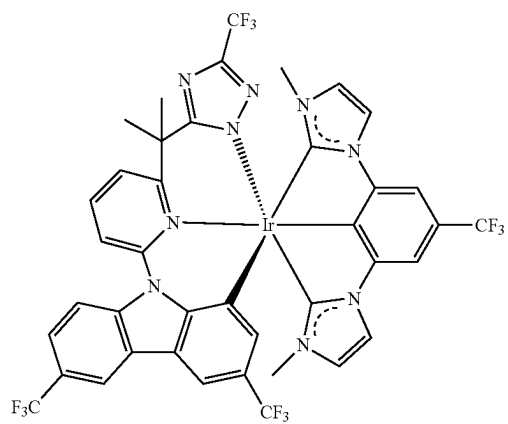
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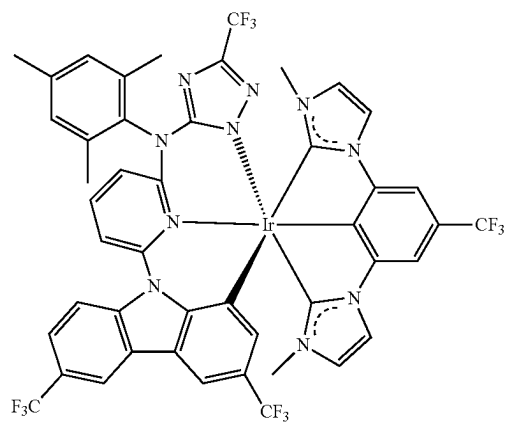
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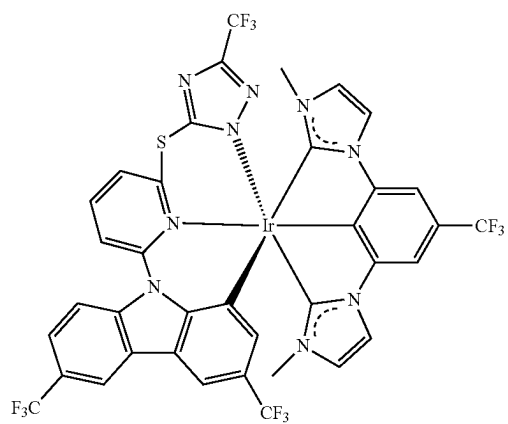
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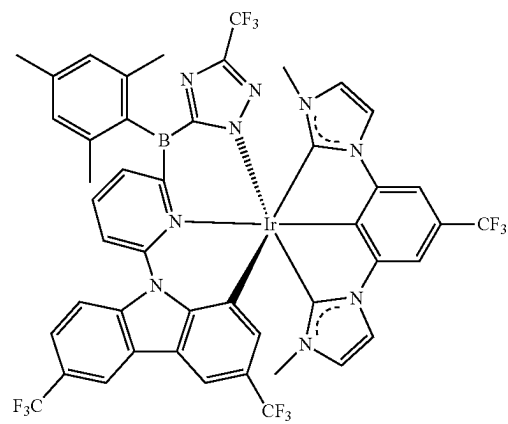
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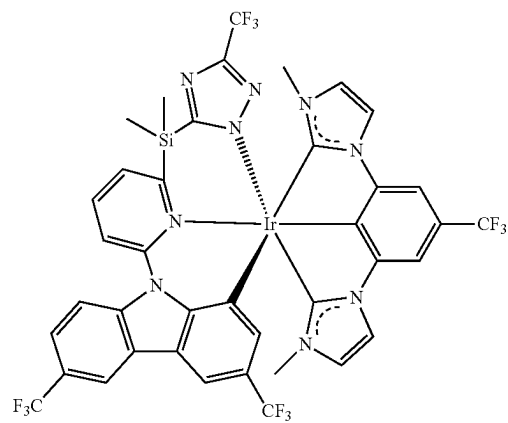
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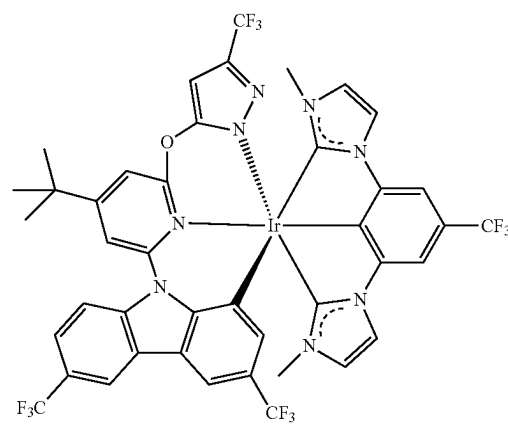
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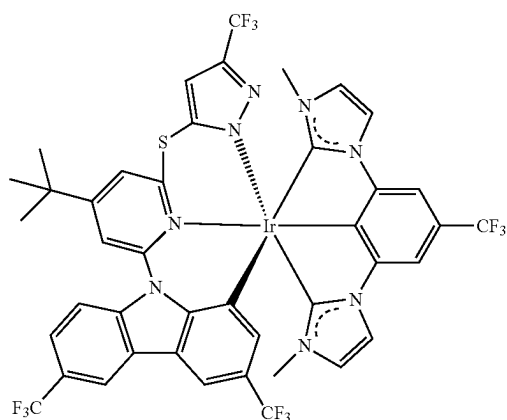
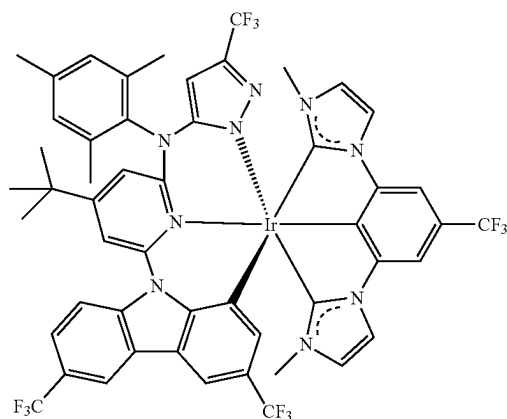
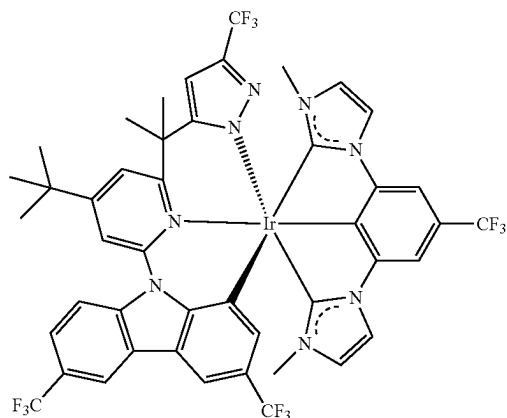
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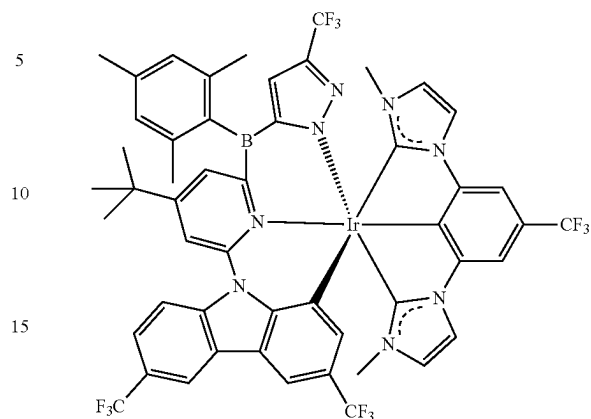
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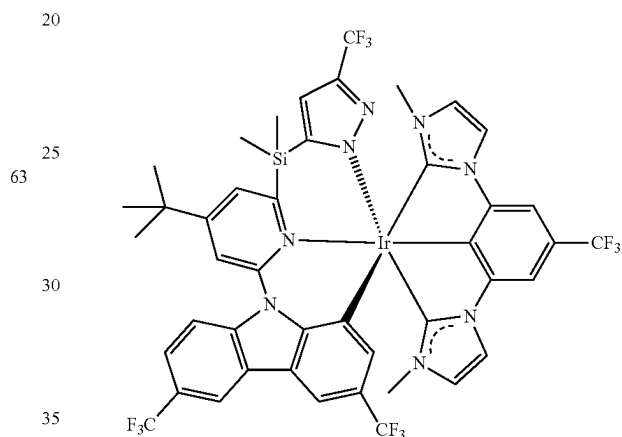
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When ring A₂ in Formula 1A is a pyridine ring in which Y₂ is N, most of the lowest unoccupied molecular orbital (LUMO) of ring A₂ (or Formula 1A) is distributed in the pyridine moiety. Because the organometallic compound represented by Formula 1 includes the tridentate ligand L_A in which ring A₁ and ring A₃ are covalently bonded on (or to) both sides of ring A₂, a stability of the bond between ring A₂ and the metal center M₁ may increase. For example, a stability of the bond between ring A₂ and the metal center M₁ may be strengthened to increase the energy required for the organometallic compound to transit (e.g., transition) from a triplet state to a metal centered state (³MC state). In some embodiments, increasing the stability of the bond between ring A₂ and the metal center M₁ increases the energy level of the ³MC state of the organometallic compound. Therefore, the lifespan of the organometallic compound may be improved, and the organic light-emitting device including the organometallic compound may have a long lifespan.

In the organometallic compound represented by Formula 1, because ring A₁ and ring A₂ are linked in the tridentate ligand L_A via a linker other than a single bond, π -conjugation between ring A₁ and ring A₂ is reduced and the adjustment of the LUMO energy level is facilitated. Therefore, the emission wavelength of the organometallic compound may be shifted to a shorter wavelength such that the organometallic compound is configured to emit blue light having high color purity. Furthermore, the adjustment of the emission wavelength of the organometallic compound may be facilitated by suitable selection of L₁ and L₂ in Formula 1A.

Because the organometallic compound represented by Formula 1 includes the tridentate ligand L_B , a highest occupied molecular orbital (HOMO) energy level is increased to shift an emission wavelength of the organometallic compound to a shorter wavelength (blue shift) such that the organometallic compound is configured to emit blue light having high color purity.

Furthermore, the organometallic compound according to one or more embodiments may include at least one electron withdrawing group in a molecule. Therefore, the emission wavelength of the organometallic compound is shifted to a shorter wavelength (blue shift) such that the organometallic compound is configured to emit blue light having high color purity.

A synthesis method for the organometallic compound represented by Formula 1 would be apparent to those of ordinary skill in the art by referring to the following examples.

The organometallic compound represented by Formula 1 may be used between a pair of electrodes of an organic light-emitting device. For example, the organometallic compound may be included in an emission layer. The organometallic compound may act as a dopant in the emission layer. In one or more embodiments, the organometallic compound of Formula 1 may be used as a material for a capping layer located outside a pair of electrodes of an organic light-emitting device.

Accordingly, another aspect of an embodiment of the present disclosure provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and including an emission layer; and at least one organometallic compound represented by Formula 1.

In one embodiment, the organic layer of the organic light-emitting device may include at least one organometallic compound represented by Formula 1.

The expression “(an organic layer) includes at least one organometallic compound,” as used herein, may include a case in which “(an organic layer) includes identical compounds represented by Formula 1” and a case in which “(an organic layer) includes two or more different organometallic compounds represented by Formula 1.”

For example, the organic layer may include, as the organometallic compound, only Compound 1. In this regard, Compound 1 may exist only in the emission layer of the organic light-emitting device. In one or more embodiments, the organic layer may include, as the organometallic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may exist in an electron transport region).

In one or more embodiments, the emission layer of the organic light-emitting device may include the organometallic compound.

In one or more embodiments, the emission layer of the organic light-emitting device may include the organometallic compound, and may further include a host. Here, an amount of the host included in the emission layer may be greater than that of the organometallic compound included in the emission layer. For example, the amount of the organometallic compound may be in a range of about 0.01 parts by weight to about 30 parts by weight based on 100 parts by weight of the emission layer.

In one embodiment, the emission layer may include the organometallic compound, and may be configured to emit blue light or green light. In one embodiment, the emission layer may include the organometallic compound, and may

be configured to emit light having a wavelength of maximum emission of about 450 nm or more and about 550 nm or less. For example, the emission layer may be configured to emit blue light having a maximum emission wavelength of about 460 nm or more and about 480 nm or less.

The wavelength of maximum emission of the organometallic compound is obtained by using a photoluminescence (PL) measuring meter.

In one embodiment, the first electrode may be an anode, and the second electrode may be a cathode, and the organic layer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, and the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

In one embodiment, the hole transport region of the organic light-emitting device may include a p-dopant having a LUMO energy level of about -3.5 eV or less.

The term “organic layer,” as used herein, refers to a single layer and/or a plurality of layers between the first electrode and the second electrode of the organic light-emitting device. A material included in the “organic layer” is not limited to an organic material. For example, the organic layer may include an inorganic material.

Description of FIG. 1

FIG. 1 is a schematic cross-sectional view of an organic light-emitting device 10 according to an embodiment. The organic light-emitting device 10 includes a first electrode 110, an organic layer 150, and a second electrode 190.

Hereinafter, the structure of the organic light-emitting device 10 according to an embodiment and a method of manufacturing the organic light-emitting device 10 will be described in connection with FIG. 1.

First Electrode 110

In FIG. 1, a substrate may be additionally under the first electrode 110 or above the second electrode 190. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

The first electrode 110 may be formed by depositing or sputtering a material for forming the first electrode 110 on the substrate. When the first electrode 110 is an anode, the material for forming the first electrode 110 may be selected from materials with a high work function to facilitate hole injection.

The first electrode 110 may be a reflective electrode, a semi-reflective electrode, or a transmissive electrode. When the first electrode 110 is a transmissive electrode, a material for forming a first electrode may be selected from indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO_2), zinc oxide (ZnO), and any combinations thereof, but embodiments of the present disclosure are not limited thereto. In one or more embodiments, when the first electrode 110 is a semi-transmissive electrode or a reflectable electrode, a material for forming a first electrode may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and any combinations thereof, but embodiments of the present disclosure are not limited thereto.

The first electrode 110 may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode 110 may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode 110 is not limited thereto.

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Organic Layer 150

The organic layer **150** is on the first electrode **110**. The organic layer **150** may include an emission layer.

The organic layer **150** may further include a hole transport region between the first electrode **110** and the emission layer and an electron transport region between the emission layer and the second electrode **190**.

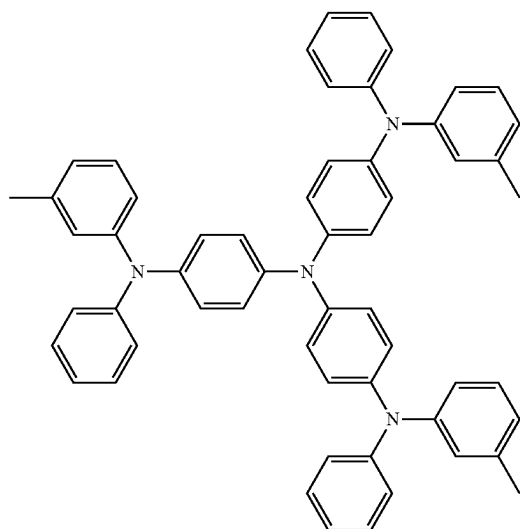
Hole Transport Region in Organic Layer 150

The hole transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers are sequentially stacked from the first electrode **110** in this stated order, but the structure of the hole transport region is not limited thereto.

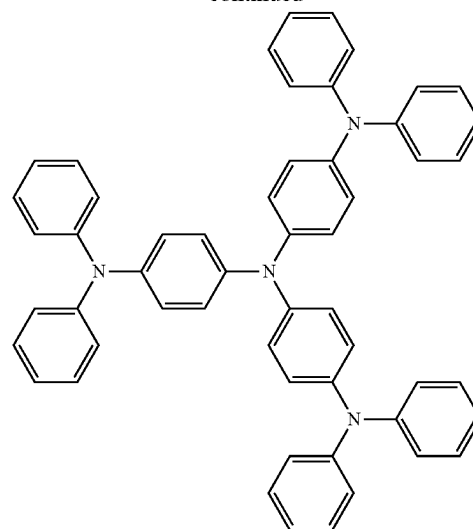
The hole transport region may include at least one selected from m-MTDATA, TDATA, 2-TNATA, NPB (NPB), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4',4"-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, and a compound represented by Formula 202:



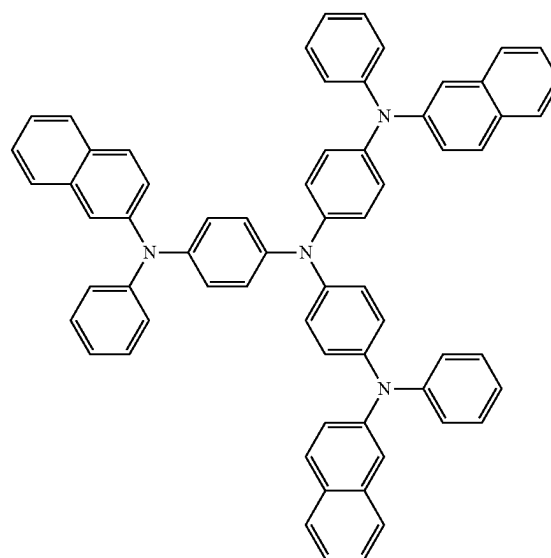
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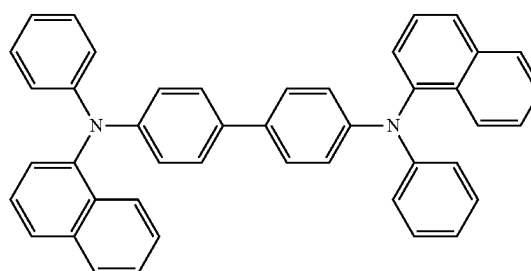
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TDATA



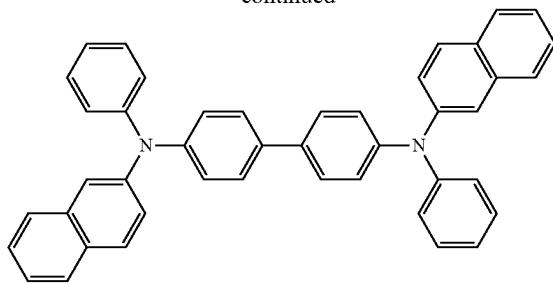
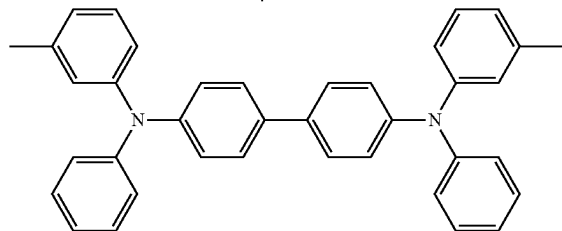
2-TNATA



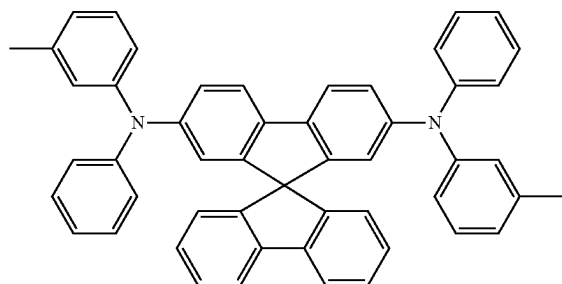
NPB

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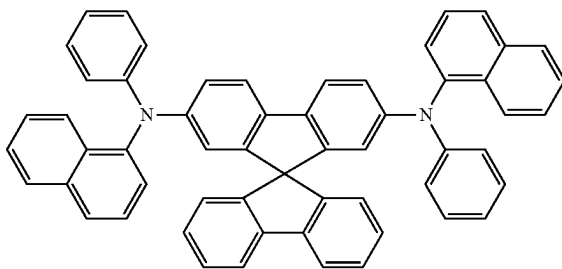
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 β -NPB

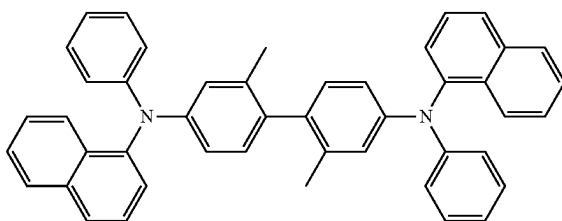
TPD



Spiro-TPD



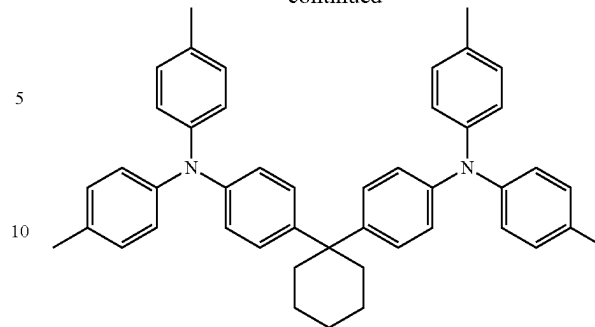
Spiro-NPB



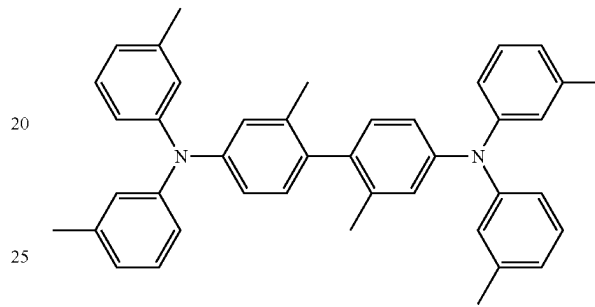
methylated NPB

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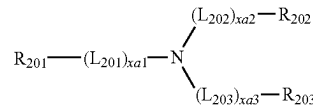


TAPC

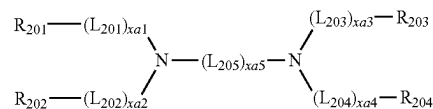


HMTPD

Formula 201



Formula 202



In Formulae 201 and 202,

L₂₀₁ to L₂₀₄ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

I_{205} may be selected from $^*O-^*$, $^*S-^*$, $^*N(Q_{201})-^*$, a substituted or unsubstituted C_1-C_{20} alkylene group, a substituted or unsubstituted C_2-C_{20} alkenylene group, a substituted or unsubstituted C_3-C_{10} cycloalkylene group, a substituted or unsubstituted C_1-C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3-C_{10} cycloalkenylene group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6-C_{60} arylene group, a substituted or unsubstituted C_1-C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

xa1 to xa4 may each independently be an integer from 0 to 3,

xa5 may be an integer from 1 to 10, and

R₂₀₁ to R₂₀₄ and Q₂₀₁ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

For example, in Formula 202, R₂₀₁ and R₂₀₂ may optionally be linked to each other via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group, R₂₀₃ and R₂₀₄ may optionally be linked to each other via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

In one embodiment, in Formulae 201 and 202,

L₂₀₁ to L₂₀₅ may each independently be selected from:

a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and

a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl

group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), and —N(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In one or more embodiments, xa1 to xa4 may each independently be 0, 1, or 2.

In one or more embodiments, xa5 may be 1, 2, 3, or 4.

In one or more embodiments, R₂₀₁ to R₂₀₄ and Q₂₀₁ may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br,

51

—I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuran group, a benzothiophenyl group, a dibenzofuran group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), and —N(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ are the same as described herein above.

In one or more embodiments, at least one selected from R₂₀₁ to R₂₀₃ in Formulae 201 may each independently be selected from:

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuran group, and a dibenzothiophenyl group; and

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuran group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuran group, and a dibenzothiophenyl group;

but embodiments of the present disclosure are not limited thereto.

In one or more embodiments, in Formula 202, i) R₂₀₁ and R₂₀₂ may be linked to each other via a single bond, and/or ii) R₂₀₃ and R₂₀₄ may be linked to each other via a single bond.

In one or more embodiments, at least one selected from R₂₀₁ to R₂₀₄ in Formula 202 may be selected from:

a carbazolyl group; and

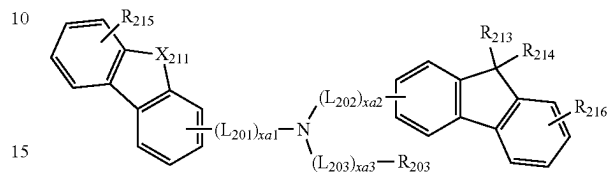
a carbazolyl group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuran group, and a dibenzothiophenyl group,

52

but embodiments of the present disclosure are not limited thereto.

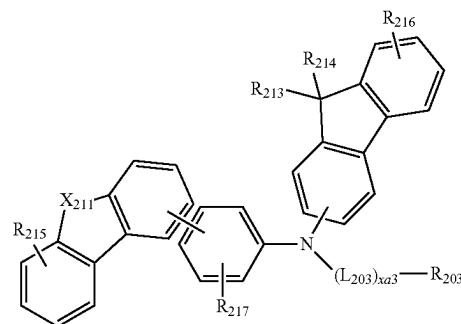
The compound represented by Formula 201 may be represented by Formula 201-1 below, but embodiments of the present disclosure are not limited thereto:

Formula 201-1



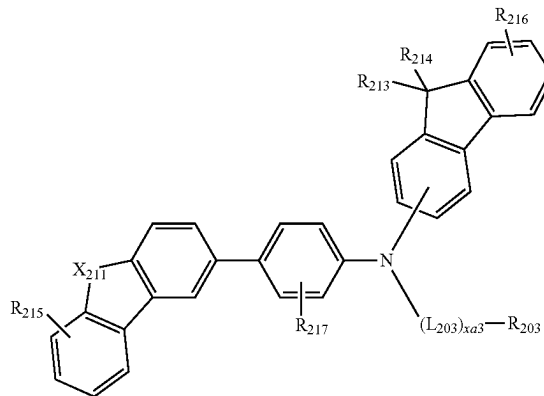
In one embodiment, the compound represented by Formula 201 may be represented by Formula 201-2 below, but embodiments of the present disclosure are not limited thereto:

Formula 201-2



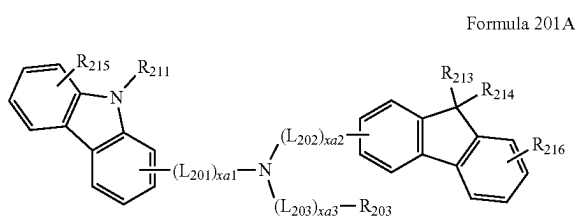
In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201-2(1) below, but embodiments of the present disclosure are not limited thereto:

Formula 201-2(1)

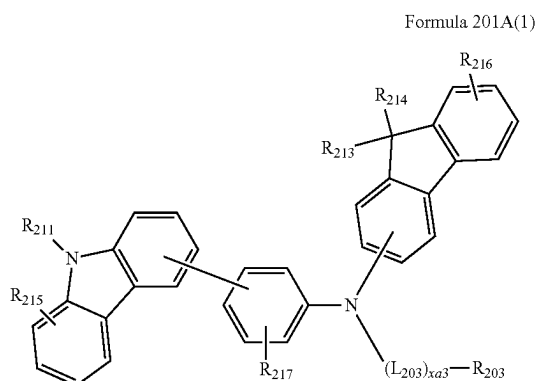


In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A, but embodiments of the present disclosure are not limited thereto:

53

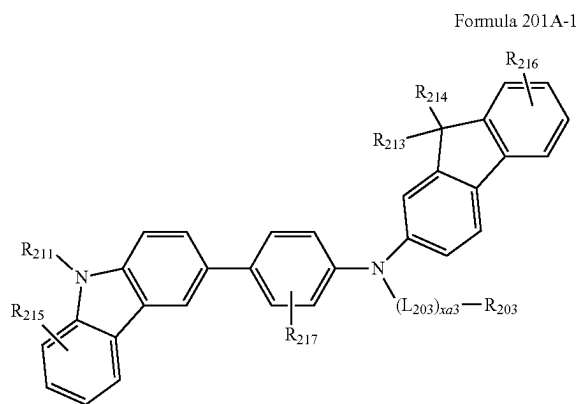


In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A(1) below, but embodiments of the present disclosure are not limited thereto:

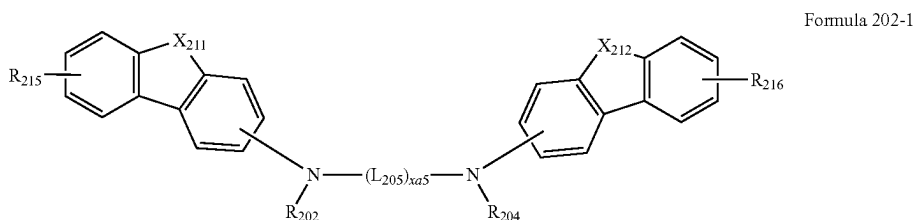


54

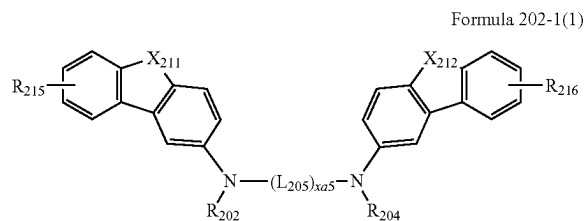
In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A-1 below, but embodiments of the present disclosure are not limited thereto:



In one embodiment, the compound represented by Formula 202 may be represented by Formula 202-1 below, but embodiments of the present disclosure are not limited thereto:



In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202-1(1) below, but embodiments of the present disclosure are not limited thereto:

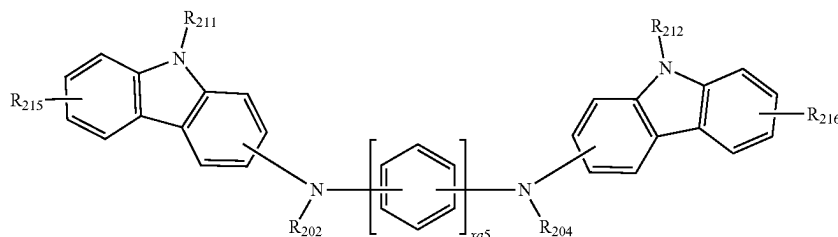


In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202A below, but embodiments of the present disclosure are not limited thereto:

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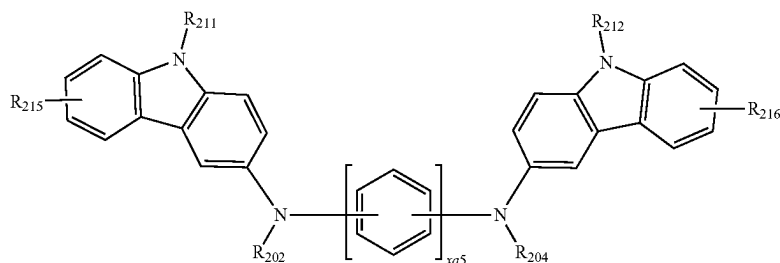
56

Formula 202A



In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202A-1 below, but embodiments of the present disclosure are not limited thereto:

Formula 202A-1



In Formulae 201-1, 201-2, 201-2(1), 201A, 201A(1), 201A-1, 202-1, 202-1(1), 202A, and 202A-1,

L₂₀₁ to L₂₀₃, xa1 to xa3, xa5, and R₂₀₂ to R₂₀₄ are the same as described herein above,

L₂₀₅ may be selected from a phenylene group and a fluorenylene group,

X₂₁₁ may be selected from O, S, and N(R₂₁₁),

X₂₁₂ may be selected from O, S, and N(R₂₁₂),

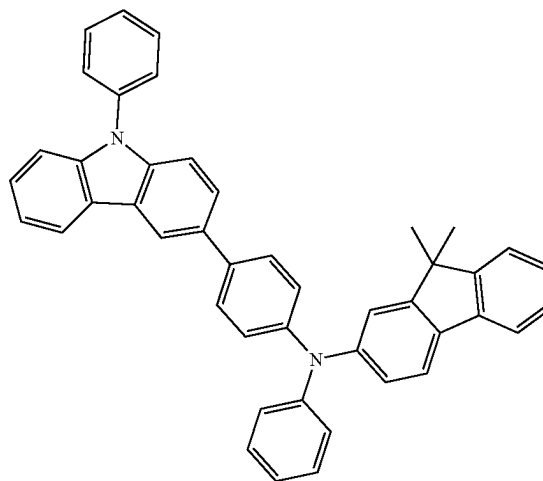
R₂₁₁ and R₂₁₂ may each independently be the same as defined in connection with R₂₀₃, and

R₂₁₃ to R₂₁₇ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thio-

phenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

The hole transport region may include at least one compound selected from Compounds HT1 to HT48, but embodiments of the present disclosure are not limited thereto:

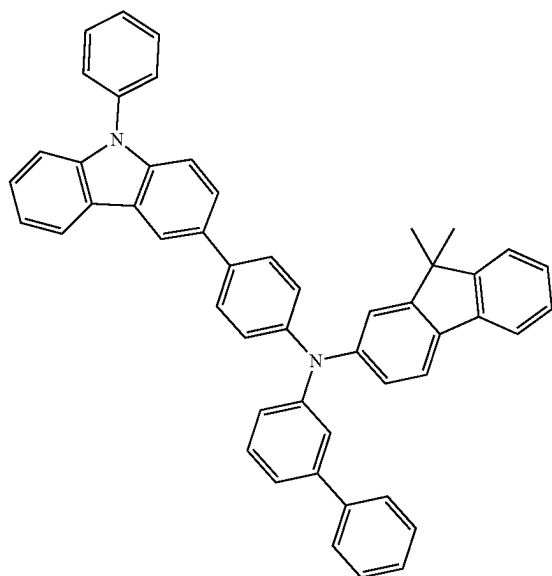
HT1



57

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HT2



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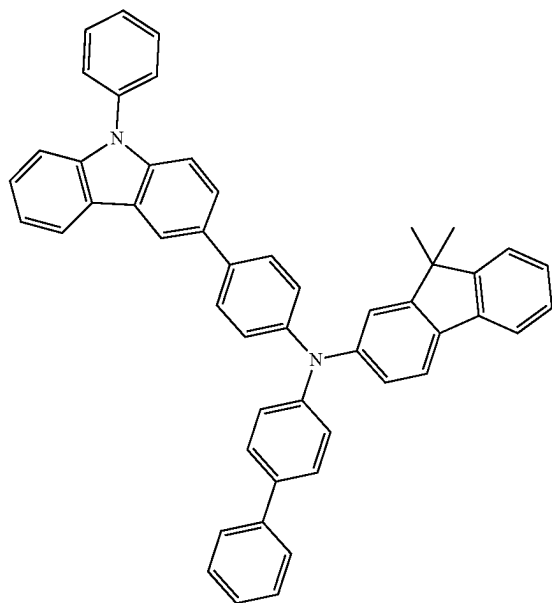
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HT3



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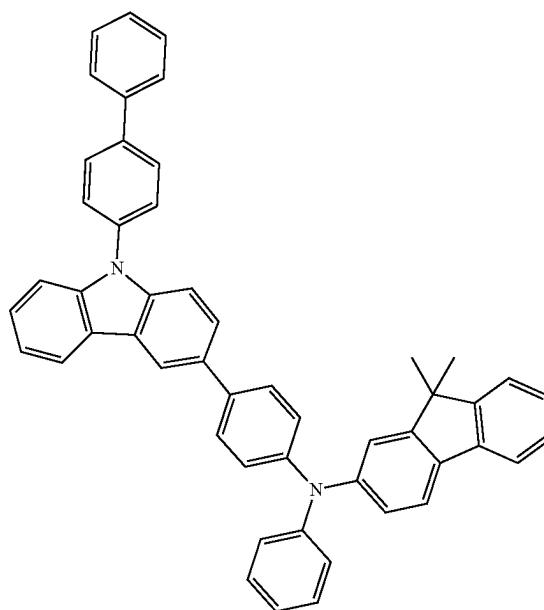
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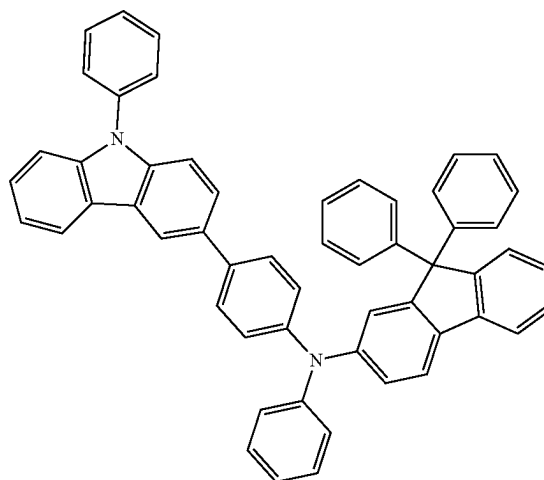
58

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HT4



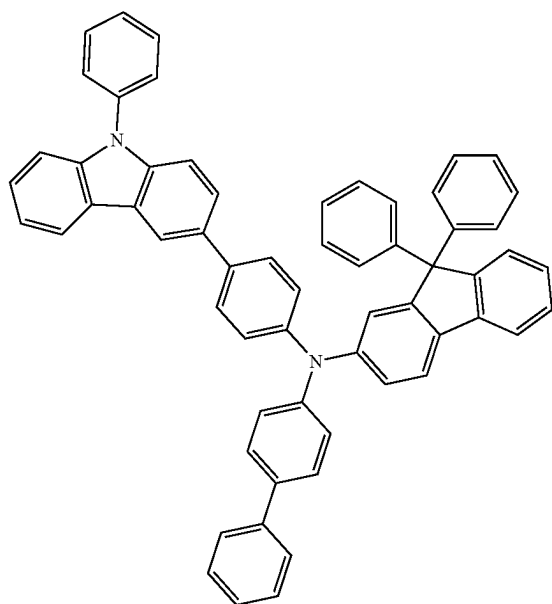
HT5



59

-continued

HT6



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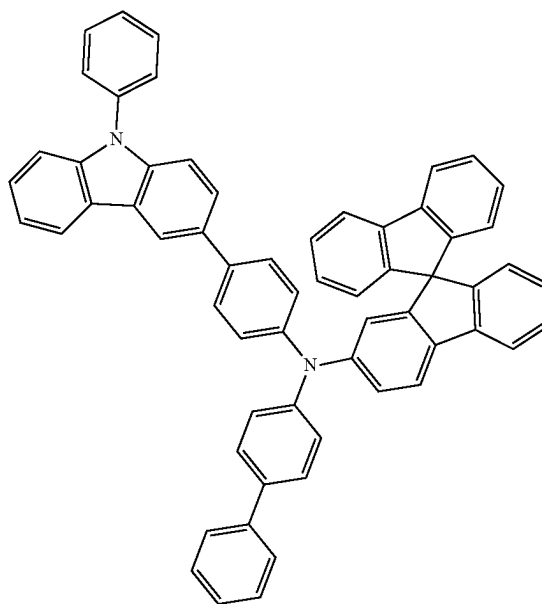
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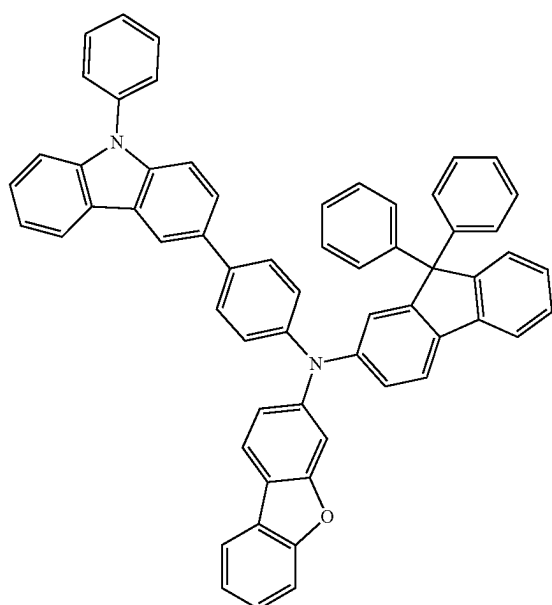
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HT8



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HT7



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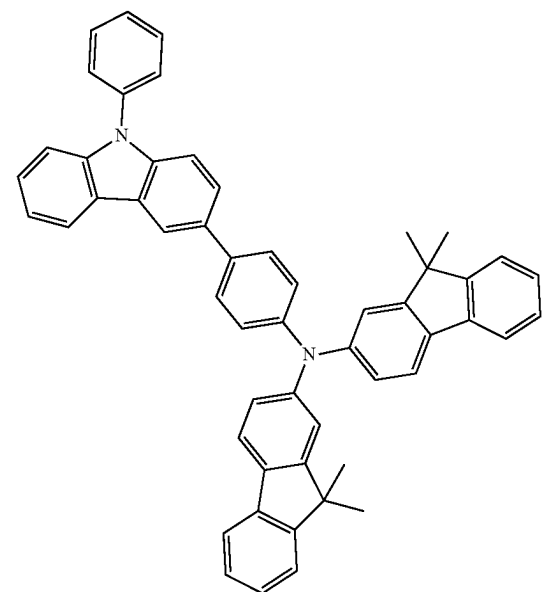
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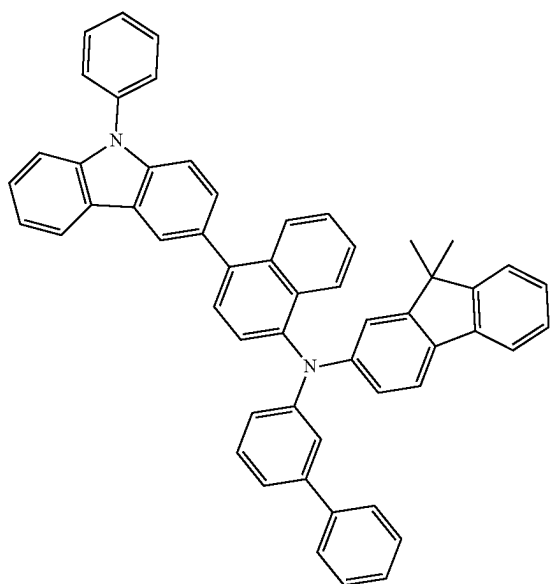
HT9



61

-continued

HT10



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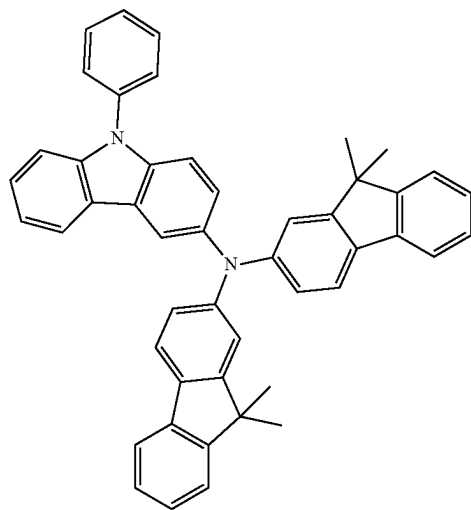
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HT11



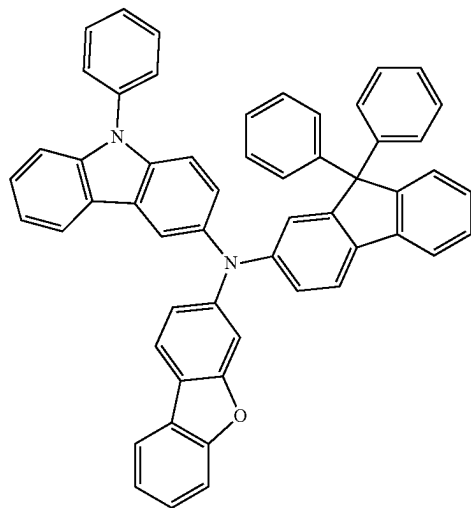
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HT12



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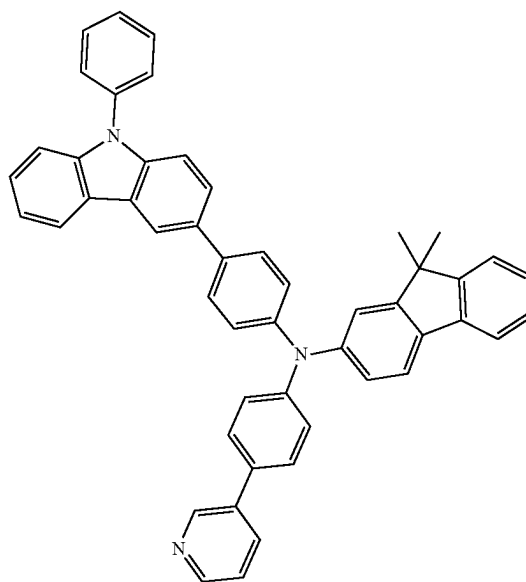
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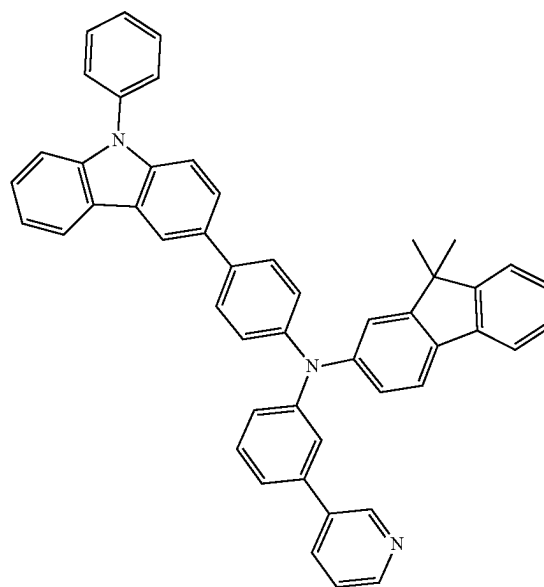
62

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HT13

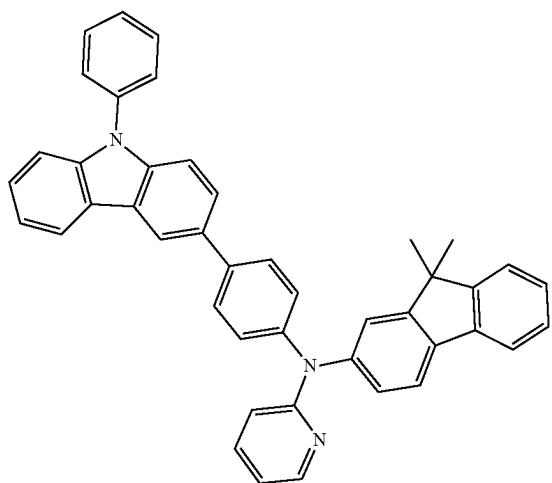


HT14



63

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HT15

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HT16

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HT17

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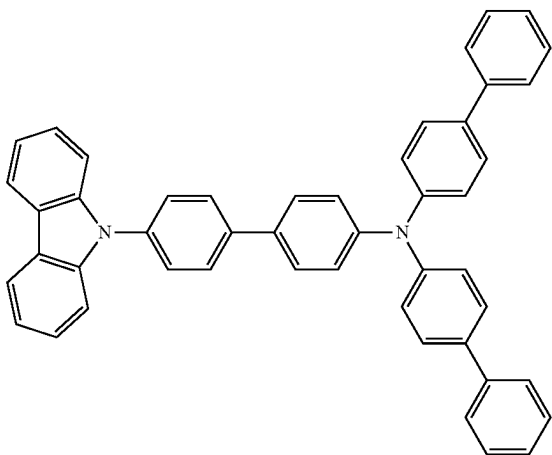
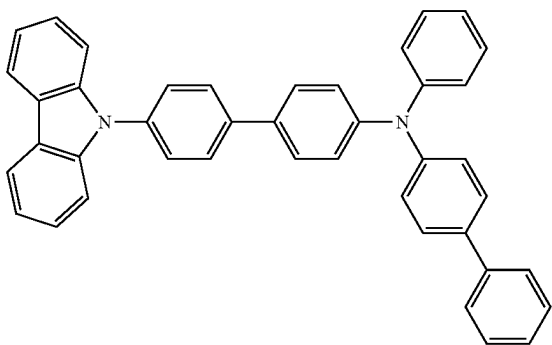
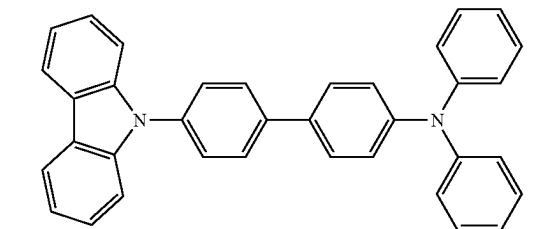
HT18

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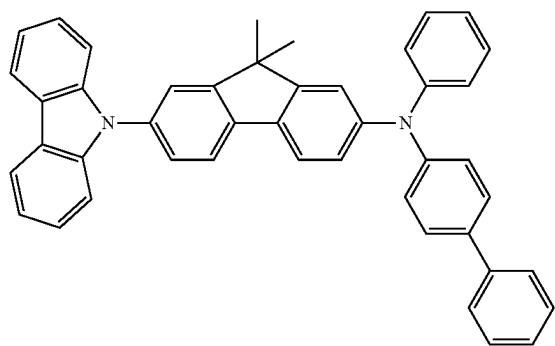
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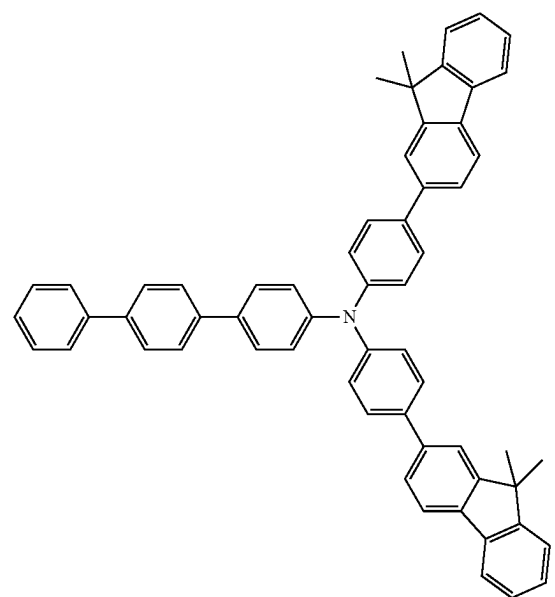
**64**

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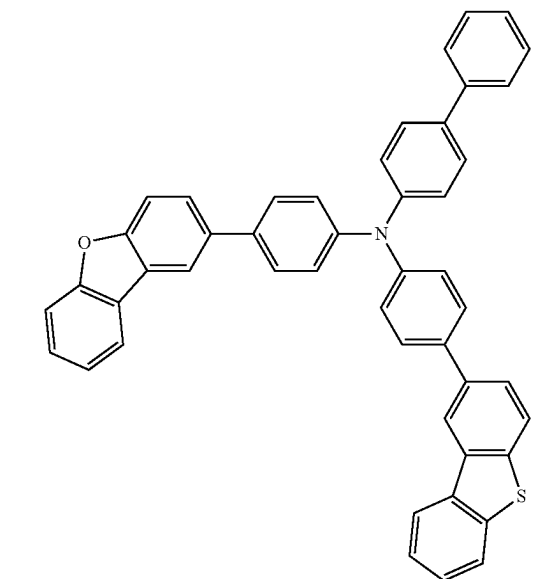


HT19

HT20



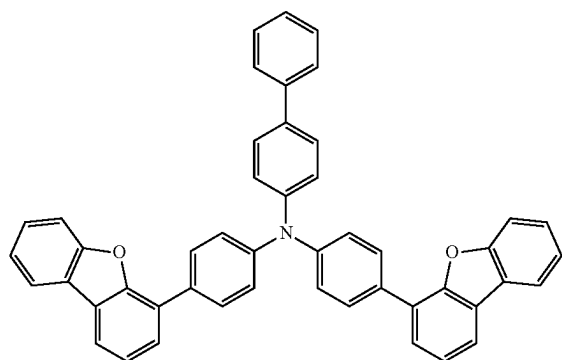
HT21



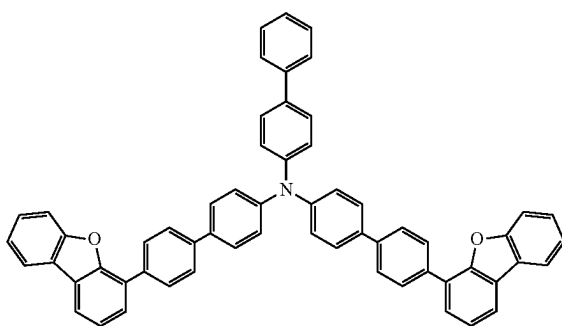
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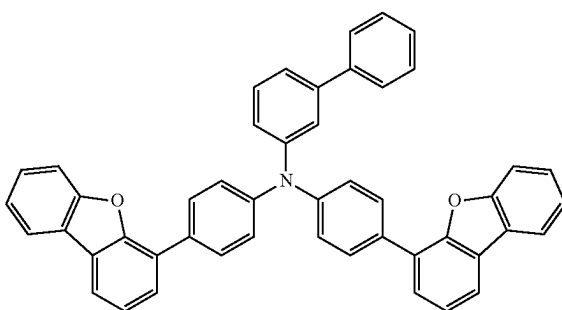
HT22



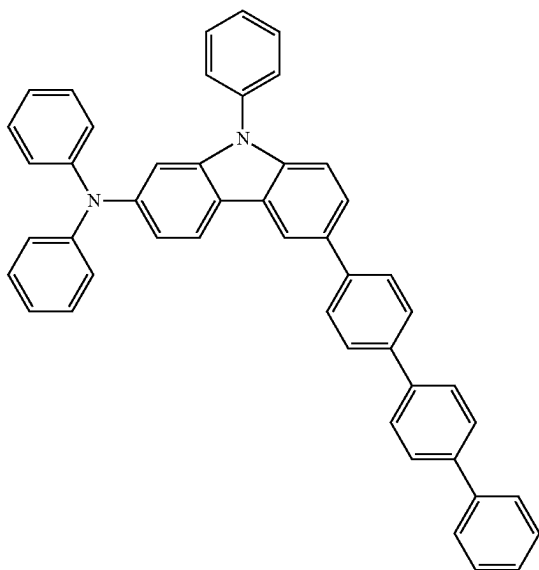
HT23



HT24

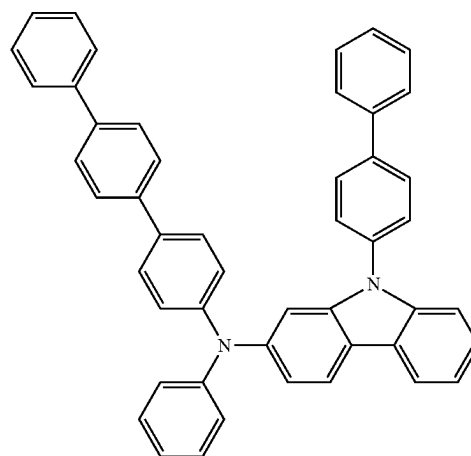


HT25

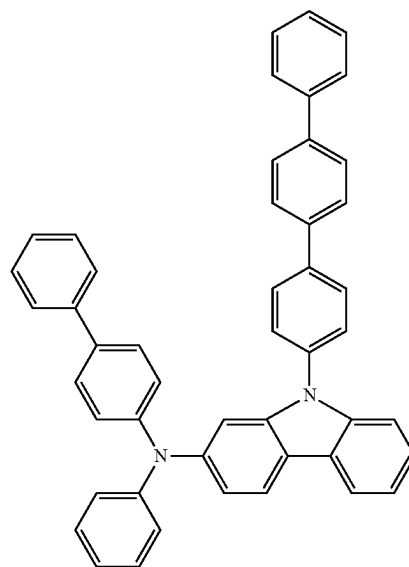
**66**

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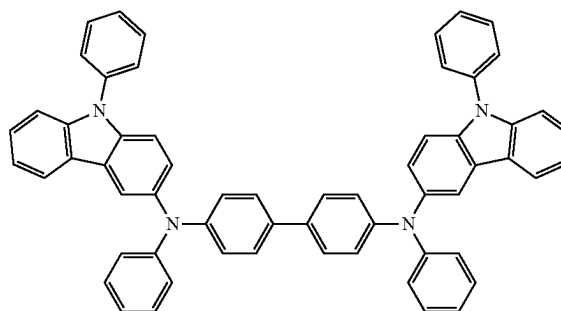
HT26



HT27



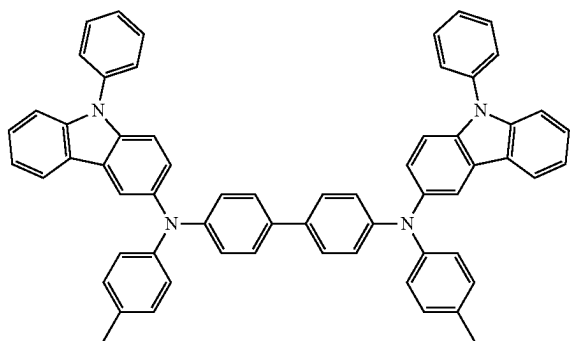
HT28



67

-continued

HT29



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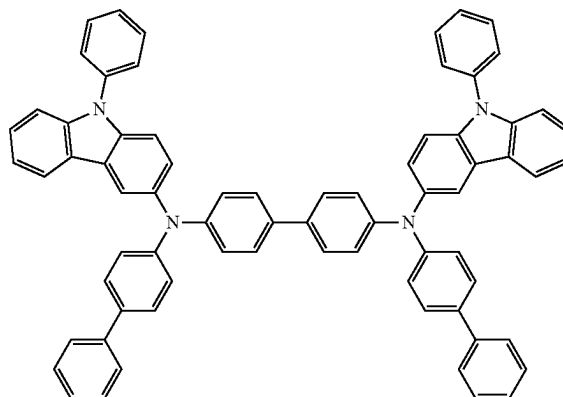
15

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68

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HT33



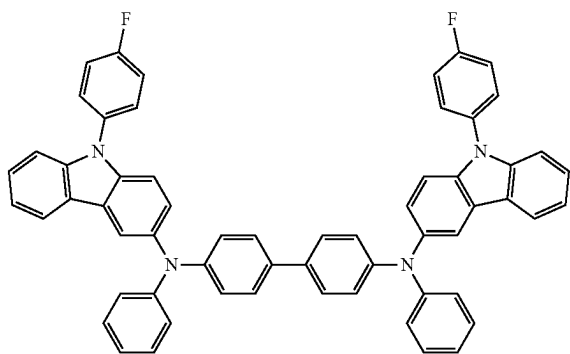
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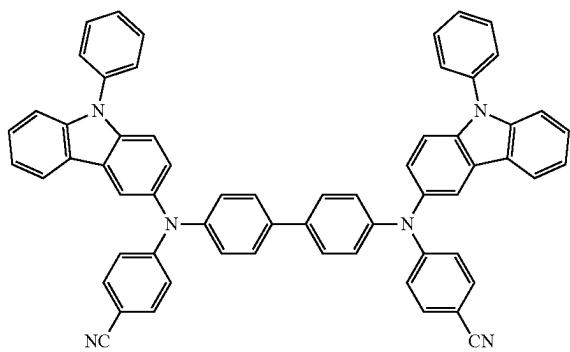
HT30



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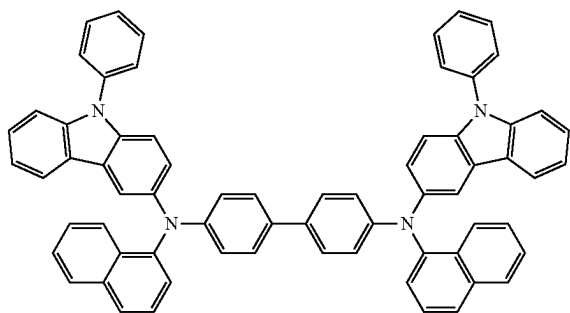
HT31



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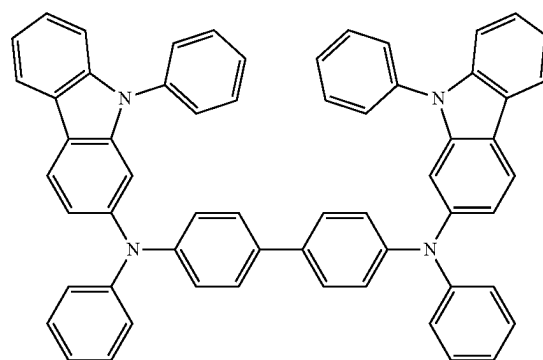
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HT32

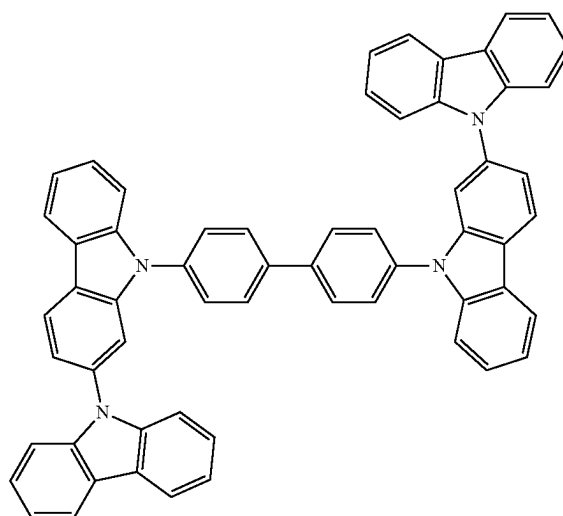


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HT34



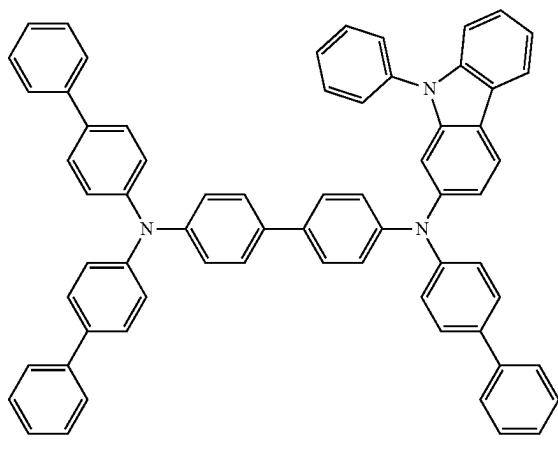
HT35



69

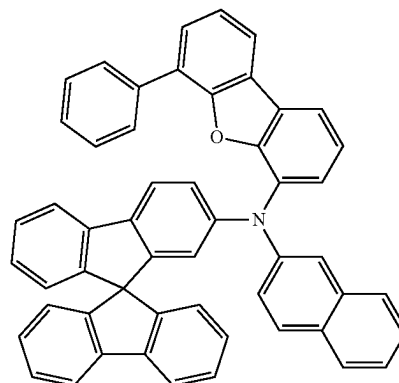
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HT36

**70**

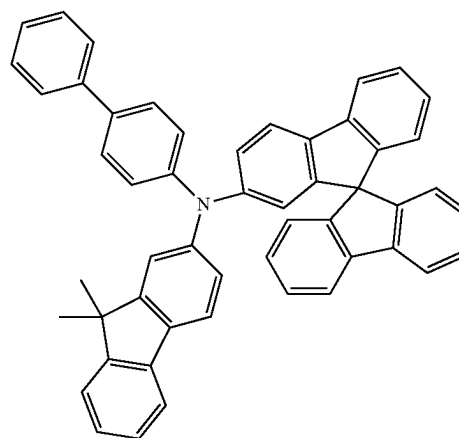
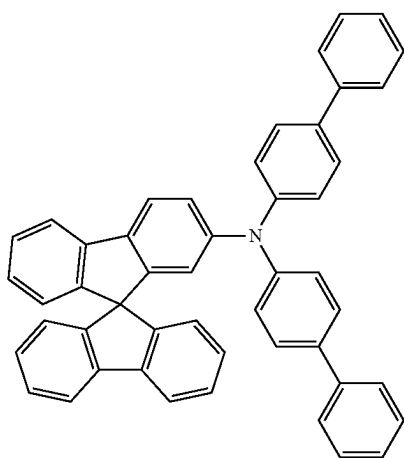
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HT39



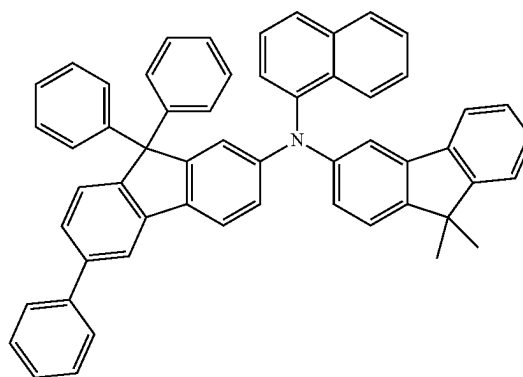
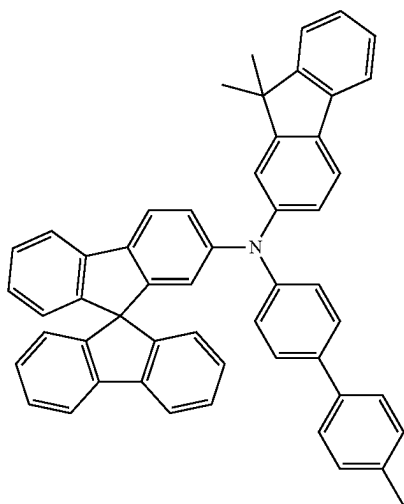
HT37

HT40



HT38

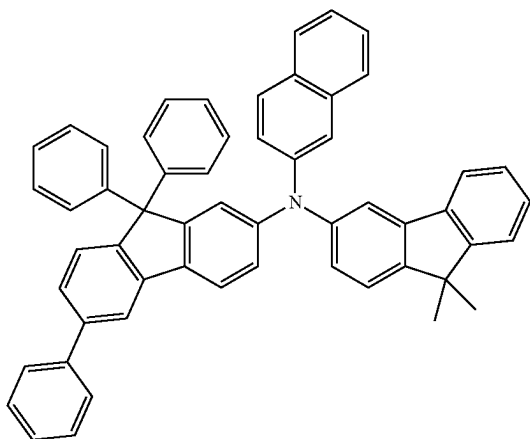
HT41



71

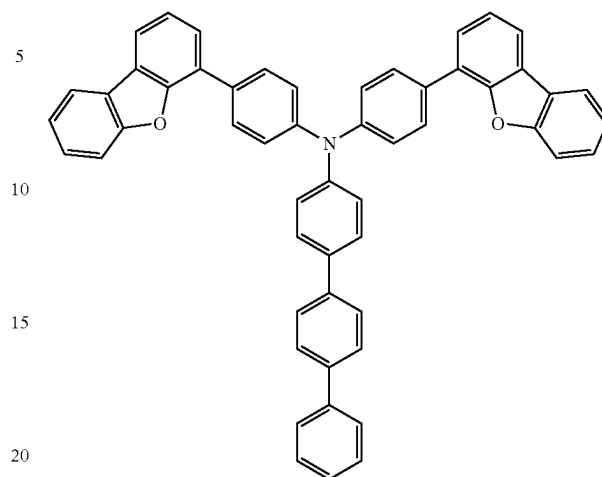
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HT42

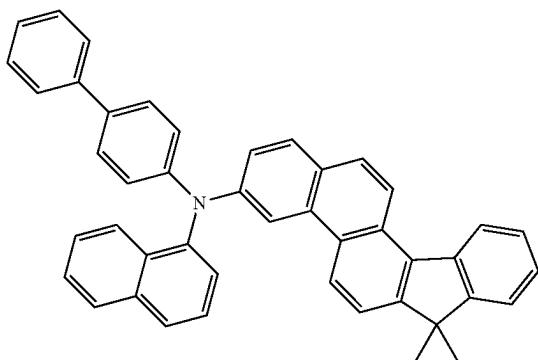
**72**

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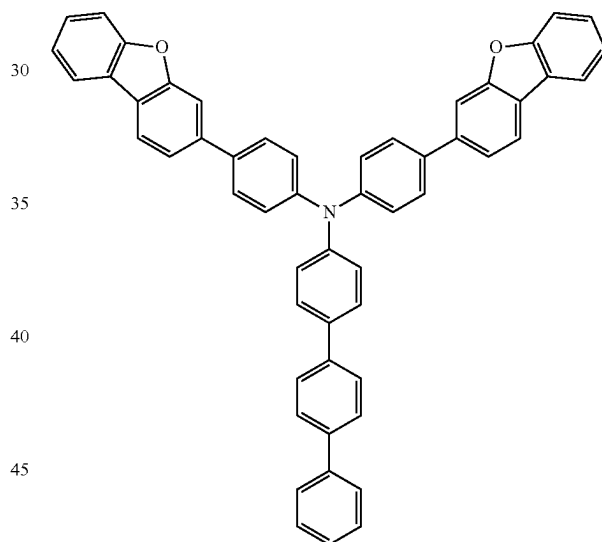
HT45



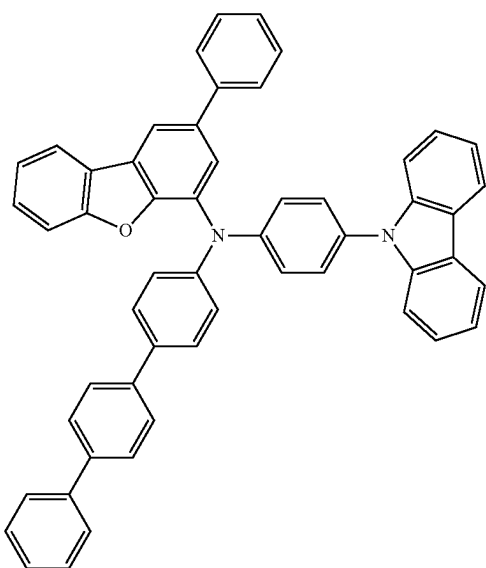
HT43



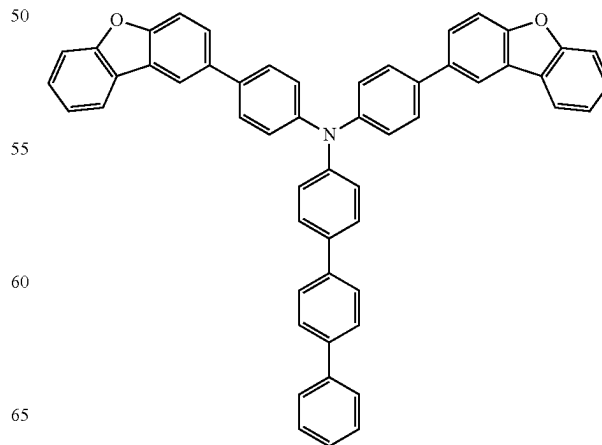
HT46



HT44



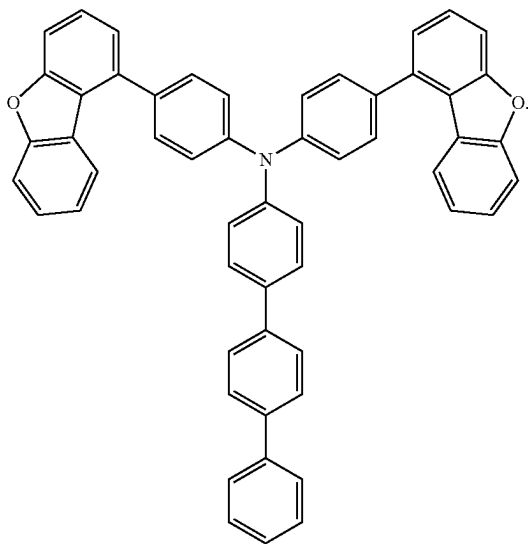
HT47



73

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HT48



A thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one selected from a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, and for example, about 100 Å to about 1500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, suitable or satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described herein above.

p-Dopant

The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

The charge-generation material may be, for example, a p-dopant.

In one embodiment, the p-dopant may have a LUMO energy level of about -3.5 eV or less.

The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound, but embodiments of the present disclosure are not limited thereto.

In one embodiment, the p-dopant may include at least one selected from:

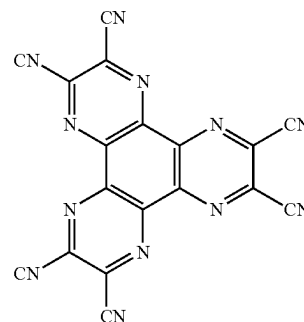
- a quinone derivative, such as tetracyanoquinodimethane (TCNQ) or 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ);
- a metal oxide, such as tungsten oxide or molybdenum oxide;

74

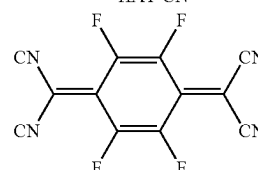
1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and

a compound represented by Formula 221 below:

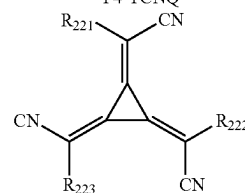
but embodiments of the present disclosure are not limited thereto:



HAT-CN



F4-TCNQ



Formula 221

In Formula 221,

R₂₂₁ to R₂₂₃ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R₂₂₁ to R₂₂₃ may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C₁-C₂₀ alkyl group substituted with —F, a C₁-C₂₀ alkyl group substituted with —Cl, a C₁-C₂₀ alkyl group substituted with —Br, and a C₁-C₂₀ alkyl group substituted with —I.

Emission Layer in Organic Layer 150

When the organic light-emitting device 10 is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a sub-pixel. In one or more embodiments, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In one or more embodiments, the emission layer may include two or more materials selected

75

from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

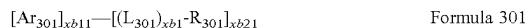
The emission layer may include a host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant.

In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host, but embodiments of the present disclosure are not limited thereto.

A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

Host in Emission Layer

In one or more embodiments, the host may include a compound represented by Formula 301 below.



In Formula 301,

Ar_{301} may be a substituted or unsubstituted C_5-C_{60} carbocyclic group or a substituted or unsubstituted C_1-C_{60} heterocyclic group,

$xb11$ may be 0, 1, 2, or 3;

L_{301} may be selected from a substituted or unsubstituted C_3-C_{10} cycloalkylene group, a substituted or unsubstituted C_1-C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3-C_{10} cycloalkenylene group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6-C_{60} arylene group, a substituted or unsubstituted C_1-C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

$xb1$ may be an integer from 0 to 5,

R_{301} may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted

76

C_6-C_{60} aryl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_{301})(Q_{302})(Q_{303}), —N(Q_{301})(Q_{302}), —B(Q_{301})(Q_{302}), —C(=O)(Q_{301}), —S(=O)₂(Q_{301}), and —P(=O)(Q_{301})(Q_{302}),

$xb21$ may be an integer from 1 to 5, and

Q_{301} to Q_{303} may each independently be selected from a C_1-C_{10} alkyl group, a C_1-C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited thereto.

In one embodiment, Ar_{301} in Formula 301 may be selected from:

a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and

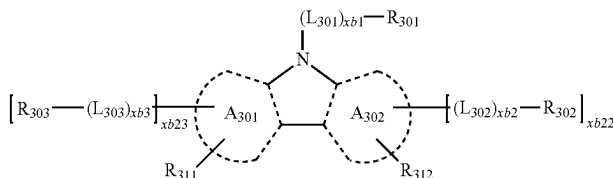
a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —N(Q_{31})(Q_{32}), —B(Q_{31})(Q_{32}), —C(=O)(Q_{31}), —S(=O)₂(Q_{31}), and —P(=O)(Q_{31})(Q_{32}), and

Q_{31} to Q_{33} may each independently be selected from a C_1-C_{10} alkyl group, a alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited thereto.

When $xb11$ in Formula 301 is 2 or more, two or more Ar_{301} (s) may be linked to each other via a single bond.

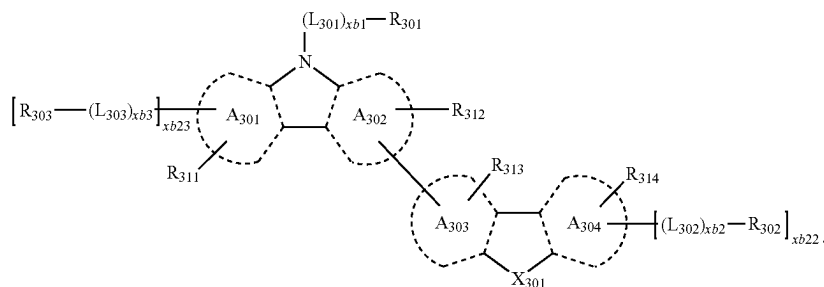
In one or more embodiments, the compound represented by Formula 301 may be represented by Formula 301-1 or 301-2 below:

Formula 301-1



-continued

Formula 301-2



In Formulae 301-1 and 301-2,

A₃₀₁ to A₃₀₄ may each independently be selected from a benzene ring, a naphthalene ring, a phenanthrene ring, a fluoranthene ring, a triphenylene ring, a pyrene ring, a chrysene ring, a pyridine ring, a pyrimidine ring, an indene ring, a fluorene ring, a spiro-bifluorene ring, a benzofluorene ring, a dibenzofluorene ring, an indole ring, a carbazole ring, a benzocarbazole ring, a dibenzocarbazole ring, a furan ring, a benzofuran ring, a dibenzofuran ring, a naphthofuran ring, a benzonaphthofuran ring, a dinaphthofuran ring, a thiophene ring, a benzothiophene ring, a dibenzothiophene ring, a naphthothiophene ring, a benzonaphthothiophene ring, and a dinaphthothiophene ring.

X₃₀₁ may be O, S, or N-[(L₃₀₄)_{xb4}-R₃₀₄].

R₃₁₁ to R₃₁₄ may each independently be selected from hydrogen, deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, -Si(Q₃₁)(Q₃₂)(Q₃₃), -N(Q₃₁)(Q₃₂), -B(Q₃₁)(Q₃₂), -C(=O)(Q₃₁), -S(=O)₂(Q₃₁), and -P(=O)(Q₃₁)(Q₃₂).

xb22 and xb23 may each independently be 0, 1, or 2,

L₃₀₁, xb1, R₃₀₁, and Q₃₁ to Q₃₃ are the same as described herein above,

L₃₀₂ to L₃₀₄ may each independently be the same as described in connection with L₃₀₁,

xb2 to xb4 may each independently be the same as described in connection with xb1, and

R₃₀₂ to R₃₀₄ may each independently be the same as described in connection with R₃₀₁.

For example, L₃₀₁ to L₃₀₄ in Formulae 301, 301-1, and 301-2 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a

pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hex-

79

aceny group, a pentaceny group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyrimidinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, and $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, and

Q_{31} to Q_{33} are the same as described herein above.

In one embodiment, R_{301} to R_{304} in Formulae 301, 301-1, and 301-2 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexaceny group, a pentaceny group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyrimidinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexaceny group, a pentaceny group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a

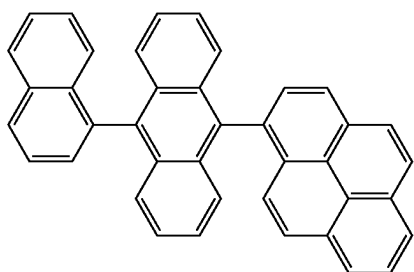
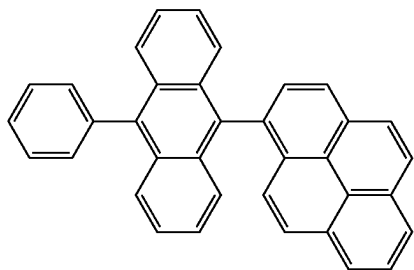
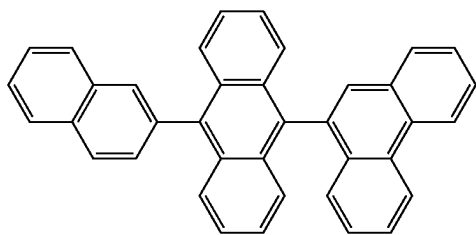
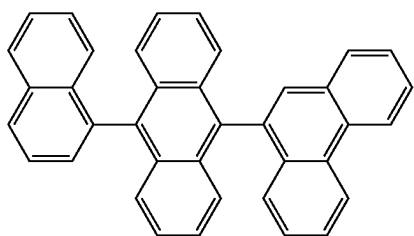
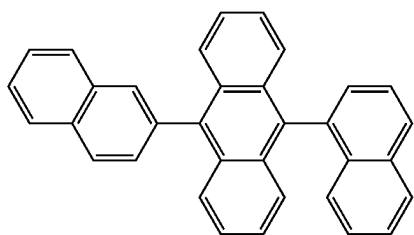
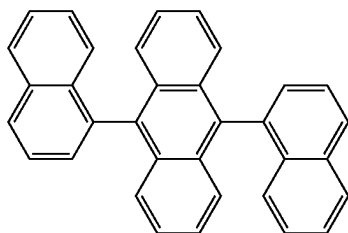
80

dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyrimidinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexaceny group, a pentaceny group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyrimidinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, and $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, and

Q_{31} to Q_{33} are the same as described herein above.

In one or more embodiments, the host may include an alkaline earth metal complex. For example, the host may be selected from a Be complex (for example, Compound H55), a Mg complex, and a Zn complex.

The host may include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55, but embodiments of the present disclosure are not limited thereto:

81**82**

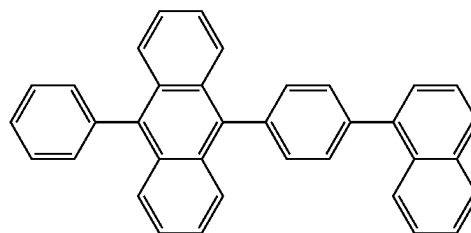
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H1

H7

5

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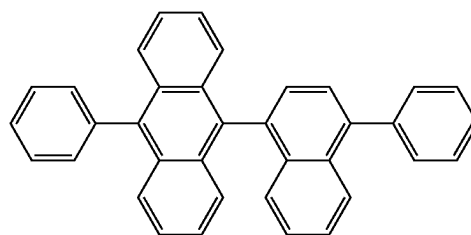


H2

H8

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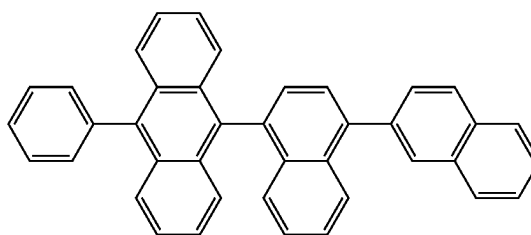


H3

H9

25

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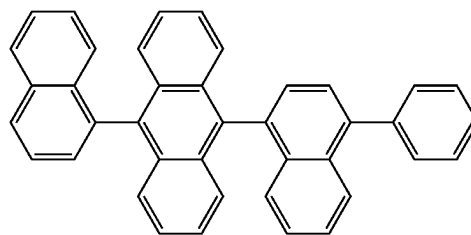


H4

H10

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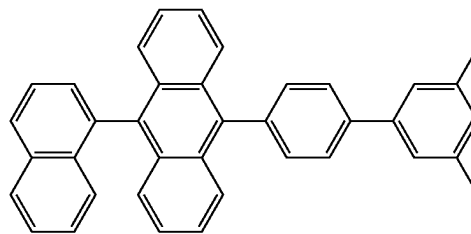


H5

H11

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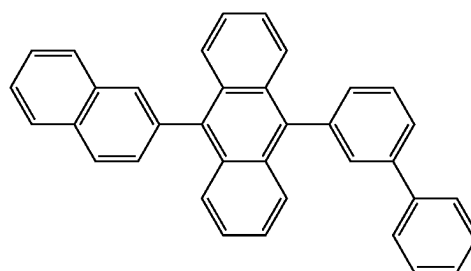
H6

H12

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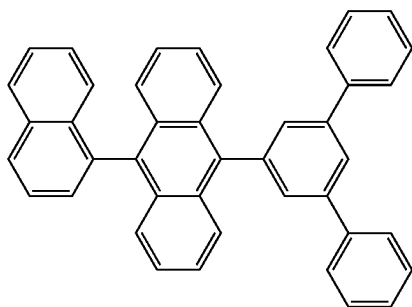
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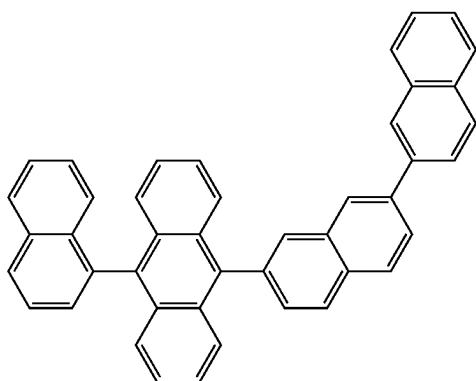
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H13

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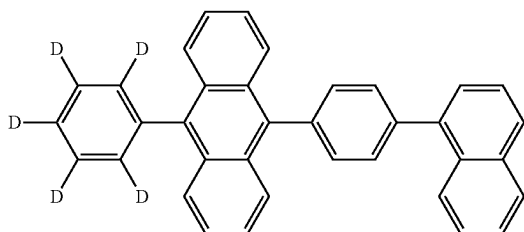
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H14 15

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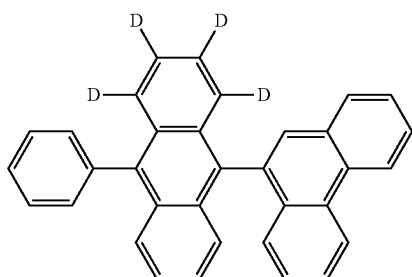
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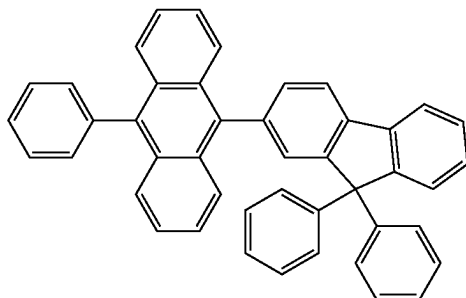
H16

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H17

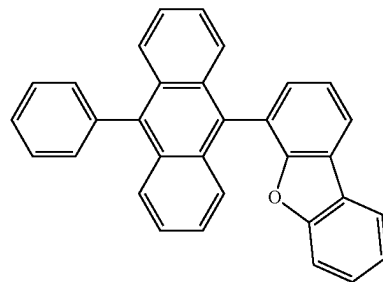
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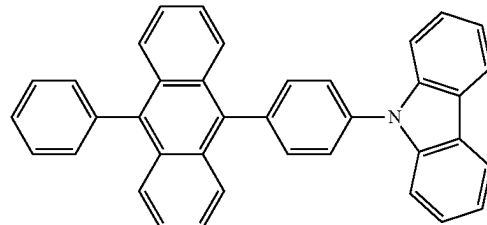
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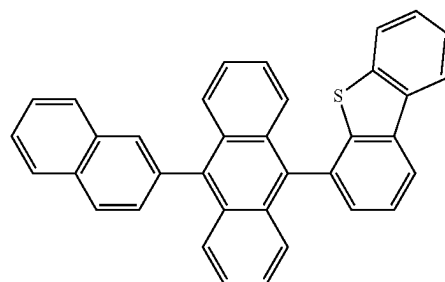
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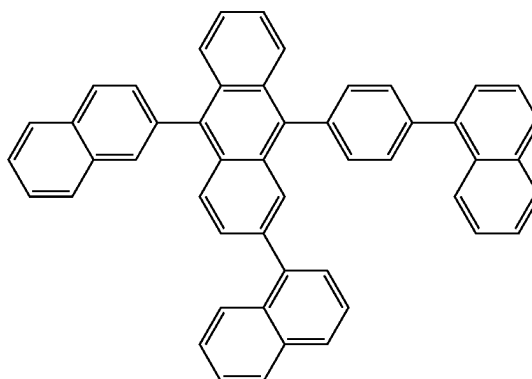
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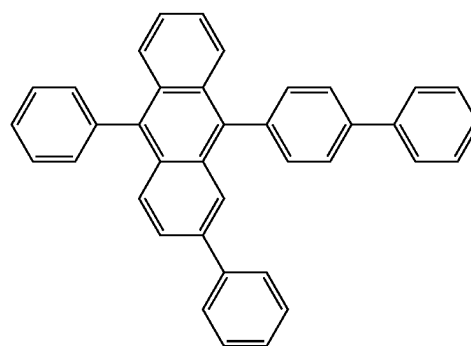
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HT20



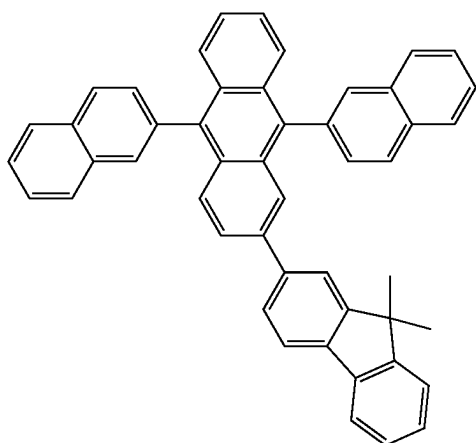
HT21



HT22

85

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H23

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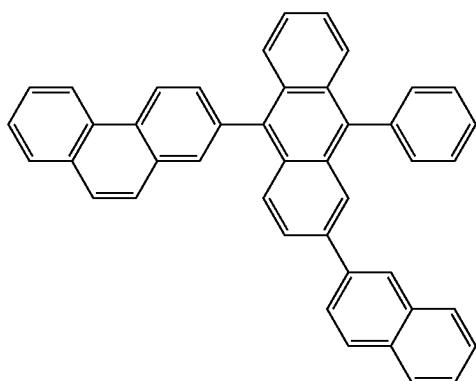
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H24

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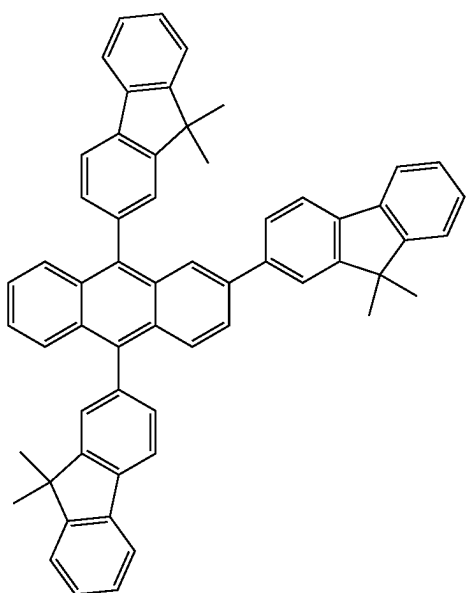
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H25

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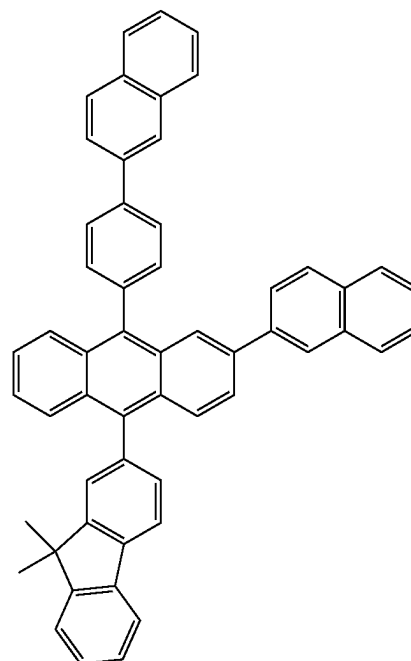
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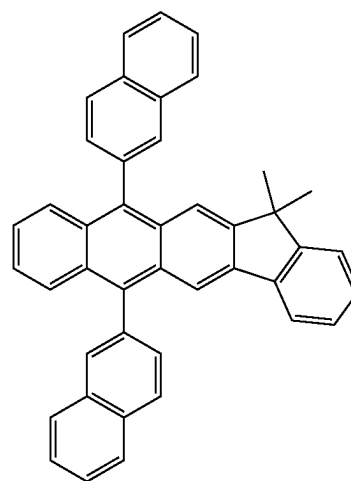
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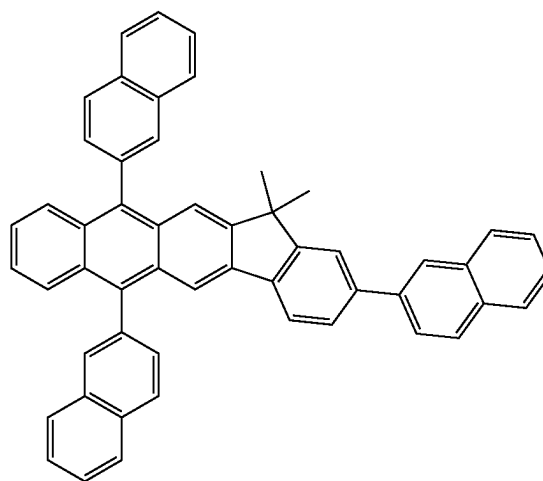
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H26



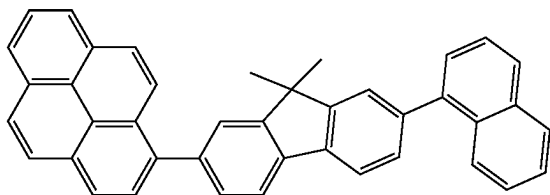
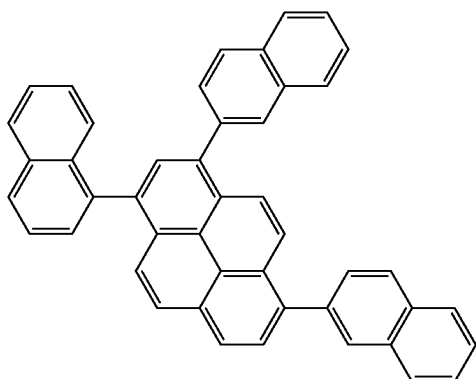
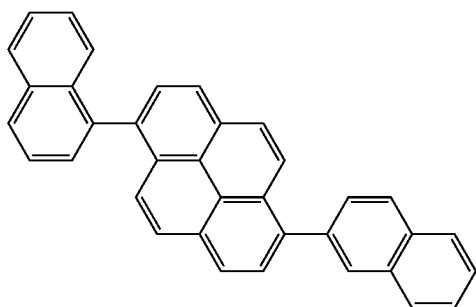
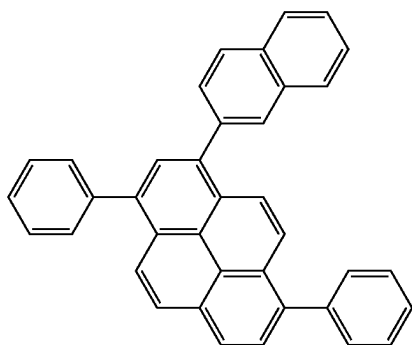
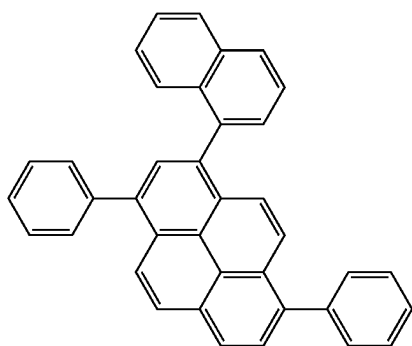
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H28

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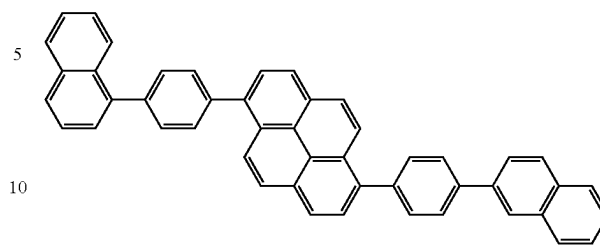
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**88**

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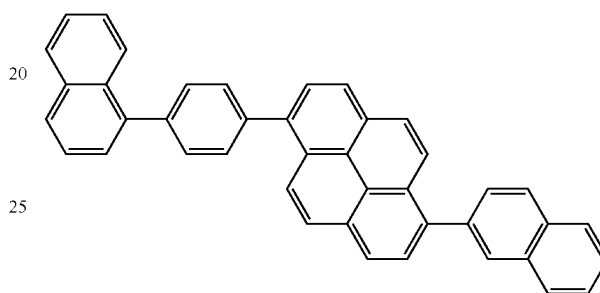
H29

H34



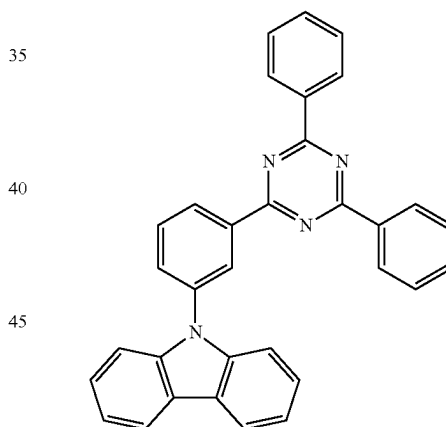
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H35



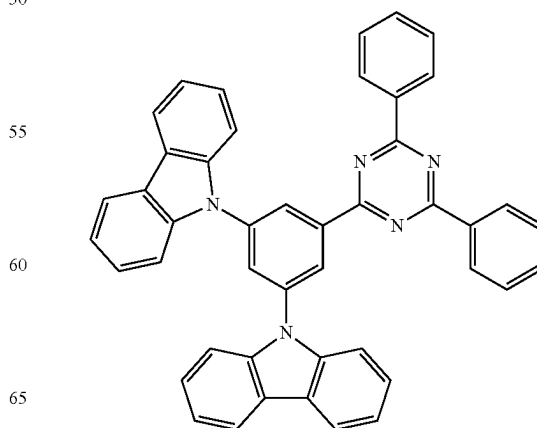
H31

H36



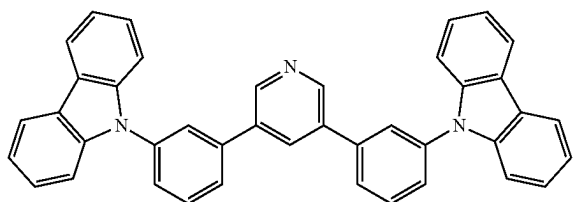
H32

H37



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H38



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H39

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H40

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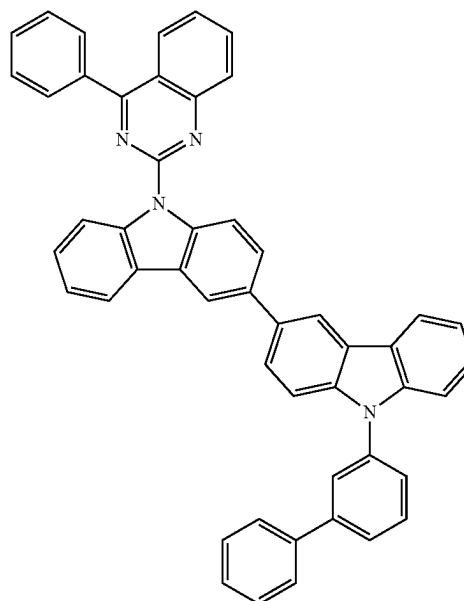
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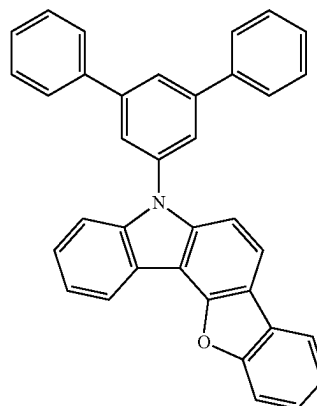
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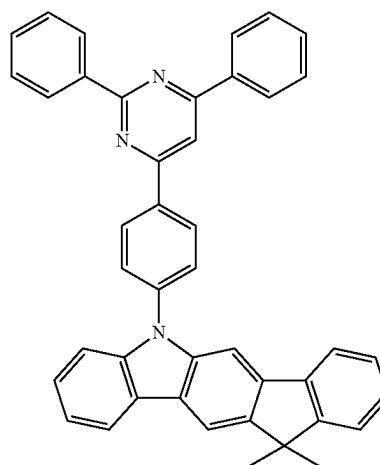
H41



H42



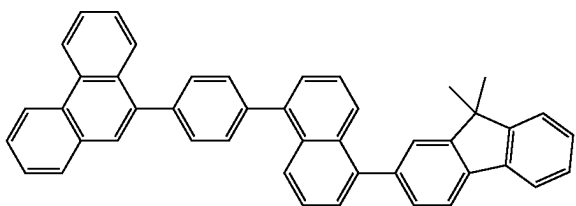
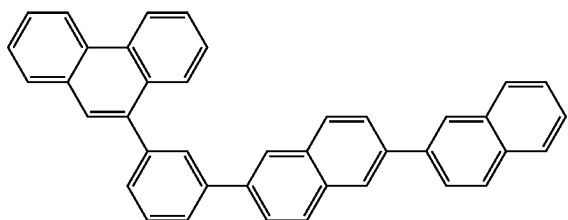
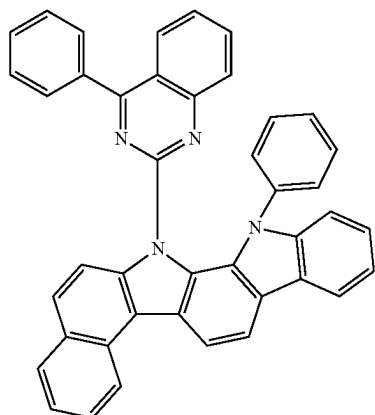
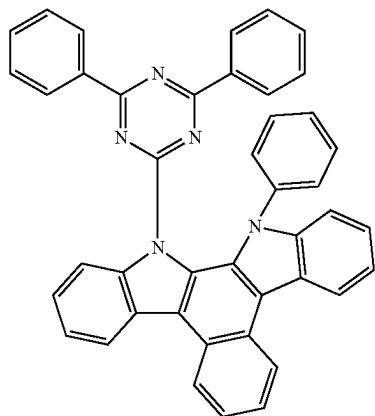
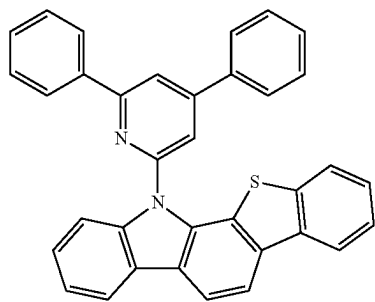
H43



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**92**

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H44

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H45

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H46

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H47

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H48

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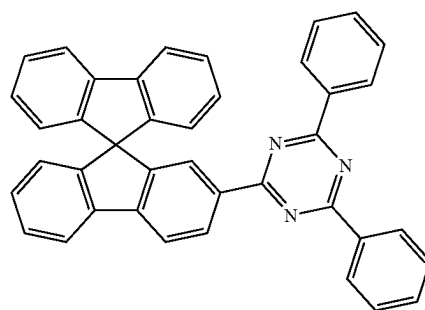
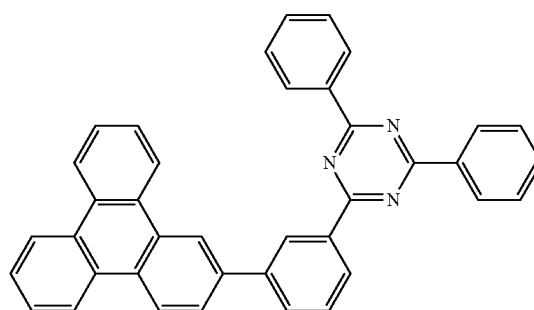
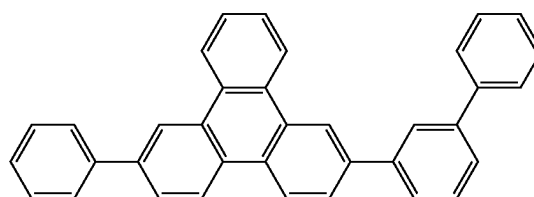
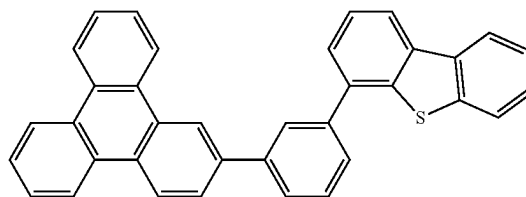
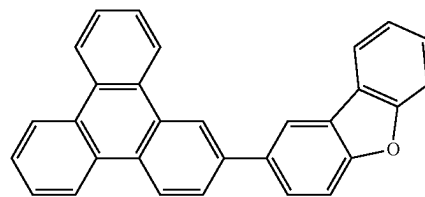
H49

H50

H51

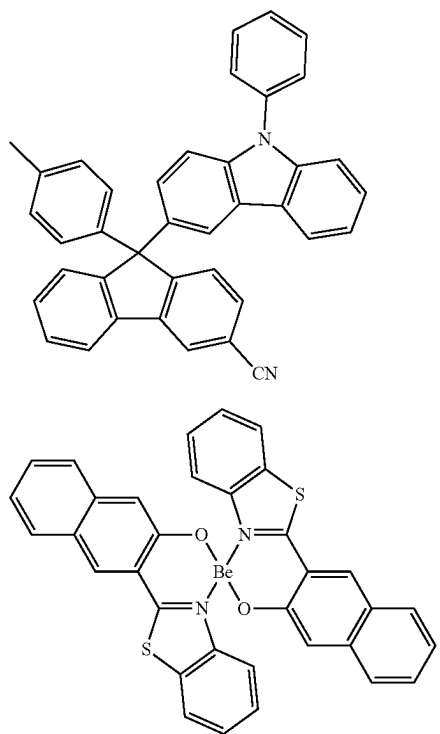
H52

H53



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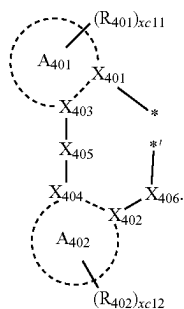
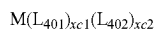
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Phosphorescent Dopant Included in Emission Layer in Organic Layer 150

The dopant may include the organometallic compound represented by Formula 1.

The phosphorescent dopant may include an organometallic complex represented by Formula 401 below:



In Formulae 401 and 402,

M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm),

L_{401} may be selected from ligands represented by Formula 402, and xc1 may be 1, 2, or 3, wherein, when xc1 is 2 or more, two or more $L_{401}(s)$ may be identical to or different from each other,

L_{402} may be an organic ligand, and xc2 may be an integer from 0 to 4, wherein, when xc2 is 2 or more, two or more $L_{402}(s)$ may be identical to or different from each other,

94

X_{401} to X_{404} may each independently be nitrogen or carbon,

X_{401} and X_{403} may be linked via a single bond or a double bond, and X_{402} and X_{404} may be linked via a single bond or a double bond,

A_{401} and A_{402} may each independently be selected from a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group,

X_{405} may be a single bond, $*-O-$, $*-S-$, $*-C(=O)-$, $*-N(Q_{411})-$, $*-C(Q_{411})(Q_{412})-$, $*-C(Q_{411})=C(Q_{412})-$, $*-C(Q_{411})=$, or $*=C(Q_{411})=$, wherein Q_{411} and Q_{412} may be hydrogen, deuterium, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group,

X_{406} may be a single bond, O, or S,

R_{401} and R_{402} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C_1 - C_{20} alkyl group, a substituted or unsubstituted C_1 - C_{20} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_{401})(Q_{402})(Q_{403})$, $-N(Q_{401})(Q_{402})$, $-B(Q_{401})(Q_{402})$, $-C(=O)(Q_{401})$, $-S(=O)_2(Q_{401})$, and $-P(=O)(Q_{401})(Q_{402})$, wherein Q_{401} to Q_{403} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a C_6 - C_{20} aryl group, and a C_1 - C_{20} heteroaryl group,

xc1 and xc2 may each independently be an integer from 0 to 10,

and $*$ in Formula 402 each indicate a binding site to M in Formula 401.

In one embodiment, A_{401} and A_{402} in Formula 402 may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, an indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

In one or more embodiments, in Formula 402, i) X_{401} may be nitrogen, and X_{402} may be carbon, or ii) X_{401} and X_{402} may each be nitrogen at the same time.

In one or more embodiments, R_{401} and R_{402} in Formula 501 may each independently be selected from:

hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

95

a C₁-C₂₀ alkyl group and a C₁-C₂₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxaliny group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxaliny group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxaliny group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and —Si(Q₄₀₁)(Q₄₀₂)(Q₄₀₃), —N(Q₄₀₁)(Q₄₀₂), —B(Q₄₀₁)(Q₄₀₂), —C(=O)(Q₄₀₁), —S(=O)₂(Q₄₀₁), and —P(=O)(Q₄₀₁)(Q₄₀₂), and Q₄₀₁ to Q₄₀₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited thereto.

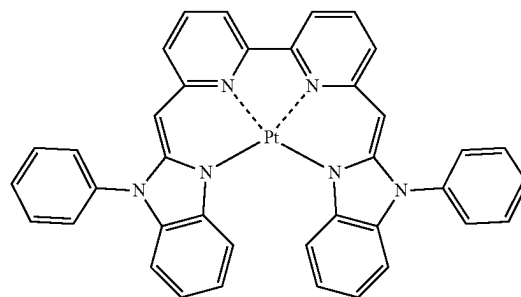
In one or more embodiments, in Formula 401, when xc1 is 2 or more, two A₄₀₁(s) in two or more L₄₀₁(s) may optionally be linked via X₄₀₇, which is a linking group, or two A₄₀₂(s) in two or more L₄₀₁(s) may optionally be linked via X₄₀₈, which is a linking group (see Compounds PD1 to PD4 and PD7). X₄₀₇ and X₄₀₈ may each independently be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q₄₁₃)—*, *—C(Q₄₁₃)(Q₄₁₄)—*, or *—C(Q₄₁₃)=C(Q₄₁₄)—* (wherein Q₄₁₃ and Q₄₁₄ may each independently be hydrogen, deuterium, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group), but embodiments of the present disclosure are not limited thereto.

L₄₀₂ in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For example, L₄₀₂ may be selected from halogen, diketone (for example, acetylacetonate), carboxylic acid (for example, picolinate), —C(=O), isonitrile, —CN, and phosphorus (for example, phosphine, or phosphite), but embodiments of the present disclosure are not limited thereto.

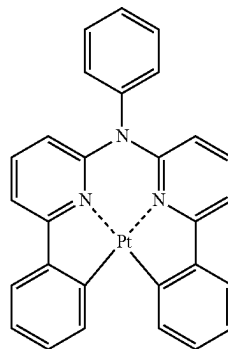
96

In one or more embodiments, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25, but embodiments of the present disclosure are not limited thereto:

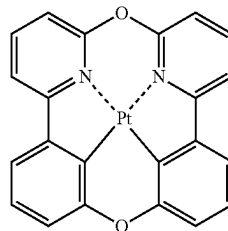
PD1



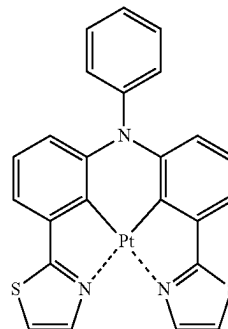
PD2



PD3

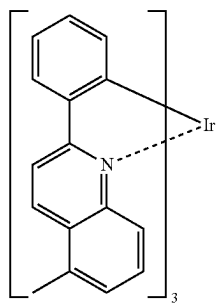
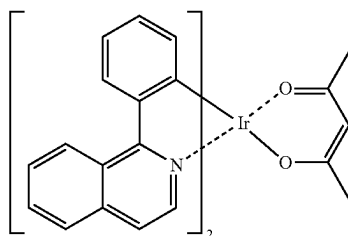
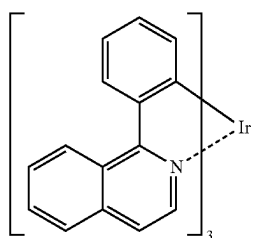
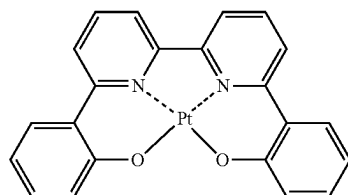
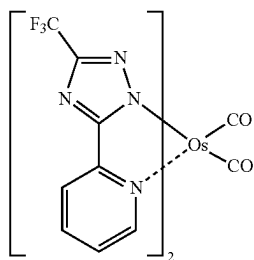
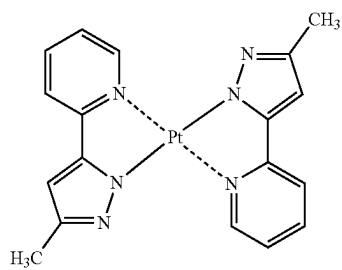


PD4



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**98**

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PD5

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PD6

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PD7

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PD8

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PD9

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PD10

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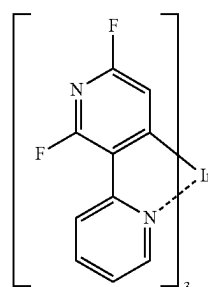
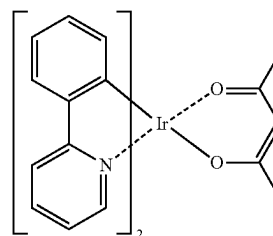
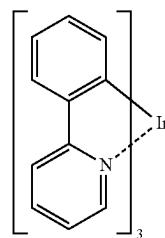
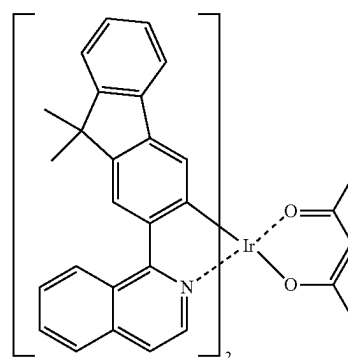
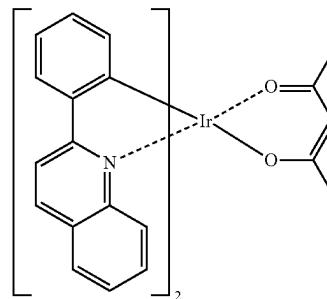
PD11

PD12

PD13

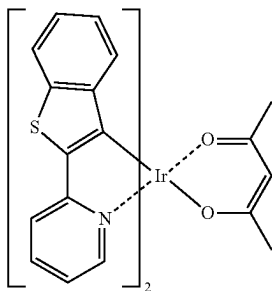
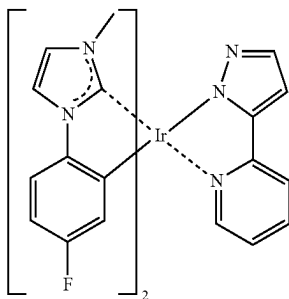
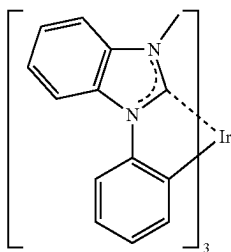
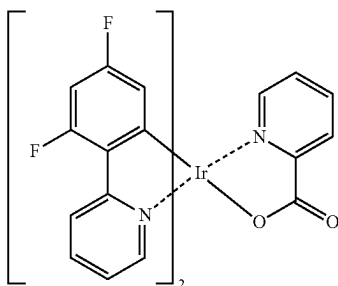
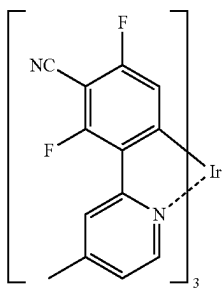
PD14

PD15



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-continued

**100**

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PD16

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PD17

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PD18

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PD19

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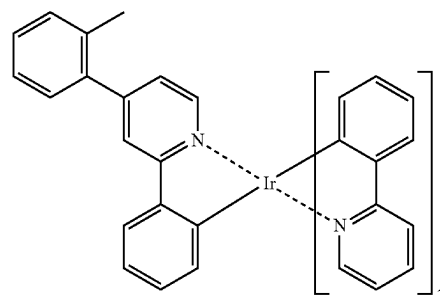
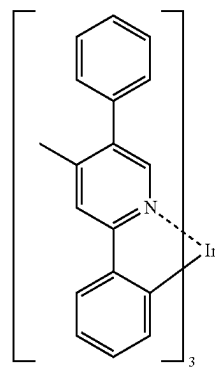
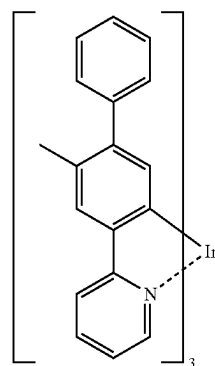
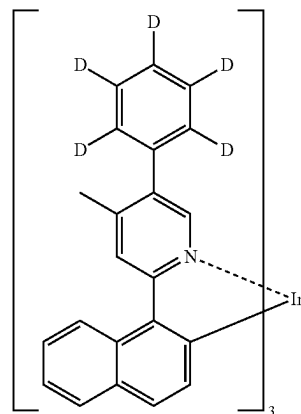
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PD20

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60

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PD21

PD22

PD23

PD24

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylene group, a hexacenylenylene group, a pentacenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazone group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothi-

103

ophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

In one or more embodiments, R_{501} and R_{502} in Formula 501 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group and —Si(Q_{31})(Q_{32})(Q_{33}), and

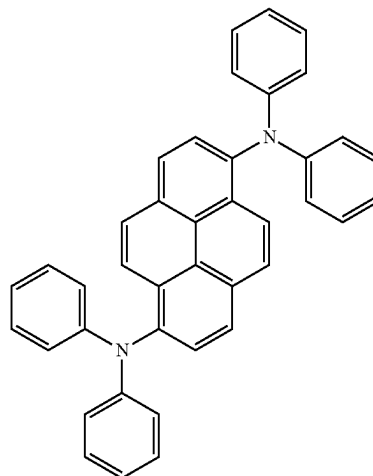
Q_{31} to Q_{33} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In one or more embodiments, xd4 in Formula 501 may be 2, but embodiments of the present disclosure are not limited thereto.

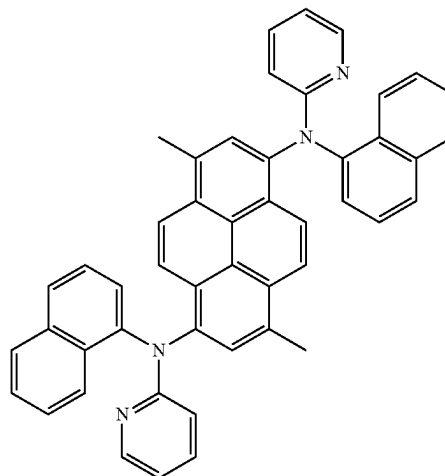
For example, the fluorescent dopant may be selected from Compounds FD1 to FD22:

104

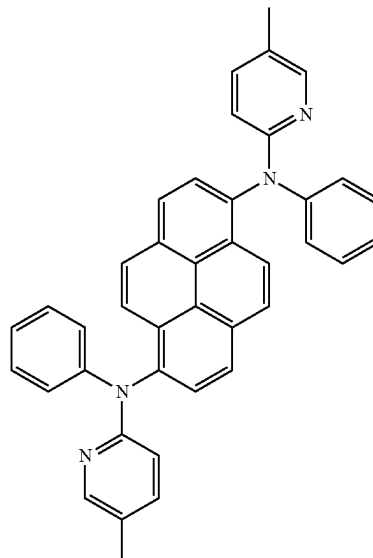
FD1



FD2

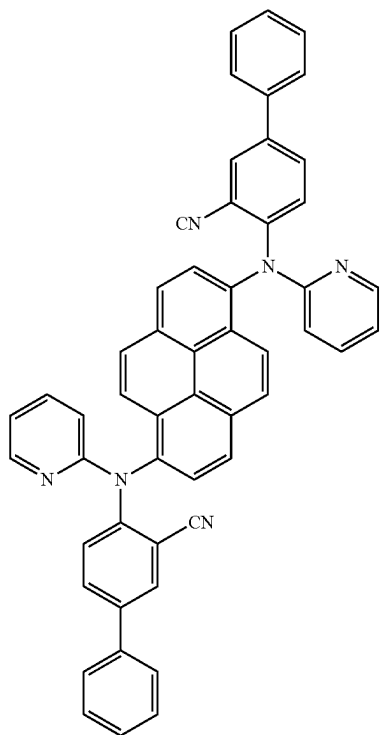


FD3



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106

-continued

FD4

FD6

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FD5 45

FD7

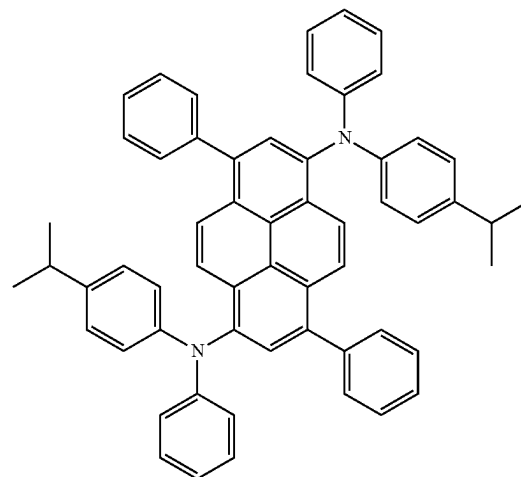
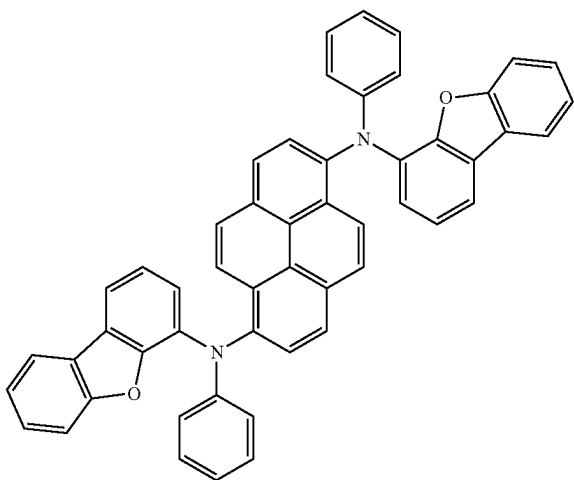
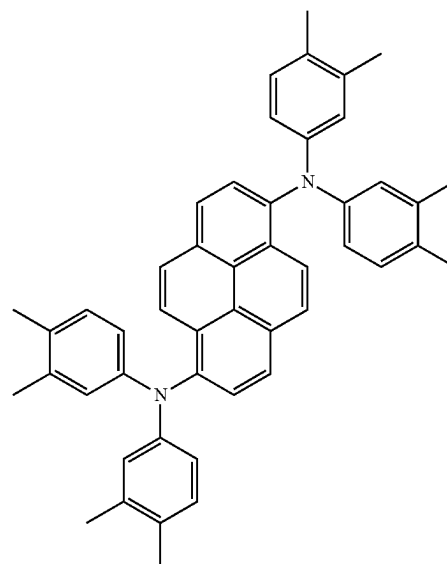
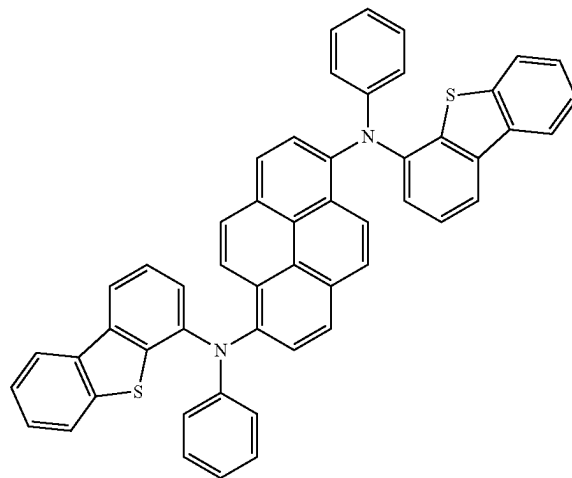
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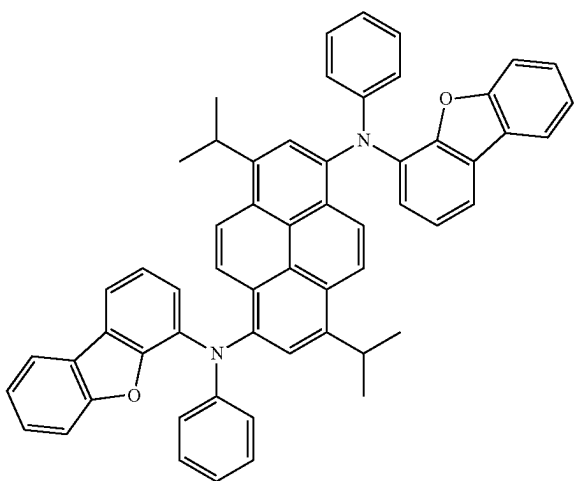
FD8



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FD9

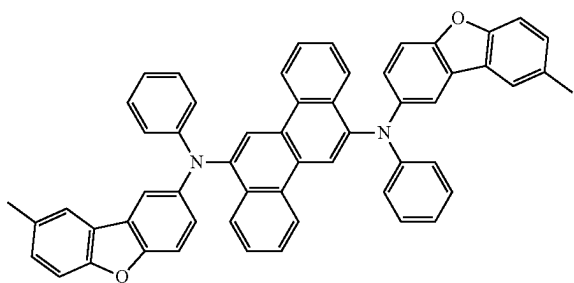


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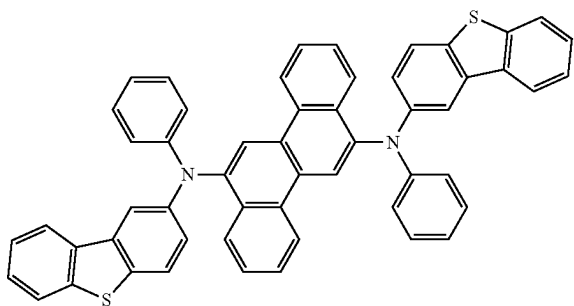
FD10



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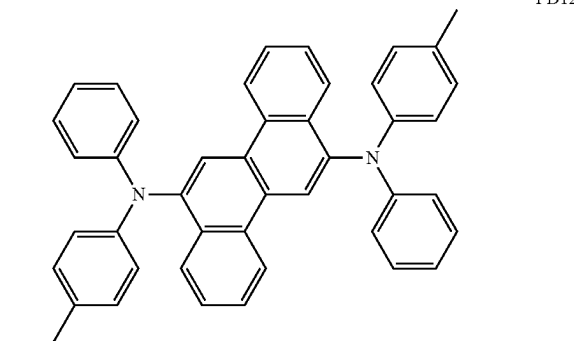
FD11



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FD12



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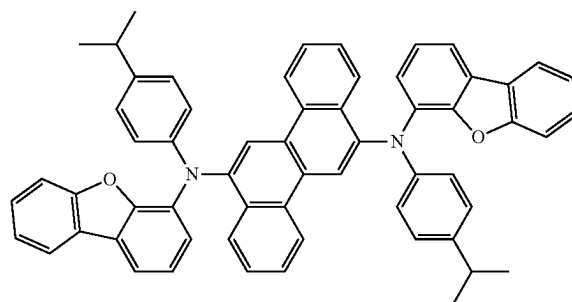
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108

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FD13



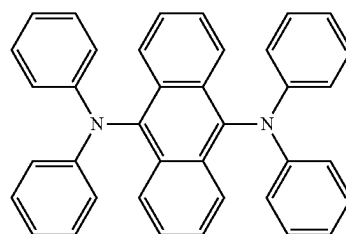
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FD14

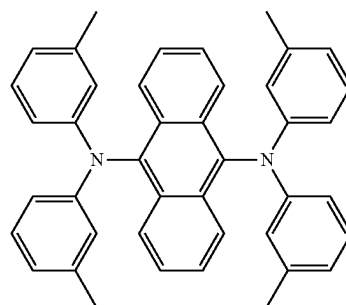
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FD15

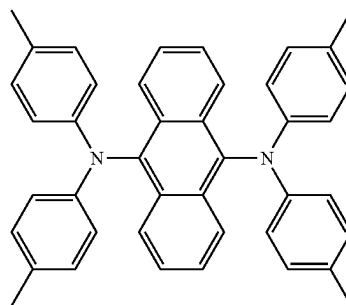
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FD16

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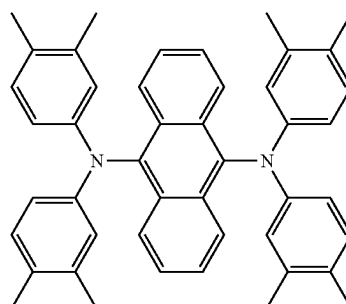
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FD17

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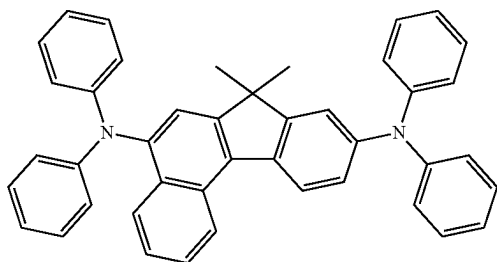
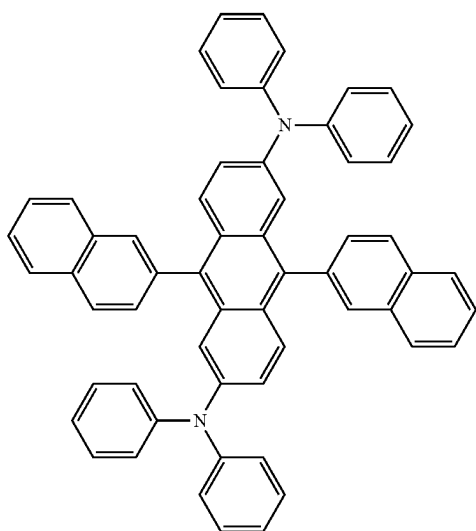
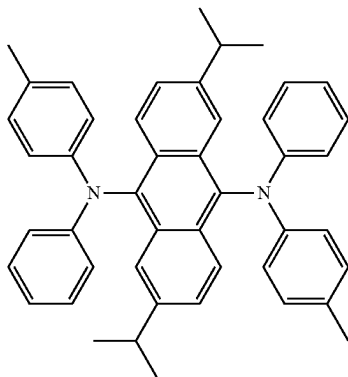
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**110**

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FD18

FD21

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FD19

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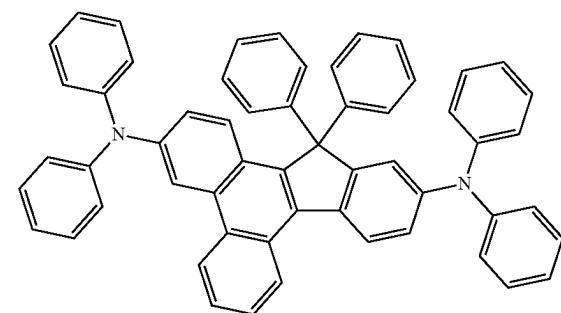
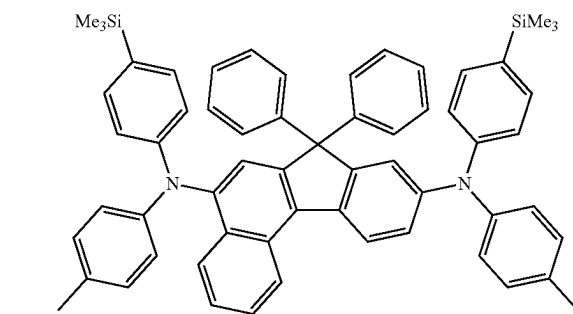
FD22

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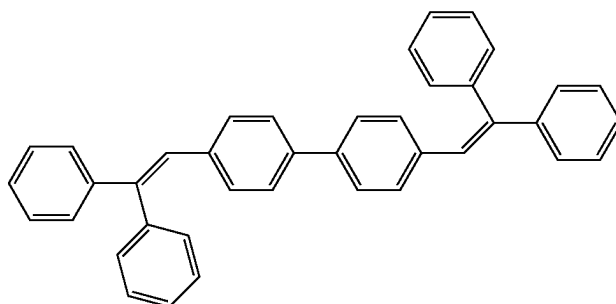
FD20

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In one or more embodiments, the fluorescent dopant may be selected from the following compounds, but embodiments of the present disclosure are not limited thereto:

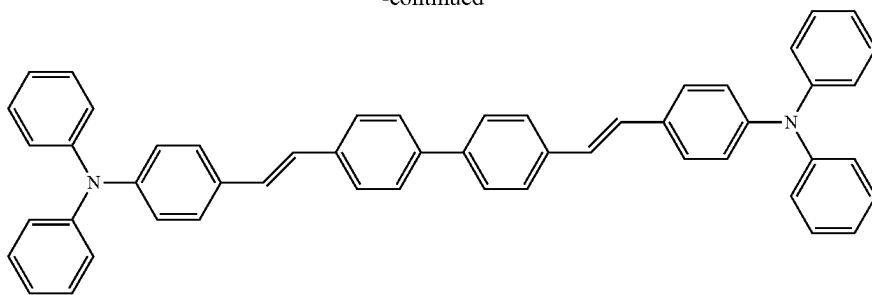


DPVBi

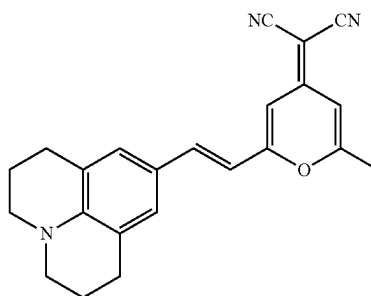
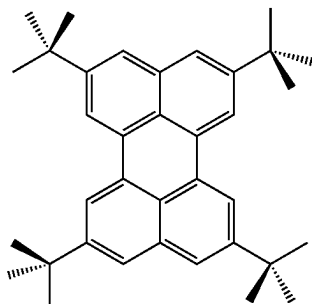
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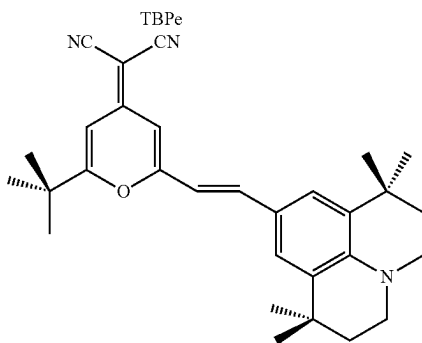
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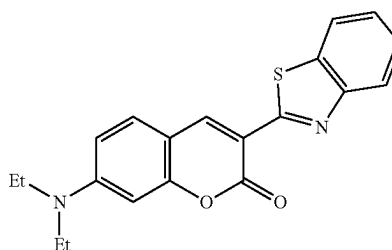
DPAVBi



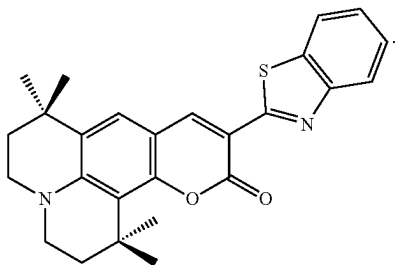
DCM



DCJTB



Coumarin 6



C545T

Electron Transport Region in Organic Layer 150

The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer, but embodiments of the present disclosure are not limited thereto.

For example, the electron transport region may have an electron transport layer/electron injection layer structure, a

hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers are sequentially stacked from an emission layer. However, embodiments of the structure of the electron transport region are not limited thereto.

The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

The “ π electron-depleted nitrogen-containing ring” indicates a C_1 - C_{60} heterocyclic group having at least one $*-N=*$ moiety as a ring-forming moiety.

For example, the “ π electron-depleted nitrogen-containing ring” may be i) a 5-membered to π -membered heteromonocyclic group having at least one $*-N=*$ moiety, ii) a heteropolycyclic group in which two or more 5-membered to π -membered heteromonocyclic groups each having at least one $*-N=*$ moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to π -membered heteromonocyclic groups, each having at least one $*-N=*$ moiety, is condensed with at least one C_5 - C_{60} carbocyclic group.

Examples of the π electron-depleted nitrogen-containing ring include an imidazole, a pyrazole, a thiazole, an isothiazole, an oxazole, an isoxazole, a pyridine, a pyrazine, a pyrimidine, a pyridazine, an indazole, a purine, a quinoline, an isoquinoline, a benzoquinoline, a phthalazine, a naphthyridine, a quinoxaline, a quinazoline, a cinnoline, a phenanthridine, an acridine, a phenanthroline, a phenazine, a benzimidazole, an isobenzothiazole, a benzoxazole, an isobenzoxazole, a triazole, a tetrazole, an oxadiazole, a triazine, a thiadiazole, an imidazopyridine, an imidazopyrimidine, and an azacarbazole, but are not limited thereto.

For example, the electron transport region may include a compound represented by Formula 601:



In Formula 601,

Ar_{601} may be a substituted or unsubstituted C_5 - C_{60} carbocyclic group or a substituted or unsubstituted C_1 - C_{60} heterocyclic group,

$xe11$ may be 0, 1, 2, or 3,

L_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

$xe1$ may be an integer from 0 to 5,

R_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_{601})(Q_{602})(Q_{603})$, $-C(=O)(Q_{601})$, $-S(=O)_2(Q_{601})$, and $-P(=O)(Q_{601})(Q_{602})$, Q_{601} to Q_{603} may each independently be a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and

$xe21$ may be an integer from 1 to 5.

In one embodiment, at least one of $Ar_{601}(s)$ in the number of $xe11$ and $R_{601}(s)$ in the number of $xe21$ may include the π electron-depleted nitrogen-containing ring.

In one embodiment, Ar_{601} in Formula 601 may be selected from:

a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, a dibenzofuran group, a dibenzothienophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group; and

a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, a dibenzofuran group, a dibenzothienophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, $-Si(Q_{31})(Q_{32})(Q_{33})$, $-S(=O)_2(Q_{31})$, and $-P(=O)(Q_{31})(Q_{32})$, and Q_{31} to Q_{33} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

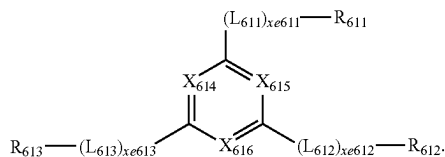
When $xe11$ in Formula 601 is 2 or more, two or more $Ar_{601}(s)$ may be linked to each other via a single bond.

In one or more embodiments, Ar_{601} in Formula 601 may be an anthracene group.

In one or more embodiments, a compound represented by Formula 601 may be represented by Formula 601-1 below:

115

Formula 601-1



In Formula 601-1,

X_{614} may be N or C(R_{614}), X_{615} may be N or C(R_{615}), and X_{616} may be N or C(R_{616}), wherein at least one selected from X_{614} to X_{616} may be N,

L_{611} to L_{613} may each independently be the same as described in connection with L_{601} ,

xe611 to xe613 may each independently be defined the same as xe1,

R_{611} to R_{613} may each independently be the same as described in connection with R_{601} , and

R_{614} to R_{616} may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In one embodiment, L_{601} and L_{611} to L_{613} in Formulae 601 and 601-1 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofura-

116

nylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinoliny group, an isoquinoliny group, a benzoquinoliny group, a phthalazinyl group, a naphthyridinyl group, a quinoxaliny group, a quinazoliny group, a cinnoliny group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group,

but embodiments of the present disclosure are not limited thereto.

In one or more embodiments, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

In one or more embodiments, R_{601} and R_{611} to R_{613} in Formulae 601 and 601-1 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group,

117

a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzo-furanyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a 5
dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quina-
zolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenaziny group, a benzimidazolyl group, an isoben-
zothiazolyl group, a benzoxazolyl group, an isobenzo-
xazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; 20
a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluo-
ranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a penta-
phenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzo-
furanyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a 30
dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quina-
zolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenaziny group, a benzimidazolyl group, an isoben-
zothiazolyl group, a benzoxazolyl group, an isobenzo-
xazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at
least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, 50
a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluo-
ranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a penta-
phenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzo-
furanyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a 60
dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl

118

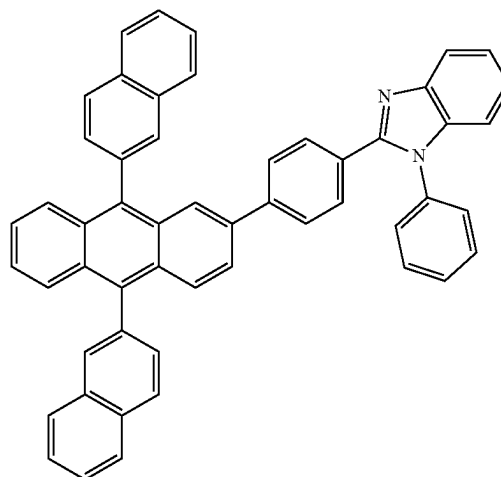
group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quina-
zolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenaziny group, a benzimidazolyl group, an isoben-
zothiazolyl group, a benzoxazolyl group, an isobenzo-
xazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and

—S(=O)₂(Q₆₀₁) and —P(=O)(Q₆₀₁)(Q₆₀₂), and

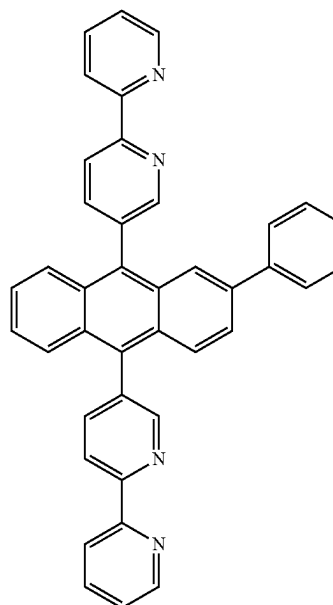
Q₆₀₁ and Q₆₀₂ are the same as described herein above.

The electron transport region may include at least one compound selected from Compounds ET1 to ET36, but embodiments of the present disclosure are not limited thereto:

ET1

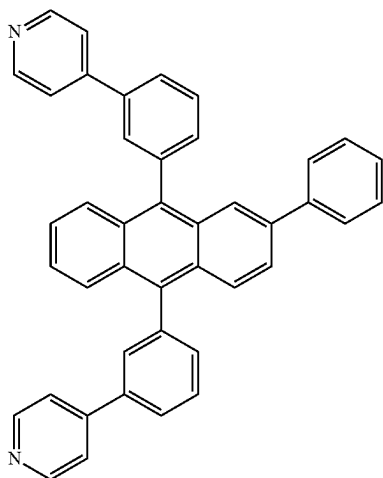


ET2



119

-continued



ET3

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120

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ET4

ET5

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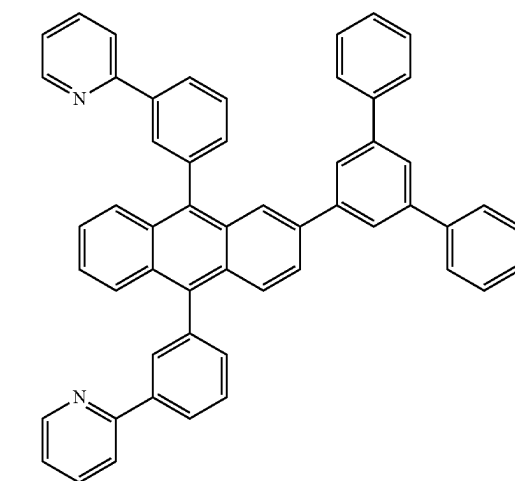
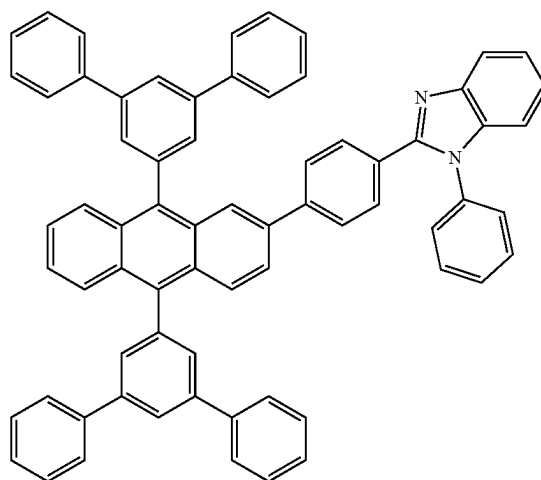
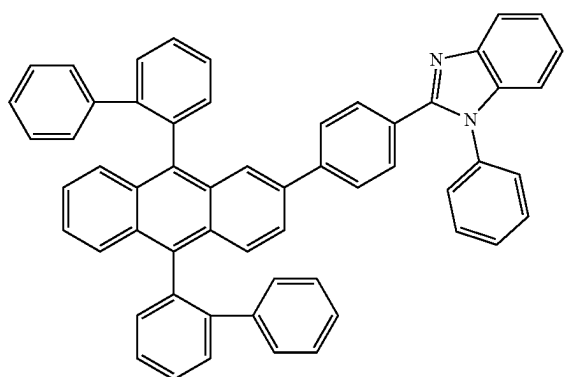
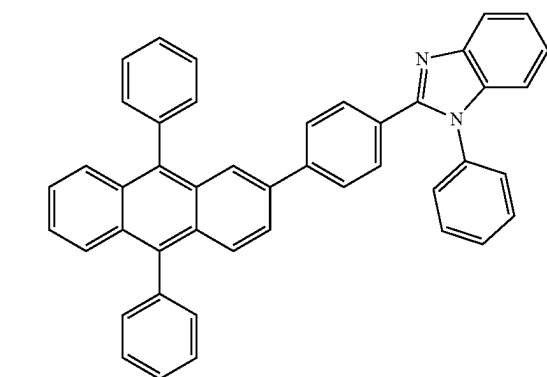
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ET6

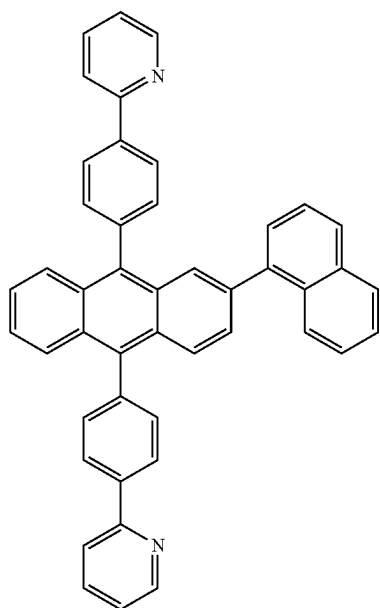
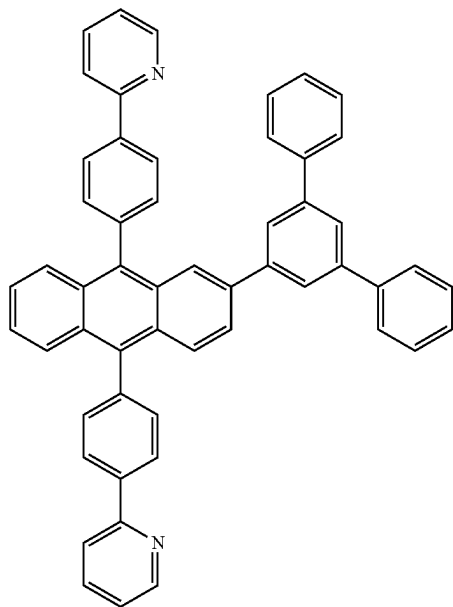
ET7

ET8



121

-continued

**122**

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ET9

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ET10

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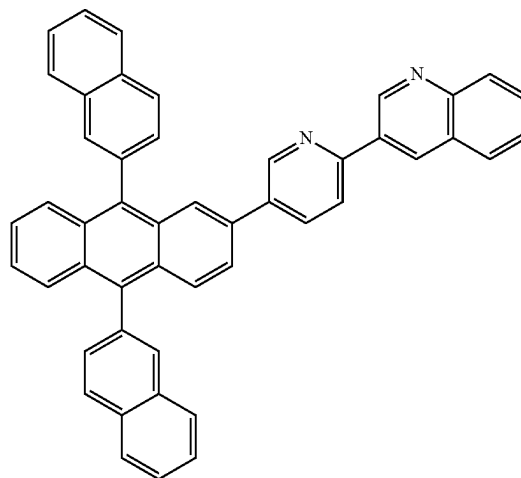
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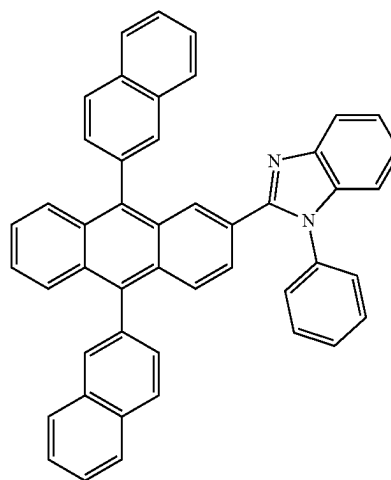
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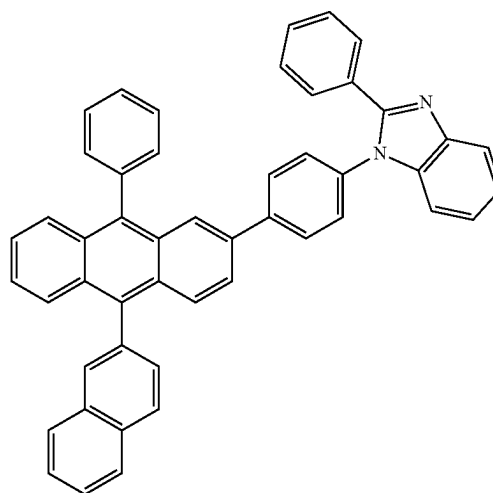
ET11



ET12

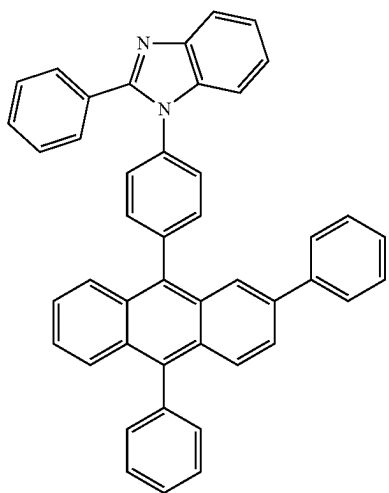
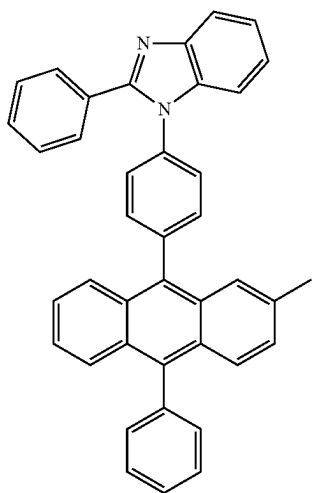
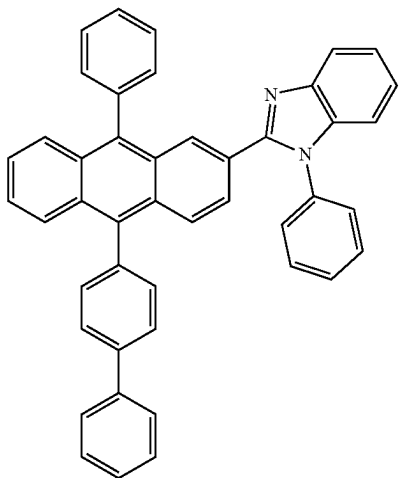


ET13



123

-continued



124

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ET14

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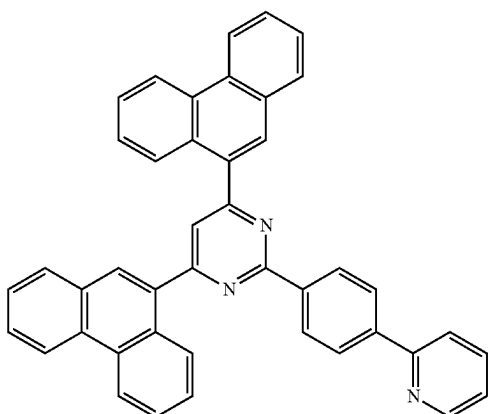
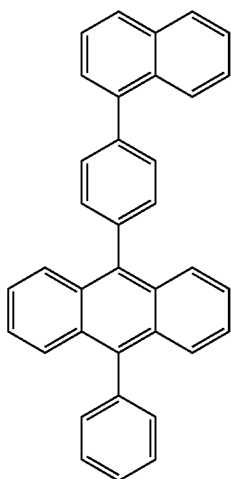
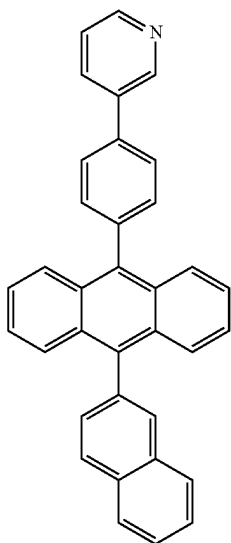
ET17

ET18

ET19

125

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**126**

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ET20

ET23

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ET22

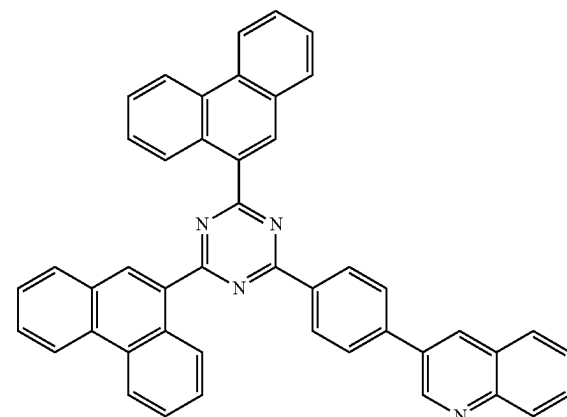
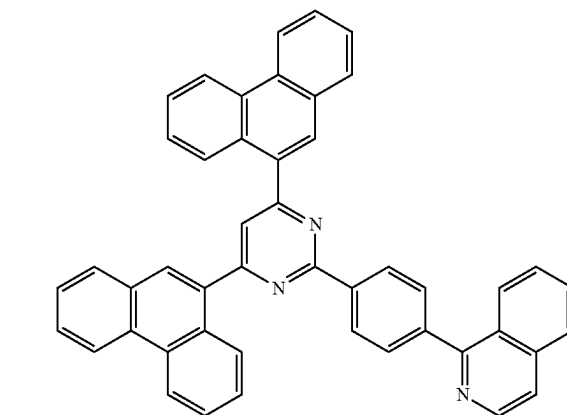
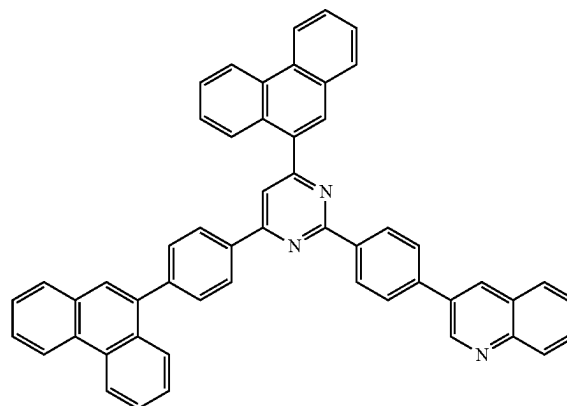
ET25

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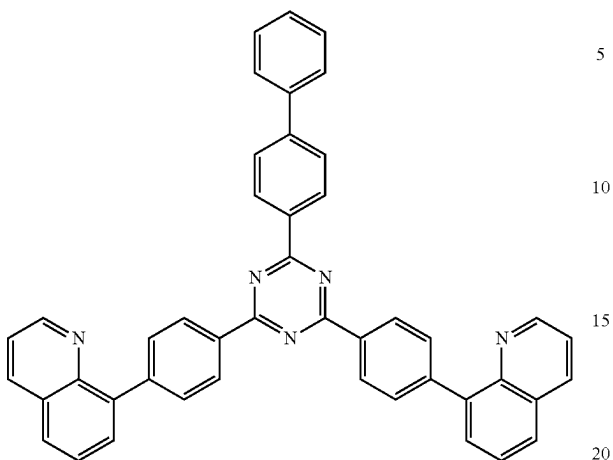
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127

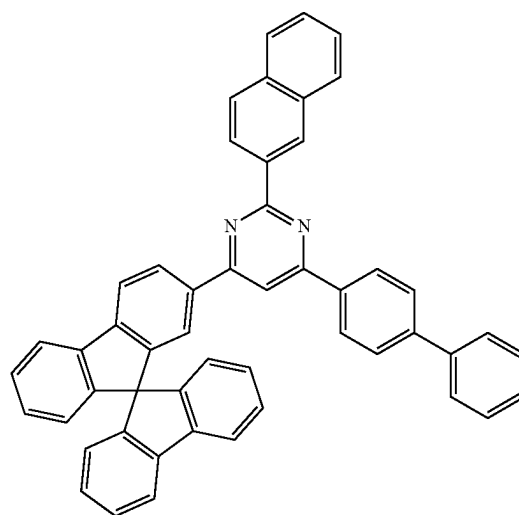
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ET26

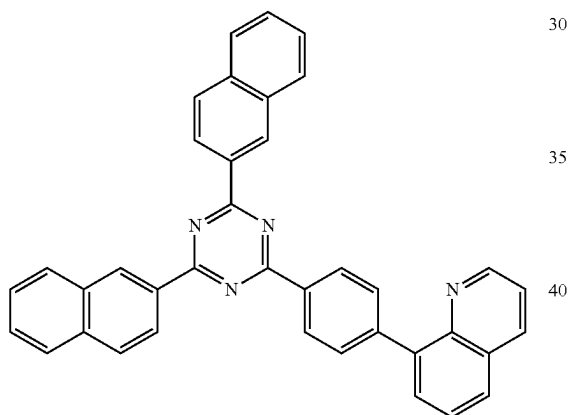
**128**

-continued

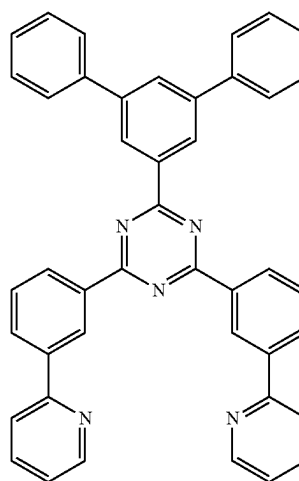
ET29



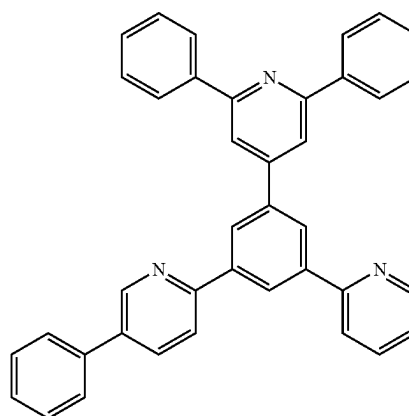
ET27



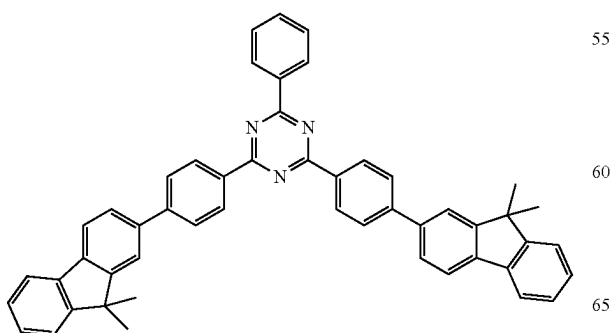
ET30



ET31

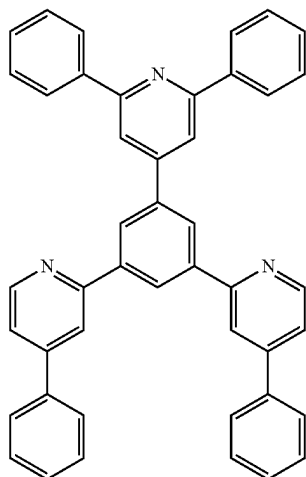


ET28



129

-continued



ET32

130

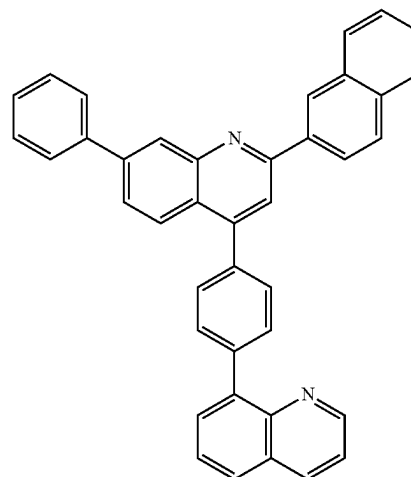
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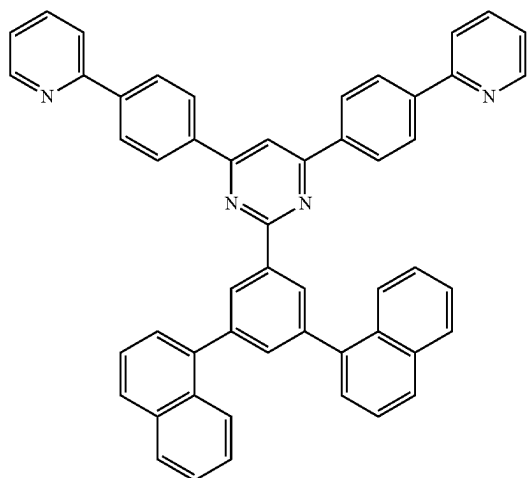


ET35

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ET33

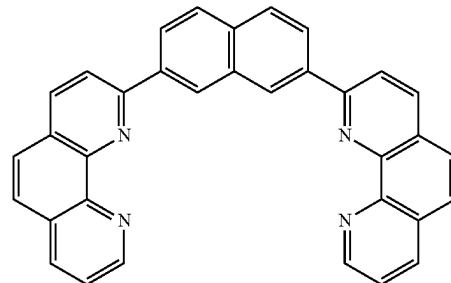
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ET36



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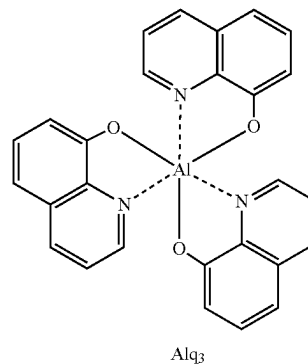
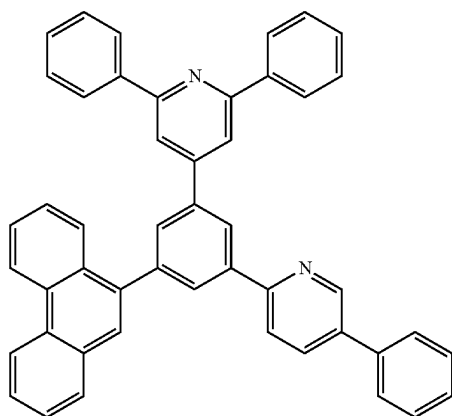
In one or more embodiments, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq₃, BAlq, 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-4H-1,2,4-triazole (TAZ), and NTAZ:

ET34

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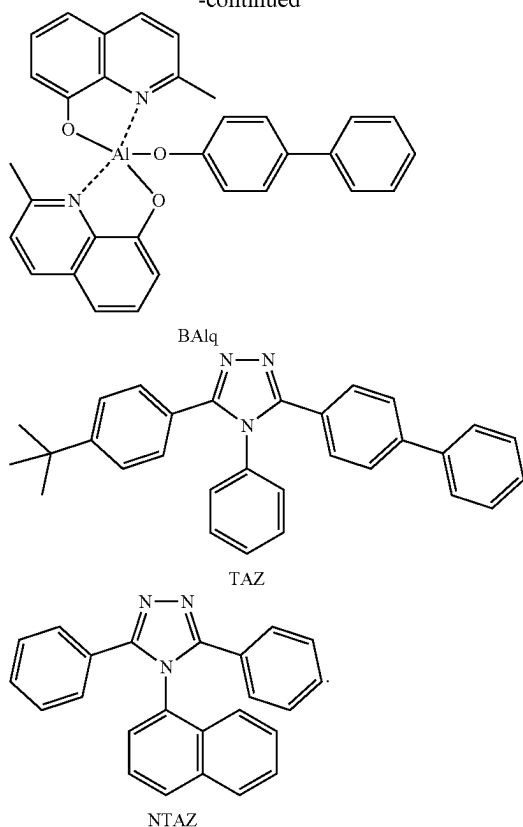
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Alq₃

131

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Thicknesses of the buffer layer, the hole blocking layer, and the electron control layer may each be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thicknesses of the buffer layer, the hole blocking layer, and the electron control layer are within these ranges, the buffer layer, the hole blocking layer, and the electron control layer may have excellent hole blocking characteristics or electron control characteristics without a substantial increase in driving voltage.

A thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described herein above, the electron transport layer may have suitable or satisfactory electron transport characteristics without a substantial increase in driving voltage.

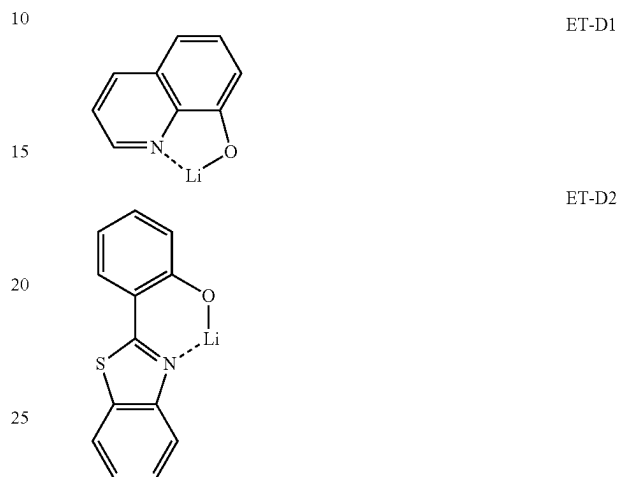
The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described herein above, a metal-containing material.

The metal-containing material may include at least one selected from alkali metal complex and alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenylloxazole, a hydroxy phenylthiazole, a hydroxy diphenylloxadiazole, a hydroxy diphenylthiadiazole, a hydroxy phenylpyri-

132

dine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene, but embodiments of the present disclosure are not limited thereto.

For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2.



The electron transport region may include an electron injection layer that facilitates electron injection from the second electrode 190. The electron injection layer may directly contact the second electrode 190.

The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof.

The alkali metal may be selected from Li, Na, K, Rb, and Cs. In one embodiment, the alkali metal may be Li, Na, or Cs. In one or more embodiments, the alkali metal may be Li or Cs, but embodiments of the present disclosure are not limited thereto.

The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

The alkali metal compound may be selected from alkali metal oxides, such as Li₂O, Cs₂O, or K₂O, and alkali metal halides, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, or KI. In one embodiment, the alkali metal compound may be selected from LiF, Li₂O, NaF, LiI, NaI, CsI, and KI, but embodiments of the present disclosure are not limited thereto.

The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO, SrO, CaO, $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ ($0 < x < 1$), or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ ($0 < x < 1$). In one embodiment, the alkaline earth-metal compound may be selected from BaO, SrO, and CaO, but embodiments of the present disclosure are not limited thereto.

The rare earth metal compound may be selected from YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , and TbF_3 . In one embodiment, the rare earth metal compound may be selected from YbF_3 , ScF_3 , TbF_3 , YbI_3 , ScI_3 , and TbI_3 , but embodiments of the present disclosure are not limited thereto.

The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described herein above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenyloxazole, hydroxy phenylthiazole, hydroxy diphenyloxadiazole, hydroxy diphenylthiadiazole, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene, but embodiments of the present disclosure are not limited thereto.

The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof, as described herein above. In one or more embodiments, the electron injection layer may further include an organic material. When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described herein above, the electron injection layer may have suitable or satisfactory electron injection characteristics without a substantial increase in driving voltage.

Second Electrode 190

The second electrode 190 may be on the organic layer 150 having such a structure. The second electrode 190 may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode 190 may be selected from metal, an alloy, an electrically conductive compound, and a combination thereof, which have a relatively low work function.

The second electrode 190 may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO, but embodiments of the present disclosure are not limited thereto. The second electrode 190 may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

The second electrode 190 may have a single-layered structure, or a multi-layered structure including two or more

Description of FIGS. 2 to 4

FIG. 2 is a schematic view of an organic light-emitting device 20 according to an embodiment. The organic light-emitting device 20 includes a first capping layer 210, the first electrode 110, the organic layer 150, and the second electrode 190, which are sequentially stacked in this stated order. FIG. 3 is a schematic view of an organic light-emitting device 30 according to an embodiment. The organic light-emitting device 30 includes the first electrode 110, the organic layer 150, the second electrode 190, and a second capping layer 220, which are sequentially stacked in this stated order. FIG. 4 is a schematic view of an organic light-emitting device 40 according to an embodiment. The organic light-emitting device 40 includes the first capping layer 210, the first electrode 110, the organic layer 150, the second electrode 190, and the second capping layer 220, which are sequentially stacked in this stated order.

Regarding FIGS. 2 to 4, the first electrode 110, the organic layer 150, and the second electrode 190 may be understood by referring to the description presented in connection with FIG. 1.

In the organic layer 150 of each of the organic light-emitting devices 20 and 40, light generated in an emission layer may pass through the first electrode 110 and the first capping layer 210 toward the outside, wherein the first electrode 110 may be a semi-transmissive electrode or a transmissive electrode. In the organic layer 150 of each of the organic light-emitting devices 30 and 40, light generated in an emission layer may pass through the second electrode 190 and the second capping layer 220 toward the outside, wherein the second electrode 190 may be a semi-transmissive electrode or a transmissive electrode.

The first capping layer 210 and the second capping layer 220 may increase external luminescent efficiency according to the principle of constructive interference.

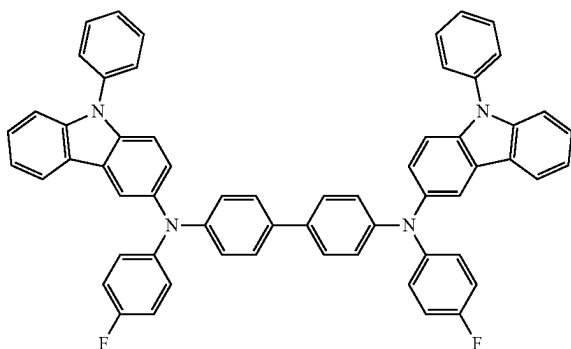
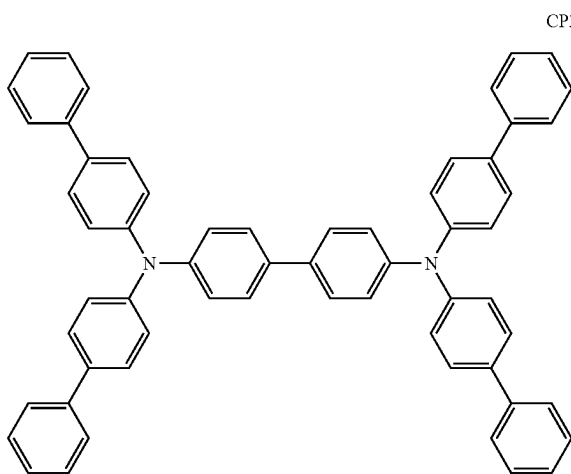
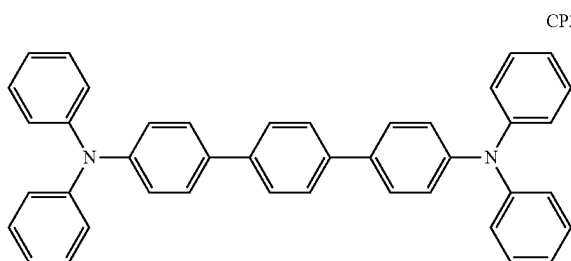
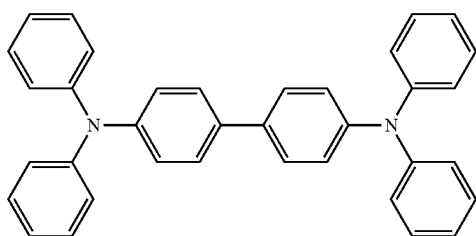
The first capping layer 210 and the second capping layer 220 may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or a composite capping layer including an organic material and an inorganic material.

At least one selected from the first capping layer 210 and the second capping layer 220 may each independently include at least one material selected from carbocyclic compounds, heterocyclic compounds, amine-based compounds, porphyrine derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, and alkaline earth-based complexes. The carbocyclic compound, the heterocyclic compound, and the amine-based compound may be optionally substituted with a substituent containing at least one element selected from O, N, S, Se, Si, F, Cl, Br, and I. In one embodiment, at least one selected from the first capping layer 210 and the second capping layer 220 may each independently include an amine-based compound.

In one embodiment, at least one selected from the first capping layer 210 and the second capping layer 220 may each independently include the compound represented by Formula 201 or the compound represented by Formula 202.

In one or more embodiments, at least one selected from the first capping layer 210 and the second capping layer 220 may each independently include a compound selected from Compounds HT28 to HT33 and Compounds CP1 to CP5, but embodiments of the present disclosure are not limited thereto.

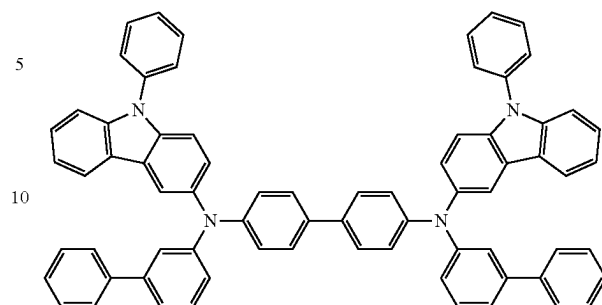
135



136

-continued

CP5



Hereinbefore, the organic light-emitting device according to an embodiment has been described in connection with FIGS. 1 to 4, but embodiments of the present disclosure are not limited thereto.

Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a set or certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10⁻⁸ torr to about 10⁻³ torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to about 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

General Definition of at Least Some of the Substituents

The term “C₁-C₆₀ alkyl group,” as used herein, refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “C₁-C₆₀ alkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₁-C₆₀ alkyl group.

The term “C₂-C₆₀ alkenyl group,” as used herein, refers to a hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C₂-C₆₀ alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C₂-C₆₀ alkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₂-C₆₀ alkenyl group.

The term “C₂-C₆₀ alkynyl group,” as used herein, refers to a hydrocarbon group having at least one carbon-carbon triple bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of the C₂-C₆₀ alkyl group, and examples thereof include an ethynyl group, and a

137

propynyl group. The term “C₂-C₆₀ alkynylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₂-C₆₀ alkynyl group.

The term “C₁-C₆₀ alkoxy group,” as used herein, refers to a monovalent group represented by —OA₁₀₁ (wherein A₁₀₁ is the C₁-C₆₀ alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

The term “C₃-C₁₀ cycloalkyl group,” as used herein, refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The term “C₃-C₁₀ cycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₃-C₁₀ cycloalkyl group.

The term “C₁-C₁₀ heterocycloalkyl group,” as used herein, refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranlyl group, and a tetrahydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₁-C₁₀ heterocycloalkyl group.

The term “C₃-C₁₀ cycloalkenyl group,” as used herein, refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity (e.g., is not aromatic), and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C₃-C₁₀ cycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₃-C₁₀ cycloalkenyl group.

The term “C₁-C₁₀ heterocycloalkenyl group,” as used herein, refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one carbon-carbon double bond in its ring. Non-limiting examples of the C₁-C₁₀ heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranlyl group, and a 2,3-dihydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C₁-C₁₀ heterocycloalkenyl group.

The term “C₆-C₆₀ aryl group,” as used herein, refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term “C₆-C₆₀ arylene group,” as used herein, refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C₆-C₆₀ aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C₆-C₆₀ aryl group and the C₆-C₆₀ arylene group each include two or more rings, the rings may be fused to each other (e.g., combined together).

The term “C₁-C₆₀ heteroaryl group,” as used herein, refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. The term “C₁-C₆₀ heteroarylene group,” as used herein, refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C₁-C₆₀ heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a

138

triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C₁-C₆₀ heteroaryl group and the C₁-C₆₀ heteroarylene group each include two or more rings, the rings may be condensed with each other (e.g., combined together).

The term “C₆-C₆₀ aryloxy group,” as used herein, refers to —OA₁₀₂ (wherein A₁₀₂ is the C₆-C₆₀ aryl group), and the term “C₆-C₆₀ arylthio group,” as used herein, indicates —SA₁₀₃ (wherein A₁₀₃ is the C₆-C₆₀ aryl group).

The term “monovalent non-aromatic condensed polycyclic group,” as used herein, refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with each other (e.g., combined together), only carbon atoms as ring-forming atoms, and containing at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed polycyclic group is a fluorenyl group. The term “divalent non-aromatic condensed polycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed polycyclic group.

The term “monovalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other (e.g., combined together), at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and containing at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed heteropolycyclic group is a carbazolyl group. The term “divalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

The term “C₅-C₆₀ carbocyclic group,” as used herein, refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The term “C₅-C₆₀ carbocyclic group,” as used herein, refers to an aromatic carbocyclic group or a non-aromatic carbocyclic group. The C₅-C₆₀ carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In one or more embodiments, depending on the number of substituents connected to the C₅-C₆₀ carbocyclic group, the C₅-C₆₀ carbocyclic group may be a trivalent group or a quadrivalent group.

The term “C₁-C₆₀ heterocyclic group,” as used herein, refers to a group having substantially the same structure as the C₅-C₆₀ carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

In the present specification, at least one substituent of the substituted C₁-C₆₀ heterocyclic group, the substituted C₃-C₁₀ cycloalkylene group, the substituted C₁-C₁₀ heterocycloalkylene group, the substituted C₃-C₁₀ cycloalkenylene group, the substituted C₁-C₁₀ heterocycloalkenylene group, the substituted C₆-C₆₀ arylene group, the substituted C₁-C₆₀ heteroarylene group, the substituted divalent non-aromatic condensed polycyclic group, the substituted divalent non-aromatic condensed heteropolycyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the

substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), and —P(=O)(Q₂₁)(Q₂₂);

a C₃-C₁₀ cycloalkyl group, a C₂-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₂-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₂-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), and —P(=O)(Q₂₁)(Q₂₂); and —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-a-

romatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

The term “Ph,” as used herein, represents a phenyl group, the term “Me,” as used herein, represents a methyl group, the term “Et,” as used herein, represents an ethyl group, the term “ter-Bu” or “But,” as used herein, represents a tert-butyl group, and the term “OMe,” as used herein, represents a methoxy group.

The term “biphenyl group,” as used herein, refers to a “phenyl group substituted with a phenyl group. The “biphenyl group” is a “substituted phenyl group” having a “C₆-C₆₀ aryl group” as a substituent.

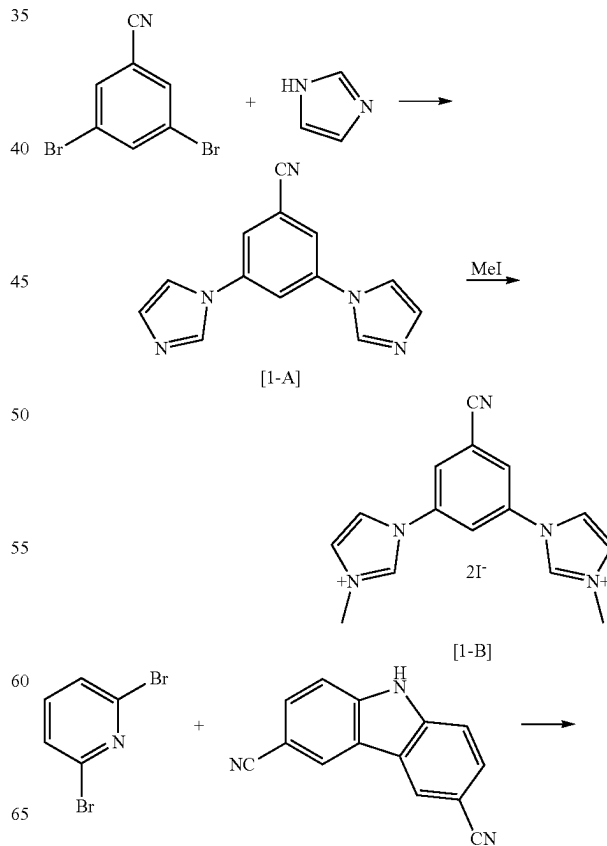
The term “terphenyl group,” as used herein, refers to a “phenyl group substituted with a biphenyl group. The “terphenyl group” is a “phenyl group” having, as a substituent, a “C₆-C₆₀ aryl group substituted with a C₆-C₆₀ aryl group”.

* and **, as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

Hereinafter, a compound according to embodiments and an organic light-emitting device according to embodiments will be described in more detail with reference to Synthesis Examples and Examples. The expression “B was used instead of A” used in describing Synthesis Examples means that an identical number of molar equivalents of B was used in place of molar equivalents of A.

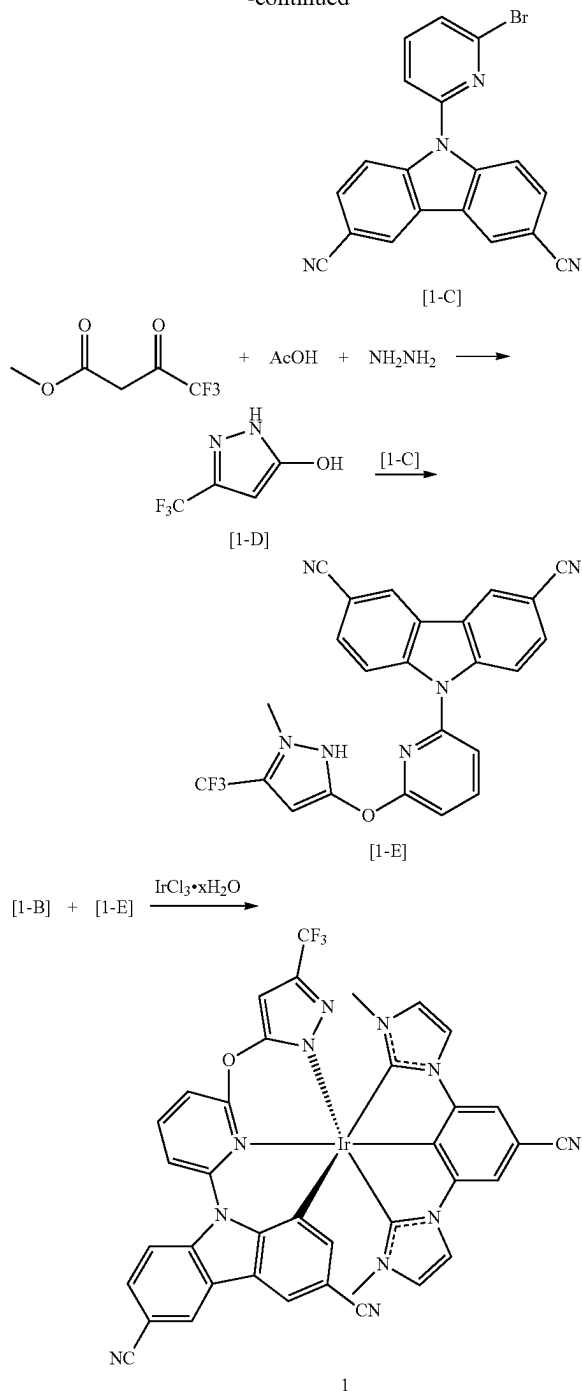
SYNTHESIS EXAMPLES

Synthesis Example 1: Synthesis of Compound 1



141

-continued



(1) Synthesis of Intermediate [1-A]

3,5-dibromobenzonitrile (1.0 eq), imidazole (3.0 eq), CuI (0.02 eq), K_2CO_3 (3.0 eq), and L-Proline (0.04 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-A] (yield: 42%).

142

(2) Synthesis of Intermediate [1-B]

Intermediate [1-A] (1.0 eq) and diiodomethane (3.0 eq) were dissolved in toluene (1.0 M) and stirred at a temperature of 120° C. for 12 hours. The reaction mixture was cooled to room temperature, washed with toluene, and filtered to obtain a solid. The obtained solid was dried to synthesize Intermediate [1-B] (yield: 68%).

(3) Synthesis of Intermediate [1-C]

2,6-dibromopyridine (1.0 eq), 9H-carbazole-3,6-dicarbonitrile (1.1 eq), CuI (0.01 eq), K_2CO_3 (2.0 eq), and L-Proline (0.02 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-C] (yield: 57%).

(4) Synthesis of Intermediate [1-D]

A reaction solution including 4,4,4-trifluoroacetamide methyl acetate (1.0 eq) and acetic acid (1.1 eq) was stirred at a temperature of 6° C., and hydrazine monohydrate (1.1 eq) was slowly added thereto for 0.3 hours. The reaction mixture was slowly heated to a temperature of 80° C. for 1 hour and stirred for 3 hours. After the reaction was believed to be completed, the reaction product was cooled to obtain Intermediate [1-D] (yield: 82%) in a crystal form.

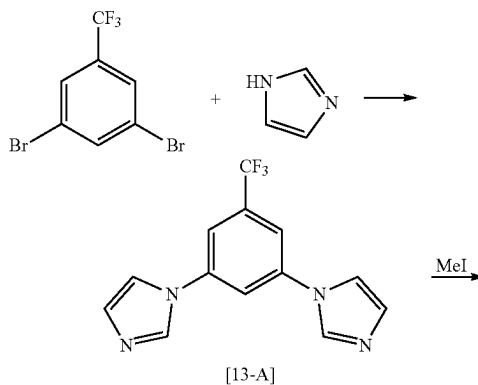
(5) Synthesis of Intermediate [1-E]

Intermediate [1-C] (1.0 eq), Intermediate [1-D] (1.2 eq), CuI (0.01 eq), K_2CO_3 (2.0 eq), and L-Proline (0.02 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-E] (yield: 47%).

(6) Synthesis of Compound 1

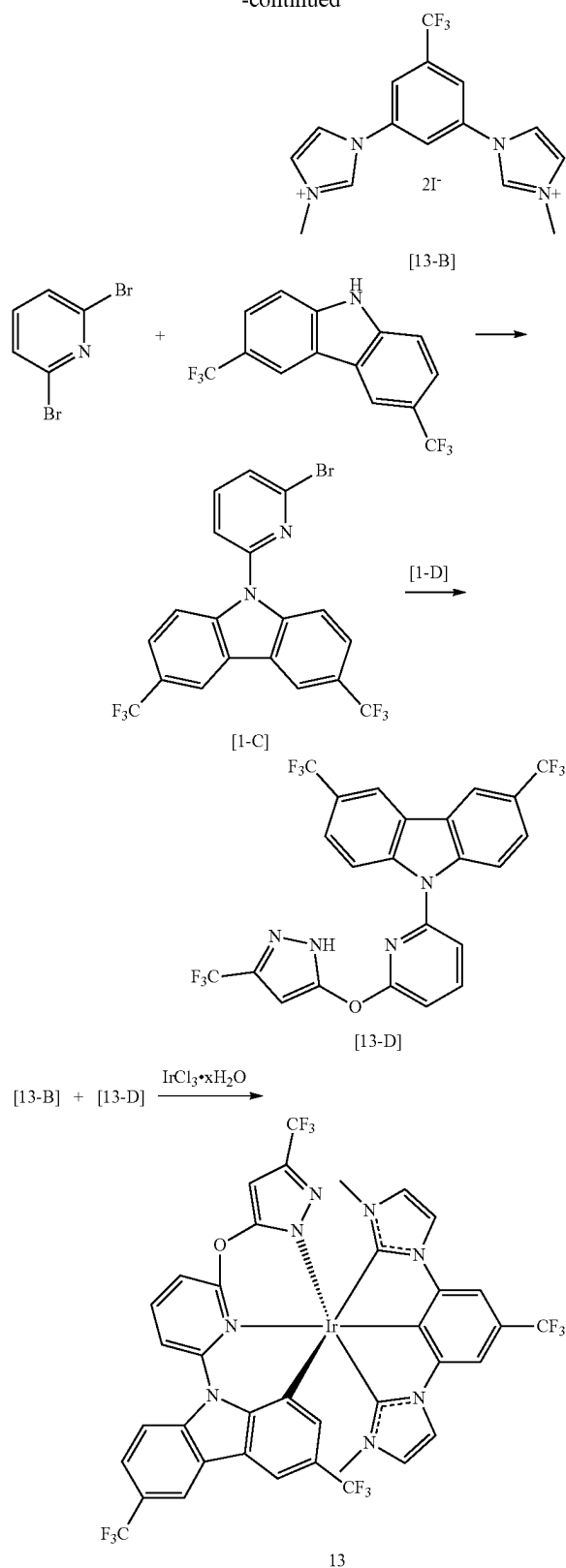
Intermediate [1-B] (1.0 eq), Intermediate [1-E] (1.0 eq), iridium chloride hydrate (1.1 eq), and K_2CO_3 (10 eq) were dissolved in propionic acid (0.2 M) and stirred at a temperature of 140° C. for 12 hours. The reaction mixture was cooled to room temperature, and a solvent was removed therefrom under reduced pressure. Then, an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Compound 1 (yield: 22%).

Synthesis Example 2: Synthesis of Compound 13



143

-continued

**(1) Synthesis of Intermediate [13-A]**

Intermediate [13-A] (yield: 52%) was synthesized in substantially the same manner as in Synthesis of Interme-

144

diate [1-A], except that 3,5-dibromotrifluoromethane was used instead of 3,5-dibromobenzonitrile.

(2) Synthesis of Intermediate [13-B]

Intermediate [13-B] (yield: 80%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-B], except that Intermediate [13-A] was used instead of Intermediate [1-A].

(3) Synthesis of Intermediate [13-C]

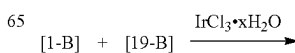
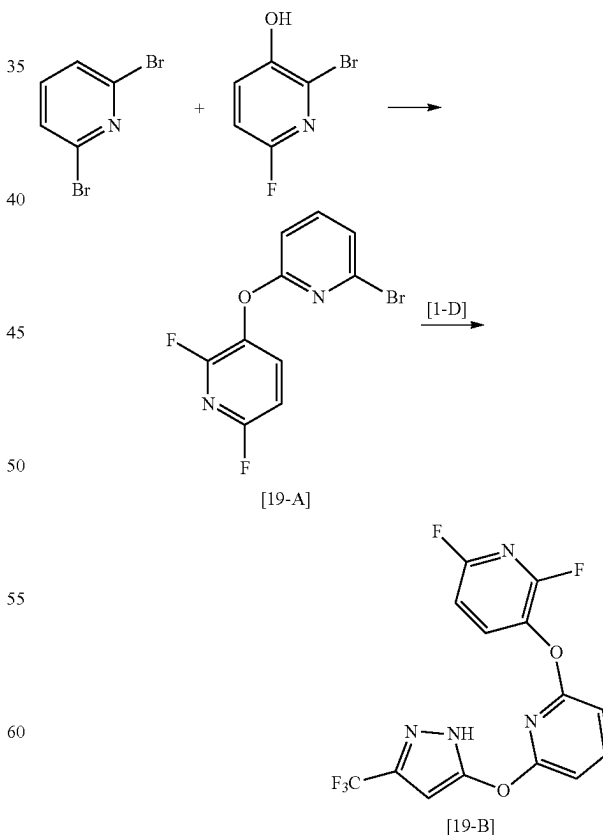
Intermediate [13-C] (yield: 67%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-C], except that 3,6-bis(trifluoromethyl)-9H-carbazole was used instead of 9H-carbazole-3,6-dicarbonitrile.

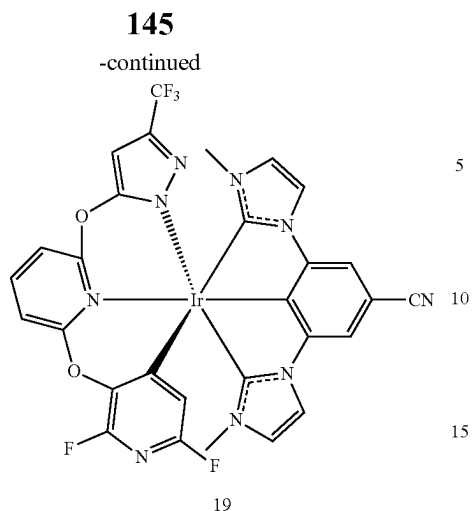
(4) Synthesis of Intermediate [13-D]

Intermediate [13-D] (yield: 76%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [13-C] was used instead of Intermediate [1-C].

(5) Synthesis of Compound 13

Compound 13 (yield: 26%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [13-B] was used instead of Intermediate [1-B], and Intermediate [13-D] was used instead of Intermediate [1-E].

Synthesis Example 3: Synthesis of Compound 19



(1) Synthesis of Intermediate [19-A]

Intermediate [19-A] (yield: 49%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-C], except that 2,6-difluoropyridin-3-ol was used instead of 9H-carbazole-3,6-dicarbonitrile.

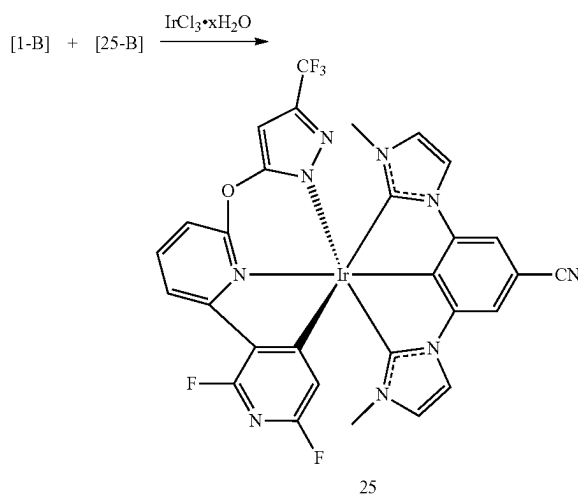
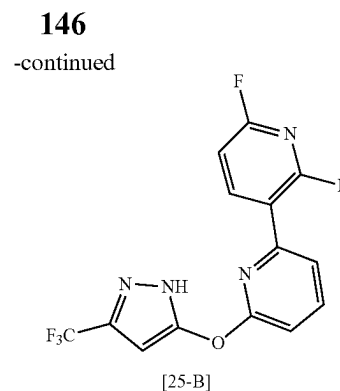
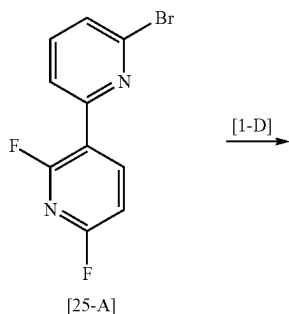
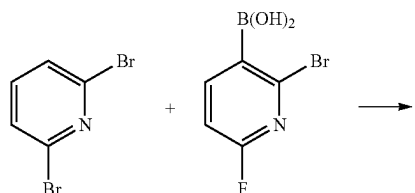
(2) Synthesis of Intermediate [19-B]

Intermediate [19-B] (yield: 72%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [19-A] was used instead of Intermediate [1-C].

(3) Synthesis of Compound 19

Compound 19 (yield: 23%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [19-B] was used instead of Intermediate [1-E].

Synthesis Example 4: Synthesis of Compound 25



(1) Synthesis of Intermediate [25-A]

2,6-dibromopyridine (1.0 eq), (2,6-difluoropyridin-3-yl) boronic acid (1.1 eq), $\text{Pd}(\text{PPh}_3)_4$ (0.002 eq), and K_2CO_3 (1.0 eq) were dissolved in tetrahydrofuran/ H_2O (2:1, 0.1 M) and stirred at a temperature of 70° C. for 12 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [25-A] (yield: 68%).

(2) Synthesis of Intermediate [25-B]

Intermediate [25-B] (yield: 76%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [25-A] was used instead of Intermediate [1-C].

(3) Synthesis of Compound 25

Compound 25 (yield: 27%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [25-B] was used instead of Intermediate [1-E].

^1H NMR and LC-MS of Compounds synthesized according to Synthesis Examples 1 to 4 are shown in Table 1.

Synthesis methods of compounds other than Compounds shown in Table 1 may also be easily recognized by those of ordinary skill in the art by referring to the synthesis mechanisms and source materials described herein above.

TABLE 1

Compound No.	¹ H NMR (CDCl ₃ , 400 MHz)	LC-MS	
		found	calc.
1	8.22-7.02 (m, 7H), 6.52-6.48 (m, 4H), 5.21 (d, 2H), 5.06 (d, 2H), 2.69 (s, 6H)	897.2	897.1
13	9.11 (s, 1H), 8.34-7.03 (m, 6H), 6.64 (s, 2H), 6.51-6.49 (m, 2H), 5.24 (d, 2H), 5.01 (d, 2H), 2.67 (s, 6H)	1025.9	1025.8
19	7.65-7.45 (m, 2H), 6.54-6.47 (m, 3H), 5.54 (d, 2H), 5.23 (d, 2H), 5.04 (d, 2H), 2.69 (s, 6H)	811.1	811.1
25	7.86-7.00 (m, 3H), 6.65 (d, 1H), 6.52-6.48 (m, 3H), 5.24 (d, 2H), 5.02 (d, 2H), 2.71 (s, 6H)	795.0	795.1

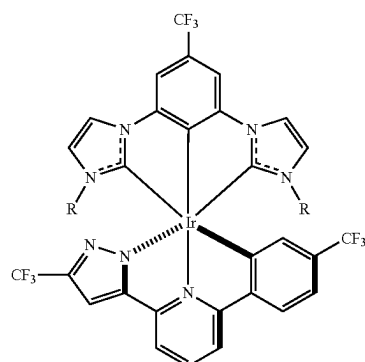
Evaluation Example

Wavelengths of maximum emission (λ_{max}) of Compounds synthesized according to Synthesis Examples 1 to 4 were measured, and ³MC state energy of Compounds synthesized according to Synthesis Examples 1 to 4 were obtained through quantum simulation. Results thereof are shown in Table 2. The wavelength of maximum emission (λ_{max}) was measured by using a photoluminescence (PL) measurement apparatus. The ³MC state energy was calculated through quantum simulation by using a density functional theory (DFT) method and the Gaussian program (available from Gaussian Inc.) where the respective compound was structurally optimized at a level of B3LYP/6-31G(d,p) (e.g., using the B3LYP hybrid functional and the 6-31G(d,p) basis set).

TABLE 2

	λ_{max} (nm)	³ MC state (eV)
Comparative Compound 1	472	0.24
1		
Comparative Compound 2	498	0.15
2		
Comparative Compound 3	502	0.21
3		
Comparative Compound 4	470	0.02
4		
Compound 1	468	0.24
Compound 13	464	0.23
Compound 19	449	0.19
Compound 25	456	0.21

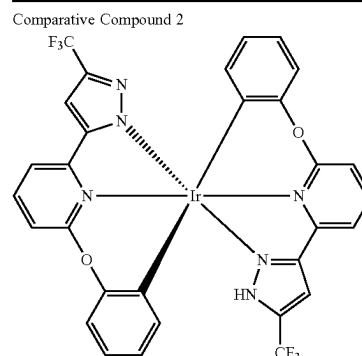
Comparative Compound 1



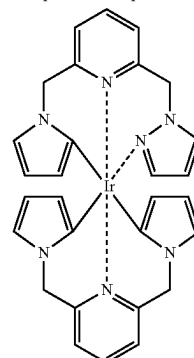
15

TABLE 2-continued

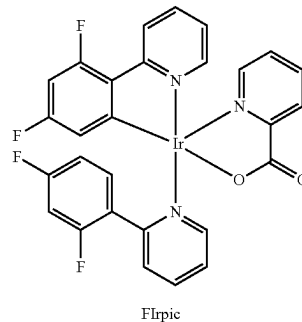
	λ_{max} (nm)	³ MC state (eV)
Comparative Compound 2		
20		
25		
30		
Comparative Compound 3		
35		
40		
45		
Comparative Compound 4		
50		
55		
60		
65		



Comparative Compound 3



Comparative Compound 4



While the present application is not limited by any particular mechanism or theory, it is believed that as a bond between a ligand and a metal center becomes stronger, energy at which the ³MC state is formed in a triplet state further increases.

From the results of Table 2, it can be seen that Compounds 1, 13, 19, and 25 according to one or more embodiments

149

have high ^3MC state energy, as compared with Comparative Example 4 including the bidentate ligand.

EXAMPLES

Example 1

As an anode, a Corning 15 Ω/cm^2 (1,200 Å) ITO glass substrate was cut to a size of 50 mm×50 mm×0.7 mm, sonicated with isopropyl alcohol and pure water each for 5 minutes, and then cleaned by exposure to ultraviolet rays and ozone for 30 minutes. Then, the ITO glass substrate was provided to a vacuum deposition apparatus. 2-TNATA was vacuum-deposited on the ITO glass substrate to form a hole injection layer having a thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenyl aminobiphenyl (NPB) was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 300 Å. Compound 1 (dopant) (weight ratio of 10%) and 3,3-di(9H-carbazol-9-yl)biphenyl (mCBP) (host) were co-deposited on the hole transport layer to form an emission layer having a thickness of 300 Å. Then, diphenyl(4-(triphenylsilyl)phenyl)-phosphine oxide (TSPO1) was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å. Then, Alq_3 was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, and an alkali metal halide LiF was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited to a thickness of 3,000 Å (cathode electrode) to form a LiF/Al electrode, thereby completing the manufacture of an organic light-emitting device.

Examples 2 to 4

Organic light-emitting devices were manufactured in substantially the same manner as in Example 1, except that Compounds shown in Table 3 were each used instead of Compound 1 as a dopant in forming an emission layer.

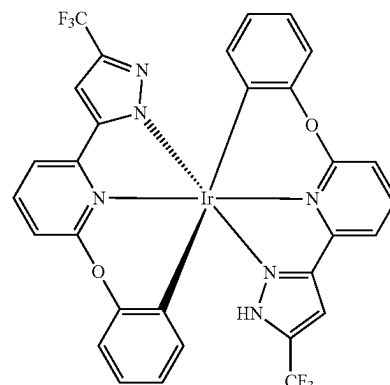
Comparative Examples 1 to 4

Organic light-emitting devices were manufactured in substantially the same manner as in Example 1, except that Comparative Compounds 1 to 4 were respectively used instead of Compound 1 as a dopant in forming an emission layer.

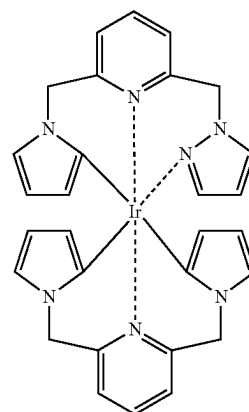
150

-continued

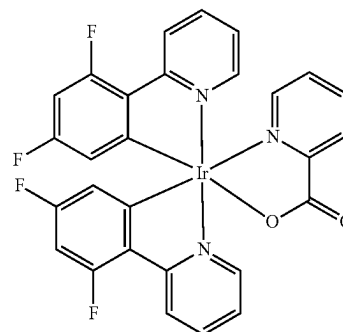
Comparative Compound 2



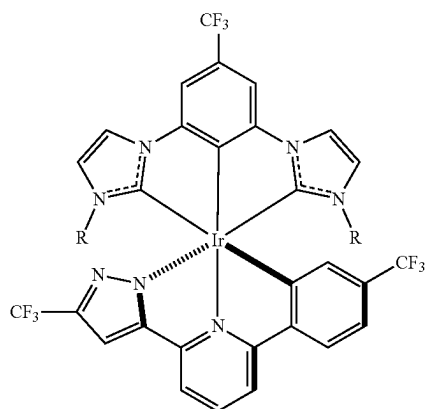
Comparative Compound 3



Comparative Compound 4



Comparative Compound 1



The driving voltage, luminance, luminescent efficiency, and maximum emission wavelength of the organic light-emitting devices manufactured according to Examples 1 to 4 and Comparative Examples 1 to 4 were measured at a current density of 50 mA/cm^2 by using a Keithley SMU 236 and a luminance meter PR650, and an amount of time (half lifespan) that lapsed when luminance was 50% of initial luminance. Results thereof are shown in Table 3.

TABLE 3

	Emission layer	Driving voltage (V)	Current density (mA/cm ²)	Luminance (cd/m ²)	Efficiency (cd/A)	Maximum emission wavelength (nm)	Half lifespan (hr)
Example 1	Compound 1	5.32	50	4125	8.25	469	13
Example 2	Compound 13	5.42	50	4025	8.05	467	15
Example 3	Compound 19	5.48	50	4033	8.07	452	10
Example 4	Compound 25	5.46	50	4024	8.05	457	11
Comparative Example 1	Comparative Compound 1	6.24	50	3785	7.57	475	12
Comparative Example 2	Comparative Compound 2	5.71	50	3570	7.14	501	14
Comparative Example 3	Comparative Compound 3	6.85	50	3860	7.72	502	9
Comparative Example 4	Comparative Compound 4	6.56	50	3870	7.74	470	3

From Table 3, it can be seen that the organic light-emitting devices of Examples 1 to 4, in which the compounds according to one or more embodiments are used as the dopant of the emission layer have a low driving voltage, high efficiency, and high color purity, as compared with the organic light-emitting devices of Comparative Examples 1 to 4. In addition, it can be seen that the lifespan is remarkably improved, as compared with the organic light-emitting devices of Comparative Examples 3 and 4.

When the compounds according to one or more embodiments are used for the organic light-emitting device, high color purity may be implemented and excellent effects may be exhibited in terms of driving voltage, efficiency, and lifespan.

The organic light-emitting device including the organo-metallic compound may have a low driving voltage, high efficiency, and a long lifespan and facilitate emission wavelength adjustment, thereby exhibiting high color purity.

It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should be considered as available for other similar features or aspects in other embodiments.

It will be understood that, although the terms “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present disclosure.

Spatially relative terms, such as “beneath,” “below,” “lower,” “under,” “above,” “upper,” and the like, may be used herein for ease of explanation to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or in operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the example terms “below” and “under” can encompass both an orientation of above and below.

The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein should be interpreted accordingly.

It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it can be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and “including,” when used in this specification, specify the presence of the stated features, integers, acts, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, acts, operations, elements, components, and/or groups thereof.

As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art. Further, the use of “may” when describing embodiments of the present disclosure refers to “one or more embodiments of the present disclosure.” As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. Also, the term “exemplary” is intended to refer to an example or illustration.

Also, any numerical range recited herein is intended to include all subranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limita-

153

tions subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein.

While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the appended claims, and equivalents thereof.

What is claimed is:

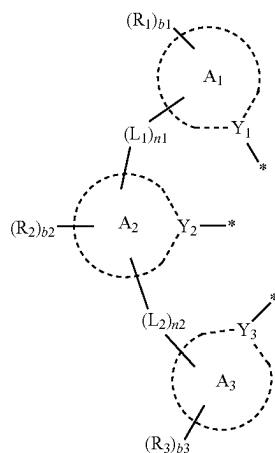
1. An organometallic compound represented by Formula 1:



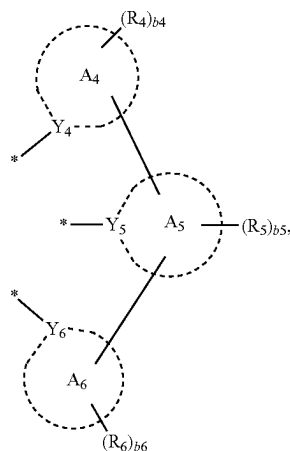
wherein, in Formula 1,

M is selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm),

L_A is a tridentate ligand represented by Formula 1A, and L_B is a tridentate ligand represented by Formula 1B:



Formula 1A



Formula 1B

wherein, in Formulae 1A and 1B, ring A_1 is a 5-membered ring including Y_1 , a C_5 - C_{60} carbocyclic ring in which a 5-membered ring including

154

Y_1 is condensed, or a C_1 - C_{60} heterocyclic ring in which a 5-membered ring including Y_1 is condensed,

ring A_2 to ring A_6 are each independently selected from a C_5 - C_{60} carbocyclic ring and a C_1 - C_{60} heterocyclic ring, Y_1 to Y_6 are each independently N or C,

L_1 is selected from $*-C\equiv C-*$, $*-B(R_7)-*$, $*-P(R_7)-*$, $*-P(R_7)(R_8)-*$, and $*-Ge(R_7)(R_8)-*$, L_2 is selected from $*-O-*$, $*-S-*$, $*-C(R_7)(R_8)-*$, $*-C(R_7)=*$, $*=C(R_7)-*$, $*-C(R_7)=C(R_8)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R_7)-*$, $*-N(R_7)-*$, $*-P(R_7)-*$, $*-Si(R_7)(R_8)-*$, $*-P(R_7)(R_8)-*$, and $*-Ge(R_7)(R_8)-*$,

n_1 is selected from 1, 2, and 3, and n_2 is selected from 0, 1, 2, and 3,

R_1 to R_6 and R_8 are each independently selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_1)(Q_2)(Q_3)$, $-N(Q_1)(Q_2)$, $-B(Q_1)(Q_2)$, $-C(=O)(Q_1)$, $-S(=O)_2(Q_1)$, and $-P(=O)(Q_1)(Q_2)$,

R_7 is selected from an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_1)(Q_2)(Q_3)$, $-N(Q_1)(Q_2)$, $-B(Q_1)(Q_2)$, $-C(=O)(Q_1)$, $-S(=O)_2(Q_1)$, and $-P(=O)(Q_1)(Q_2)$,

two neighboring $R_1(s)$ among $R_1(s)$ in the number of b_1 , two neighboring $R_2(s)$ among $R_2(s)$ in the number of b_2 , two neighboring $R_3(s)$ among $R_3(s)$ in the number of b_3 , two neighboring $R_4(s)$ among $R_4(s)$ in the number of b_4 , two neighboring $R_5(s)$ among $R_5(s)$ in the number of b_5 , two neighboring $R_6(s)$ among $R_6(s)$

155

in the number of b6, and at least one of R₁ and R₇, R₂ and R₇, and R₅ and R₇ are optionally linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group.

b1 to b6 are each independently an integer from 1 to 10, * indicates a binding site to M,

at least one substituent of the substituted C₅-C₆₀ carbocyclic group, the substituted C₁-C₆₀ heterocyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is selected from:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), and —P(=O)(Q₁₁)(Q₁₂);

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)

156

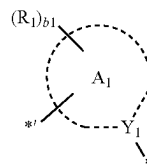
(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), and —P(=O)(Q₂₁)(Q₂₂); and

—Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

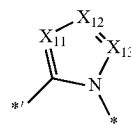
Q₁ to Q₃, Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group,

wherein, when ring A₃ is selected from a C₅-C₆₀ carbocyclic ring and a C₁-C₆₀ heterocyclic ring other than a carbazole group, R₅ is selected from deuterium, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —N(Q₁)(Q₂), —B(Q₁)(Q₂), —C(=O)(Q₁), —S(=O)₂(Q₁), and —P(=O)(Q₁)(Q₂).

2. The organometallic compound of claim 1, wherein, a group represented by



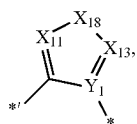
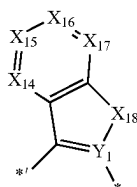
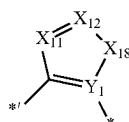
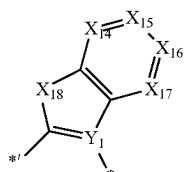
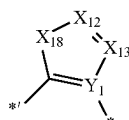
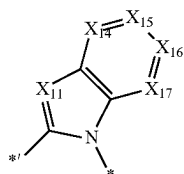
in Formula 1A is selected from groups represented by Formulae A1-1 to A1-7:



A1-1

157

-continued



wherein, in Formulae A1-1 to A1-7,

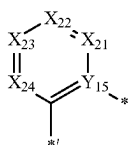
X₁₁ is C(R₁₁) or N, X₁₂ is C(R₁₂) or N, X₁₃ is C(R₁₃) or N, X₁₄ is C(R₁₄) or N, X₁₅ is C(R₁₅) or N, X₁₆ is C(R₁₆) or N, and X₁₇ is C(R₁₇) or N,

X₁₈ is C(R₁₈)(R₁₉), Si(R₁₈)(R₁₉), N(C₁₈), P(C₁₈), O, or S, R₁₁ to R₁₉ are each independently the same as defined in connection with R₁ in Formula 1A, and

* indicates a binding site to M, and * indicates a binding site to (L₁)_{a1}.

3. The organometallic compound of claim 1, wherein, ring A₃, ring A₄, and ring A₆ are each independently selected from groups represented by Formulae 2-1(1) to 2-1(27), and

ring A₂ and ring A₅ are each independently selected from groups represented by Formulae 2-2(1) to 2-2(19):

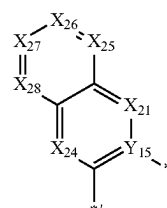


158

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A1-2

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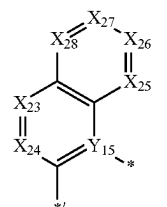


A1-3

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A1-4

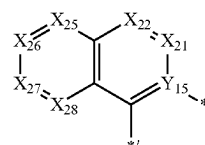
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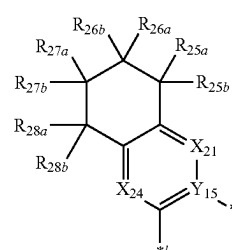
A1-5

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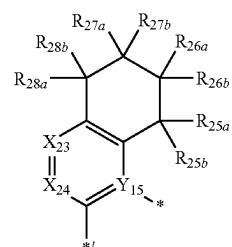
A1-6

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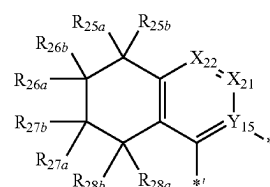


A1-7

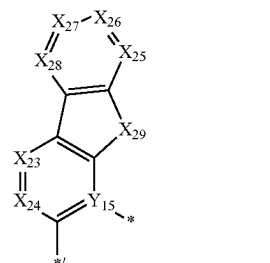
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2-1(2)

2-1(3)

2-1(4)

2-1(5)

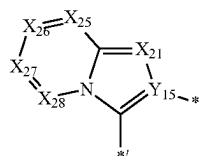
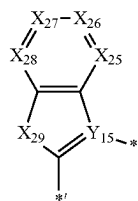
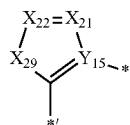
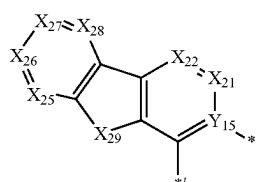
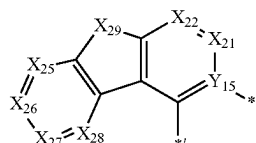
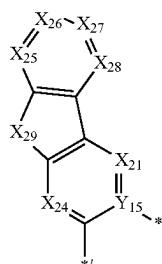
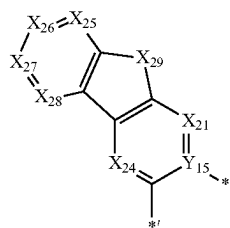
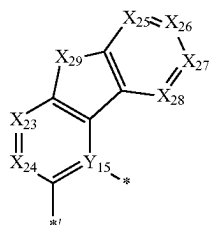
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2-1(7)

2-1(8)

159

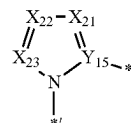
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**160**

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2-1(9)

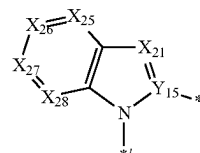
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2-1(17)

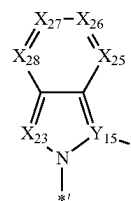
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2-1(18)

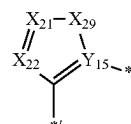
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2-1(19)

2-1(11)

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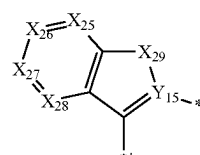


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2-1(12)

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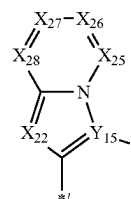


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2-1(13)

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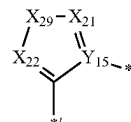


2-1(22)

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2-1(14)

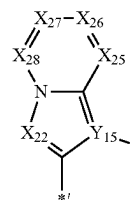
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2-1(23)

2-1(15)

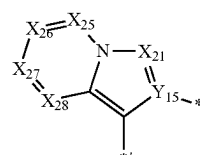
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2-1(24)

2-1(16)

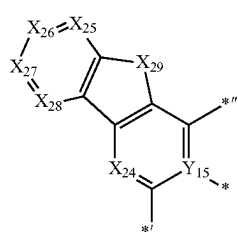
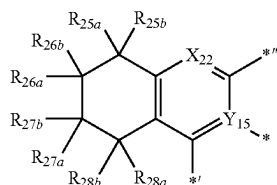
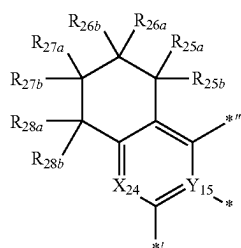
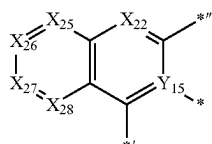
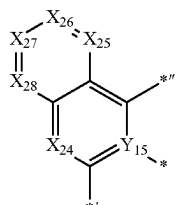
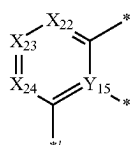
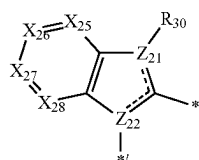
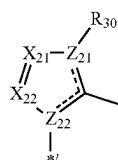
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2-1(25)

161

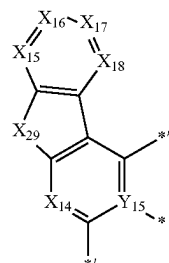
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**162**

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2-1(26)

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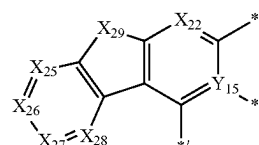
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2-1(27)

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2-2(1)

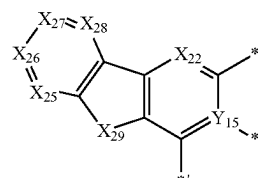


2-2(8)

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2-2(2)

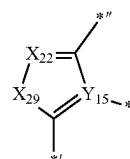
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2-2(9)

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2-2(3)

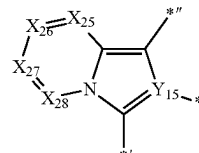


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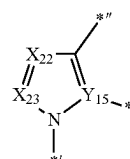
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2-2(11)

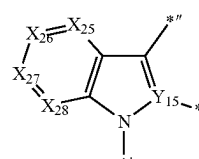
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2-2(12)

2-2(5)

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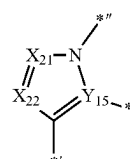


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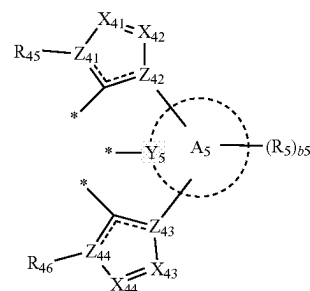
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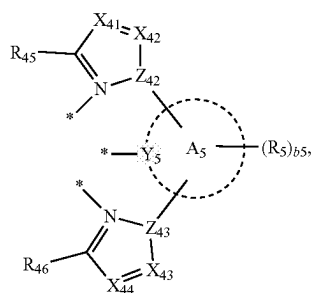
2-2(14)

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165

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wherein, in Formulae 1B-11 and 1B-12,

Y_5 , ring A_5 , R_5 , and b_5 are each independently the same as defined in claim 1,

Z_{41} to Z_{44} are each independently N or P,

X_{41} is C(R_{41}) or N, X_{42} is C(R_{42}) or N, X_{43} is C(R_{43}) or N, and X_{44} is C(R_{44}) or N,

R_{41} to R_{46} are each independently the same as defined in connection with R_4 and R_6 in Formula 1B, and

* indicates a binding site to M.

11. The organometallic compound of claim 10, wherein, Z_{41} to Z_{44} are each N.

12. The organometallic compound of claim 1, wherein, at least one selected from $R_1(s)$ in the number of b_1 , $R_2(s)$ in the number of b_2 , $R_3(s)$ in the number of b_3 , $R_4(s)$ in the number of b_4 , $R_5(s)$ in the number of b_5 , and $R_6(s)$ in the number of b_6 is selected from a cyano group, a C_1 - C_{10} alkyl group substituted with at least one —F, a C_1 - C_{10} alkyl group substituted with at least one —Cl, a C_1 - C_{10} alkyl group substituted with at least one —Br, and a C_1 - C_{10} alkyl group substituted with at least one —I.

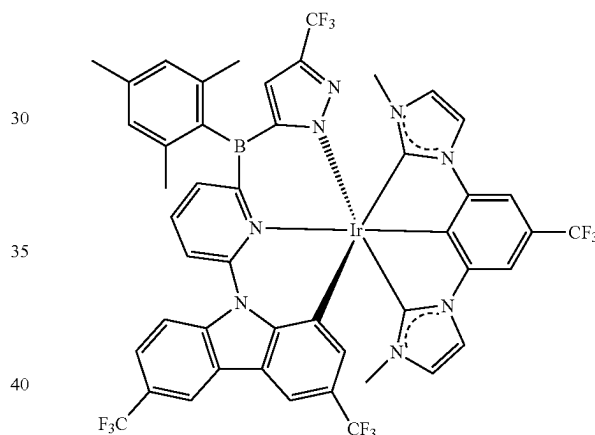
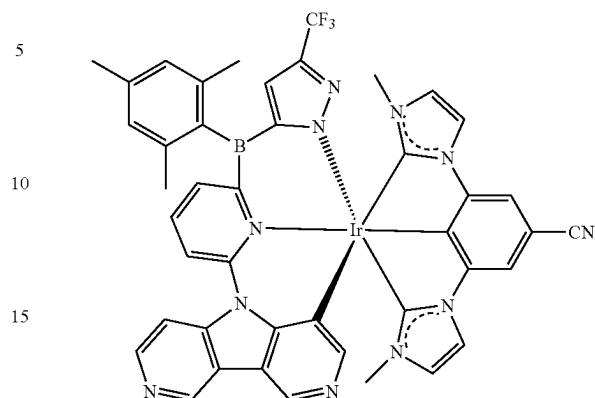
13. An organometallic compound being selected from Compounds 5, 11, 17, 23, 29, 35, 41, 47, 53, 59, and 65:

166

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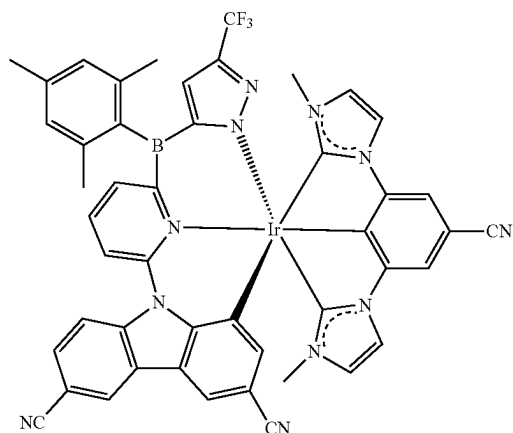
1B-12

11



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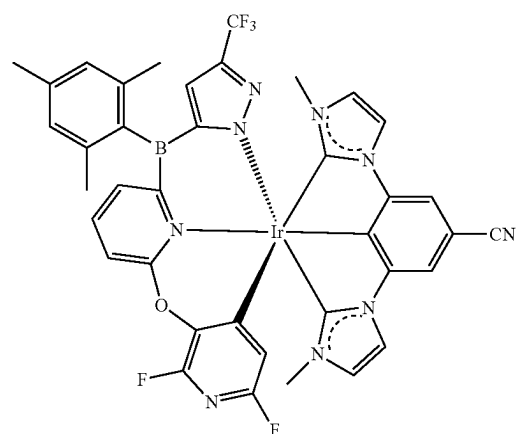


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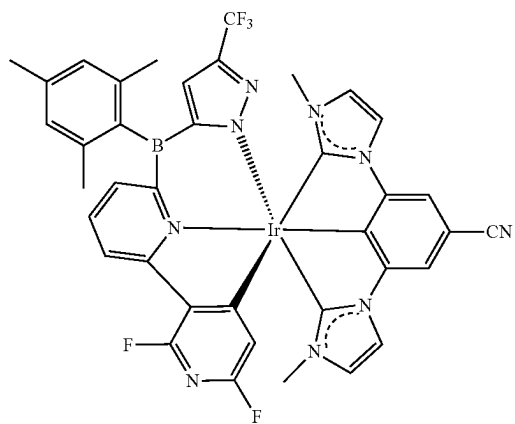
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167

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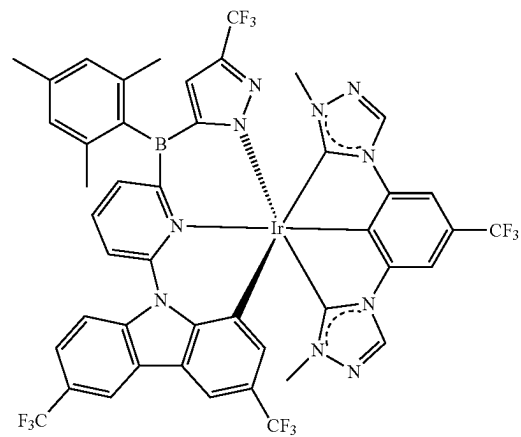
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168

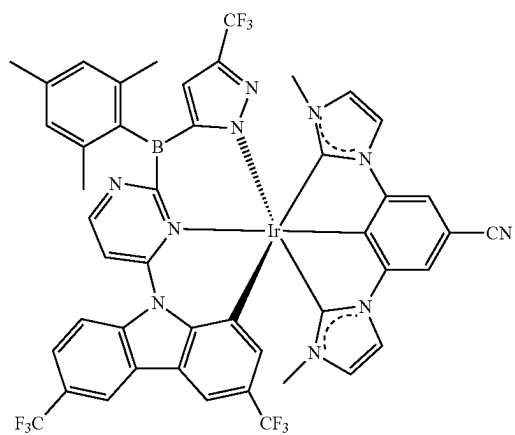
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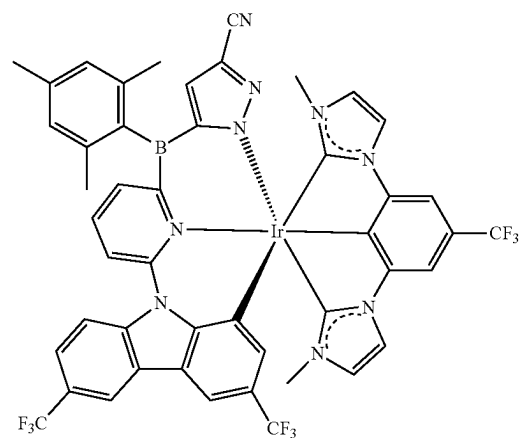


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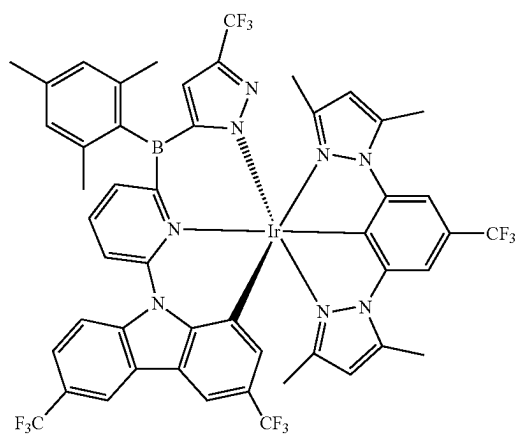
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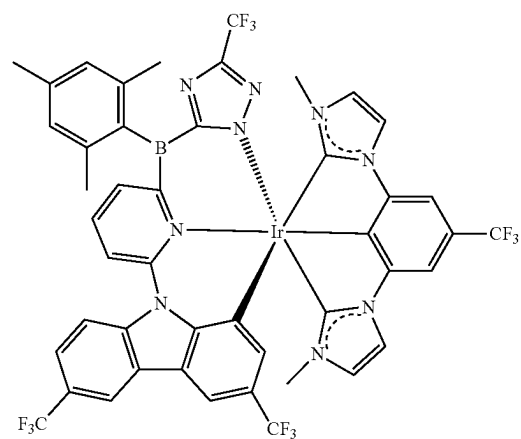


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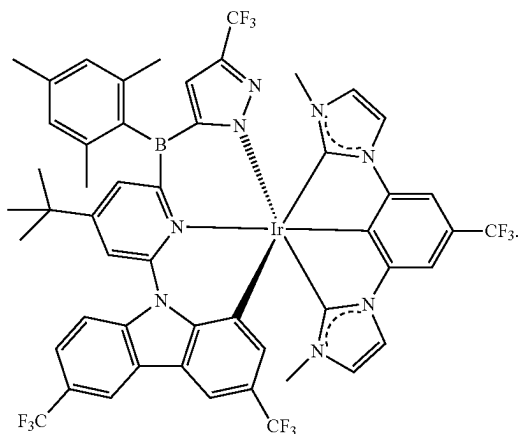
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169

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14. An organic light-emitting device comprising:
a first electrode;
a second electrode facing the first electrode;
an organic layer between the first electrode and the second
electrode and comprising an emission layer; and
at least one of the organometallic compound of claim 1.
15. The organic light-emitting device of claim 14,
wherein,
the organic layer comprises the organometallic com-
pound.
16. The organic light-emitting device of claim 14,
wherein,
the emission layer comprises the organometallic com-
pound.
17. The organic light-emitting device of claim 16,
wherein,
the emission layer further comprises a host, and an
amount of the organometallic compound comprised in
the emission layer is in a range of about 0.01 parts by
weight to about 30 parts by weight based on 100 parts
by weight of the emission layer.
18. The organic light-emitting device of claim 16,
wherein,
the emission layer emits a light having a wavelength of
maximum emission of 450 nm or more and 550 nm or
less.
19. The organic light-emitting device of claim 14,
wherein,
the first electrode is an anode,
the second electrode is a cathode,
the organic layer further comprises a hole transport region
between the first electrode and the emission layer and
an electron transport region between the emission layer
and the second electrode,
the hole transport region comprises a hole injection layer,
a hole transport layer, an emission auxiliary layer, an
electron blocking layer, or any combination thereof, and
the electron transport region comprises a buffer layer, a
hole blocking layer, an electron transport layer, an
electron injection layer, or any combination thereof.
20. The organic light-emitting device of claim 19,
wherein,
the hole transport region comprises a p-dopant having a
lowest unoccupied molecular orbital (LUMO) energy
level of -3.5 eV or less.

170

21. An organometallic compound represented by Formula

1:



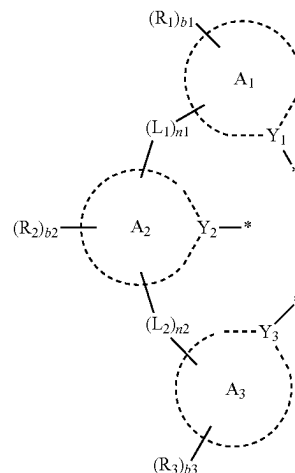
Formula 1

wherein, in Formula 1,

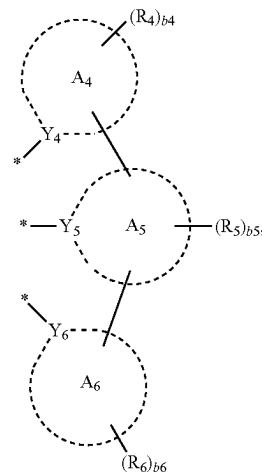
M is selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm),

L_A is a tridentate ligand represented by Formula 1A, and L_B is a tridentate ligand represented by Formula 1B:

Formula 1A



Formula 1B



wherein, in Formulae 1A and 1B,

ring A_1 is a 5-membered ring including Y_1 , a C_5-C_{60} carbocyclic ring in which a 5-membered ring including Y_1 is condensed, or a C_1-C_{60} heterocyclic ring in which a 5-membered ring including Y_1 is condensed,

ring A_2 to ring A_6 are each independently selected from a C_5-C_{60} carbocyclic ring and a C_1-C_{60} heterocyclic ring, Y_1 to Y_6 are each independently N or C,

L_1 is selected from $*-C\equiv C-*$, $*-B(R_7)-*$, $*-P(R_7)-*$, $*-P(R_7)(R_8)-*$, and $*-Ge(R_7)(R_8)-*$, L_2 is selected from $*-O-*$, $*-S-*$, $*-C(R_7)(R_8)-*$, $*-C(R_7)=*$, $*-C(R_7)-*$, $*-C(R_7)=C(R_8)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$,

171

—B(R₇)—, *—N(R₇)—*, *—P(R₇)—*, *—Si(R₇)(R₈)—*, *—P(R₇)(R₈)—*, and *—Ge(R₇)(R₈)—*

R₁ to R₆ and R_s are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —N(Q₁)(Q₂), —B(Q₁)(Q₂), —C(=O)(Q₁), —S(=O)₂(Q₁), and —P(=O)(Q₁)(Q₂),

R₇ is selected from an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —N(Q₁)(Q₂), —B(Q₁)(Q₂), —C(=O)(Q₁), —S(=O)₂(Q₁), and —P(=O)(Q₁)(Q₂),

two neighboring R₁(s) among R₁(s) in the number of b1, two neighboring R₂(s) among R₂(s) in the number of b2, two neighboring R₃(s) among R₃(s) in the number of b3, two neighboring R₄(s) among R₄(s) in the number of b4, two neighboring R₅(s) among R₅(s) in the number of b5, two neighboring R₆(s) among R₆(s) in the number of b6, and at least one of R₁ and R₇, R₂ and R₇, and R₃ and R₇ are optionally linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

172

b1 to b6 are each independently an integer from 1 to 10, * indicates a binding site to M,

at least one substituent of the substituted C₅-C₆₀ carbocyclic group, the substituted C₁-C₆₀ heterocyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is selected from:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-Coo alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-Coo heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), and —P(=O)(Q₁₁)(Q₁₂);

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group,

173

eropolycyclic group, $-\text{Si}(\text{Q}_{21})(\text{Q}_{22})(\text{Q}_{23})$, $-\text{N}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{B}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{C}(=\text{O})(\text{Q}_{21})$, $-\text{S}(=\text{O})_2(\text{Q}_{21})$, and $-\text{P}(=\text{O})(\text{Q}_{21})(\text{Q}_{22})$; and

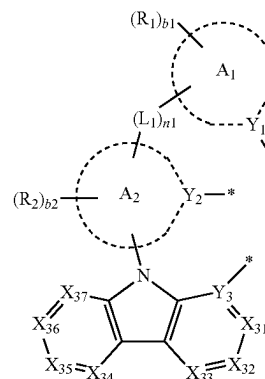
—Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂),
—C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

Q₁ to Q₃, Q₁₁to Q₁₃, Q₂₁to Q₂₃, and Q₃₁to Q₃₃ are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-Coo alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group,

n1 is selected from 1, 2, and 3, and n2 is selected from 0, 1, 2, and 3, provided that when n2 is 0, L_A in Formula 1 is a group represented by Formula 1C:

174

Formula 1C



wherein, in Formula 1C,
ring A₁, ring A₂, Y₁ to Y₃, L₁, n1, R₁, R₂, b1, and b2 are
each independently the same as described in Formula
1A.

X_{31} is $C(R_{31})$ or N , X_{32} is $C(R_{32})$ or N , X_{33} is $C(R_{33})$ or N , X_{34} is $C(R_{34})$ or N , X_{35} is $C(R_{35})$ or N , X_{36} is $C(R_{36})$ or N , and X_{37} is $C(R_{37})$ or N ,

R₃₁ to R₃₇ are each independently the same as defined in connection with R₃ in Formula 1A, and

* indicates a binding site to M.

* * * * *